



KONICA MINOLTA



DIRECT DIGITIZER

REGIUS Σ / Σ II



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Service Manual

Version 01

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Introduction

The Direct Digitizer REGIUS Σ and/or REGIUS Σ II (called the "device" hereinafter) is an X-ray image cassette reading device that uses photostimulable phosphor as an X-ray detector.

This manual states the cautions and steps for service engineers to perform repair or maintenance of this device (or a system including this device).

After reading through this manual, make sure it is readily accessible for future reference.

How to Read this Manual

Structure of This Manual

This manual is divided into the following 9 chapters describing device repair and maintenance methods, as well as safety precautions and operation.

Chapter 1: Safety Warnings & Cautions

An explanation of the safety warnings and cautions to follow while repairing or maintaining this device.

Chapter 2: Before Repairing

An explanation of device fundamentals that need to be understood before attempting to repair or maintain this device.

Chapter 3: Troubleshooting

An explanation of the various methods for dealing with device-related troubles.

Chapter 4: Using a Service Tool to Confirm Behavior

An explanation of how to confirm device operation.

Chapter 5: Disassembly & Assembly

An explanation of the steps needed to disassemble and assemble the device.

Chapter 6: Making Adjustments

An explanation of part adjustments necessary for repair and maintenance.

Chapter 7: Maintenance & Inspection

An explanation of how to maintain and inspect the device.

Chapter 8: Appendix

A list of technical information for reference during device repair or maintenance.

Chapter 9: Revision History

Cautions

- 1) Unauthorized reproduction of any part of this manual is prohibited.
- 2) The contents of this manual are subject to change without notice.
- 3) Any discrepancies, errors or omissions noted should be communicated to the manufacturer.
- 4) Notwithstanding item 3) above, the manufacturer accepts no responsibility whatsoever for any loss or decrease in profits arising from usage of the product.

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IMPORTANT

Settings and behavior confirmation of the device are performed with the service tools of REGIUS ImagePilot (hereinafter ImagePilot). Service tool operation details can be found in the ImagePilot Installation & Service Manual, so you are recommended to refer to it when necessary.

Difference between REGIUS Σ and REGIUS Σ II

Parts changed

The following REGIUS Σ parts are replaced in the REGIUS Σ II.

No.	Category	Name	Outline of the change
1	Parts	Front panel	Logo (REGIUS Σ II) is printed
2	Parts	Rating plate	-
3	Parts	Sensor on each part_relay Assy	Pin-out
4	Parts	LED_BOARD	LED2_BOARD
5	Parts	LED_DRIVE_BOARD	LED_DRIVE2_BOARD
6	Parts	FFC cable_Assy	Number of pins, Connector connection method
7	Parts	SW power supply - MAIN_BOARD_Assy	With a ferrite core
8	Unit	Eraser Unit	- LED2_BOARD - LED_DRIVE2_BOARD - FFC cable_Assy

Board Compatibility

Equipment and Board

- REGIUS Σ can use both new and old boards.
- REGIUS Σ is also be planned to be integrated in the new board.

Equipment \ Board	Old	New
	- LED_BOARD - LED_DRIVE_BOARD	- LED2_BOARD - LED_DRIVE2_BOARD
REGIUS Σ	○	○
REGIUS Σ II	×	○

LED and Drive Board

- The new and old boards are not compatible with each other when combined.

LED \ Drive Board	LED_DRIVE_BOARD	LED_DRIVE2_BOARD
REGIUS Σ	○	×
REGIUS Σ II	×	○

Chapter 1

Safety Warnings & Precautions

An explanation of the safety warnings and cautions to follow while repairing or maintaining this device.

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1.1 Symbols relating to Safety

1.1.1 Safety Alert Symbol



This is the "safety alert symbol." This symbol alerts you to matters and/or operation potentially hazardous to yourself and other people. Read these messages and follow the instructions carefully.

1.1.2 Warning Notice (Signal Words)

Signal words indicate the degree of potential hazards in the use of the product.

Signal words include the following three types, which are used according to risk of damage caused by danger and the severity of damage.

DANGER

Indicates a imminently hazardous situation which, if not avoided, will result in death or serious injury.

WARNING

Indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.

CAUTION

Indicates a potentially hazardous situation which, if not avoided, could result in minor or moderate injury. It may also be used to indicate hazardous situation where only physical damage is likely to occur.

1.1.3 Description of Graphic Symbols

	A triangle indicates a danger against which you should take precaution. e.g.)  This symbol warns against "Danger of Electrical Shocks".
	A diagonal line indicates a prohibited course of action. e.g.)  This symbol indicates you "Do Not Disassemble".
	A solid circle indicates an imperative course of action. e.g.)  This symbol indicates you must "Remove Plug from Power Outlet".

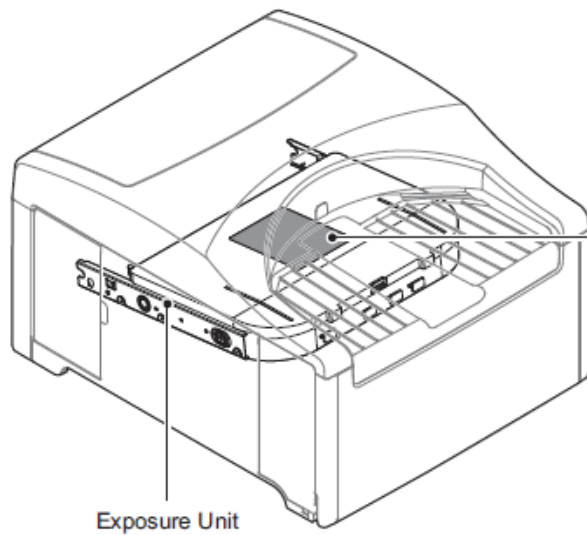
Graphic symbols	Explanation
	Indicates items that cannot be touched.
	Indicates that AC power is being used.
	Indicates the off position for the main power switch of this device.
	Indicates the on position for the main power switch of this device.
	Indicates that a plate is jammed inside the device.

1.2 Warning Labels

Warning labels are attached to the following areas of the device.

⚠ CAUTION

Before performing maintenance such as installation or repair, make sure to fully understand the meaning of each warning label, and how to handle each section they are located in.



Laser warning label

⚠ 注意 開放時クラス3B のレーザ放射。 ビームへの暴露をしないこと。
⚠ CAUTION CLASS 3B LASER RADIATION WHEN OPEN. AVOID EXPOSURE TO THE BEAM.
⚠ ATTENTION RAYONNEMENT LASER DE CLASSE 3B SI OUVERTURE. ÉVITEZ TOUTE EXPOSITION AU FAISCEAU.
⚠ PRECAUCIÓN RADIACIÓN LÁSER DE CLASE 3B AL ABRIR. EVITE LA EXPOSICIÓN AL RAYO.
⚠ 注意 开启时会导致 3B 类激光辐射。 请避免暴露于光束中。

1.3 Safety Precautions

Read all safety precautions thoroughly before using the device.
Be sure to observe the safety precautions described in this section.

1.3.1 Precautions Necessary to Follow Legal and Regulatory Requirements

CAUTION

Caution as to laser control

To prevent danger, make sure that only service engineers who received proper formal training will remove the exterior cover or touch the inside of the equipment.

Specifications of laser unit

Item	Specifications
Class	3B
Medium	Semiconductor laser
Wavelength	659 nm
Maximum output	130 mW

This device is a Class 1 laser product.

1.3.2 General Precautions

WARNING

- Do not perform any work or operations that are not described in this manual.
If you do accidentally conduct work not described in this manual, you may suffer from burns or electric shock due to touching a hot section or high voltage component of this machine.
- Be sure to follow safety precautions for warning labels.
Be sure to follow the safety precautions contained in this manual and shown on warning labels attached to this machine.
Failure to observe these precautions may result in personal injury or damage to this machine.
- This device is equipped with a laser generator (class 3B).
Avoid exposing the eyes or skin to direct laser, as it could cause a serious injury.
- Removing exterior covers
To prevent danger, make sure that only service engineers who received proper formal training will remove the exterior cover or touch the inside of the device.

1.3.3 Handling Precautions

WARNING

- If unusual noises, odor, or smoke comes from this device, stop operation.
If unusual noises, odor, or smoke coming from this device is detected, immediately stop operation. If you continue to operate it with the abnormal condition left as it is, it could cause an electric shock, fire, or damage to the device.
- Be careful not to trip over or trample on the power cable.
If you continue to use the damaged cable, it could cause an electric shock, heat generation or fire.
- Do not remove the power plug by pulling on the power cable.
Doing so may cause the power cable to break, resulting in heat generation or fire.
- Do not put drinks or foreign objects on the device.
Do not put drinks such as juice or foreign objects such as clips and pins on this machine. If you use this device after liquids or foreign objects such as metals have entered the machine, it can cause a short in the internal circuit, resulting in fire.

CAUTION

- Do not block air intakes or outlets of the device.
If air intakes or outlets of the device are blocked, it can cause the device to fail or reduce the accuracy of image reading.
- Do not leave objects standing on the device or do not step on it.
Doing so may lead to the risk of injury due to falling, or damage the device.
- Do not use equipment such as mobile phones that emit electromagnetic waves.
If you use equipment such as mobile phones that emit electromagnetic waves near the device, it can have a negative effect on the device.

1.3.4 Precautions Relating to Handling Cassettes/Plates

WARNING

- Do not scratch the plate.
The photostimulable luminescent material used on the plate can cause problems if it enters the human body. If the protective layer on the plate surface has been severely damaged and the photostimulable luminescent material is exposed, take the following actions.
 - If swallowed, consult a physician immediately.
 - If the eyes are exposed to it, flush them with clean water and consult a physician immediately.
 - If the skin is exposed to it, rinse with clean water immediately.
- Even if the photostimulable luminescent material does not enter the human body, take preventive measures against contact and dispose of the plate immediately in accordance with the applicable standards.

1.3.5 Servicing Precautions

WARNING

- Do not insert a piece of wire or metal.
Do not insert foreign objects such as metal strip or wire through vent or gap in the main device. Doing so may result in electric shock.
- Be sure to turn OFF when servicing.
Pulling the circuit board out of this machine or removing connectors and cables while the power is turned on may result in a serious accident. Make sure to turn off the power supply circuit breaker before starting these operations.

CAUTION

- When handling internal electronic components of the device, wear a wristband.
When handling internal electronic components, wear an antistatic wristband.
If you touch an electronic component such as circuit board with an electrostatically charged hand, it can cause damage.
- Before cleaning, make sure to unplug the power supply, or switch OFF the power supply circuit breaker.
Be sure to unplug or turn OFF the power switch before cleaning this machine.
You could get your finger caught in sliding or rotating parts.

1.3.6 Disposal Cautions

CAUTION

- Please follow the rules and regulations of your relevant authorities in the disposal of this device, accessories, options, consumables, media and their packing materials.

Chapter 2

Before Repairing

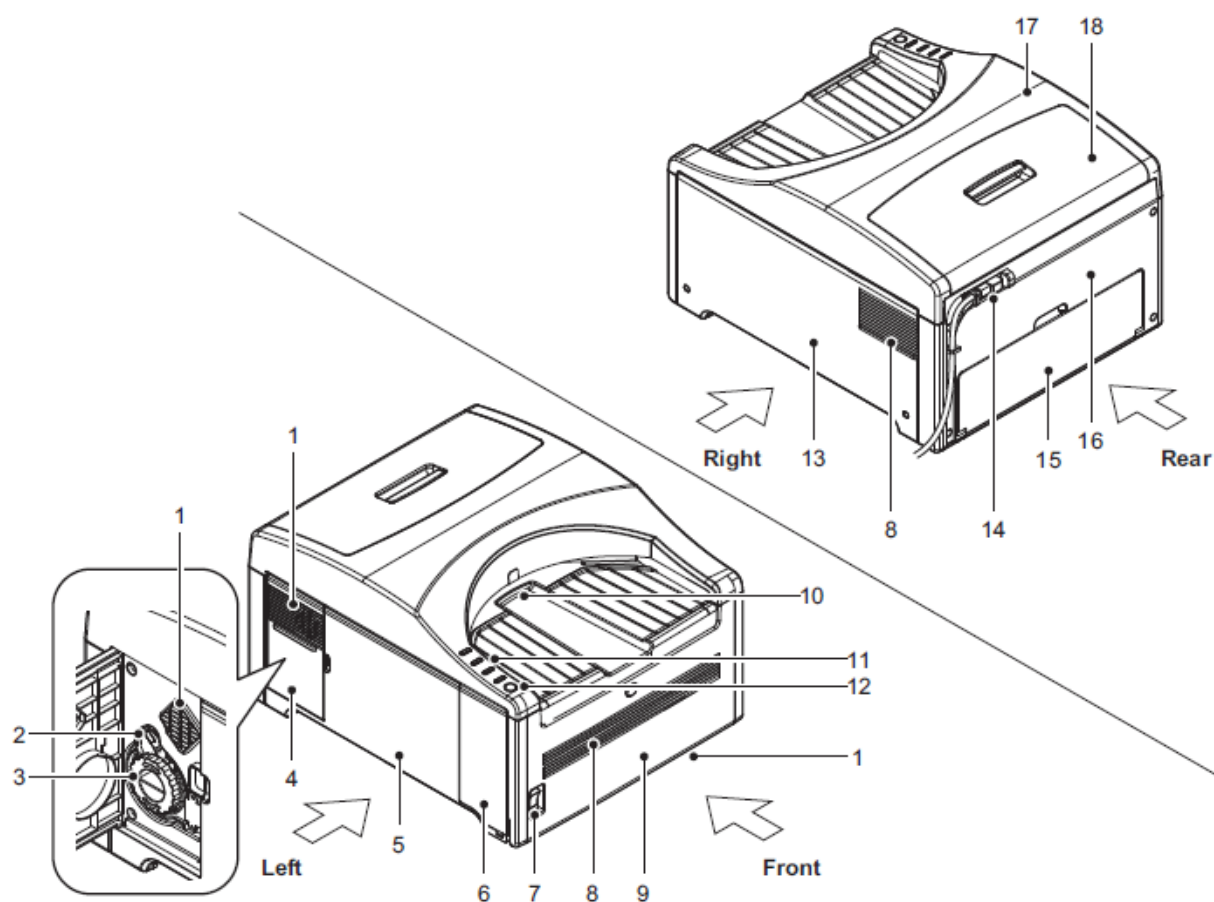
An explanation of device fundamentals that need to be understood before attempting to repair or maintain this device.

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2.1 Names of Parts

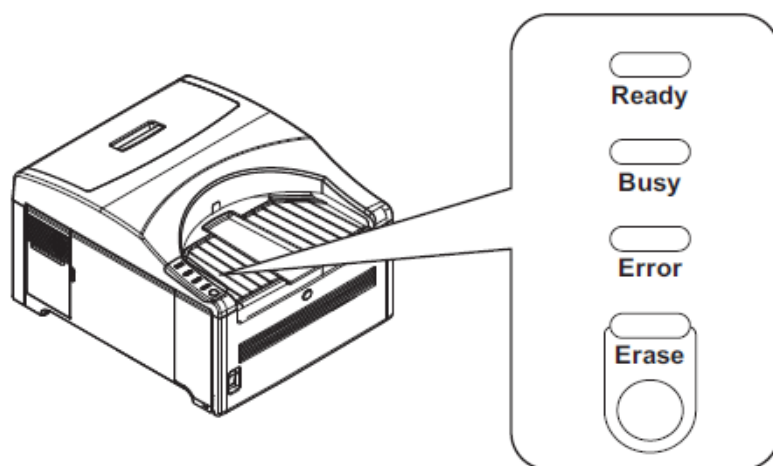
Part names are indicated here that should be known in order to repair the device.

2.1.1 External View of the device



No.	Common Name (Part Name)	No.	Common Name (Part Name)	No.	Common Name (Part Name)	No.	Common Name (Part Name)
1	Intake vent	2	Light Collection Unit Cleaning Handle (Handle)	3	Unjamming dial (Handle)	4	Left Door (Left door Assy)
5	Left side panel (Left side panel)	6	USB Cover (USB cover)	7	Power supply circuit breaker (Circuit protector)	8	Exhaust outlet
9	Front panel (Front panel Assy)	10	Cassette insertion slot	11	Status lamps • Refer to Page 2-3 .	12	Erase button (Erase button)
13	Right side panel (Right side panel)	14	Power supply port (Inlet-NF wiring Assy)	15	Rear door (Rear Assy)	16	Rear panel (Rear panel patch Assy)
17	Upper cover (Upper cover Assy)	18	Upper Door (Upper door Assy)				

2.1.2 Status lamps

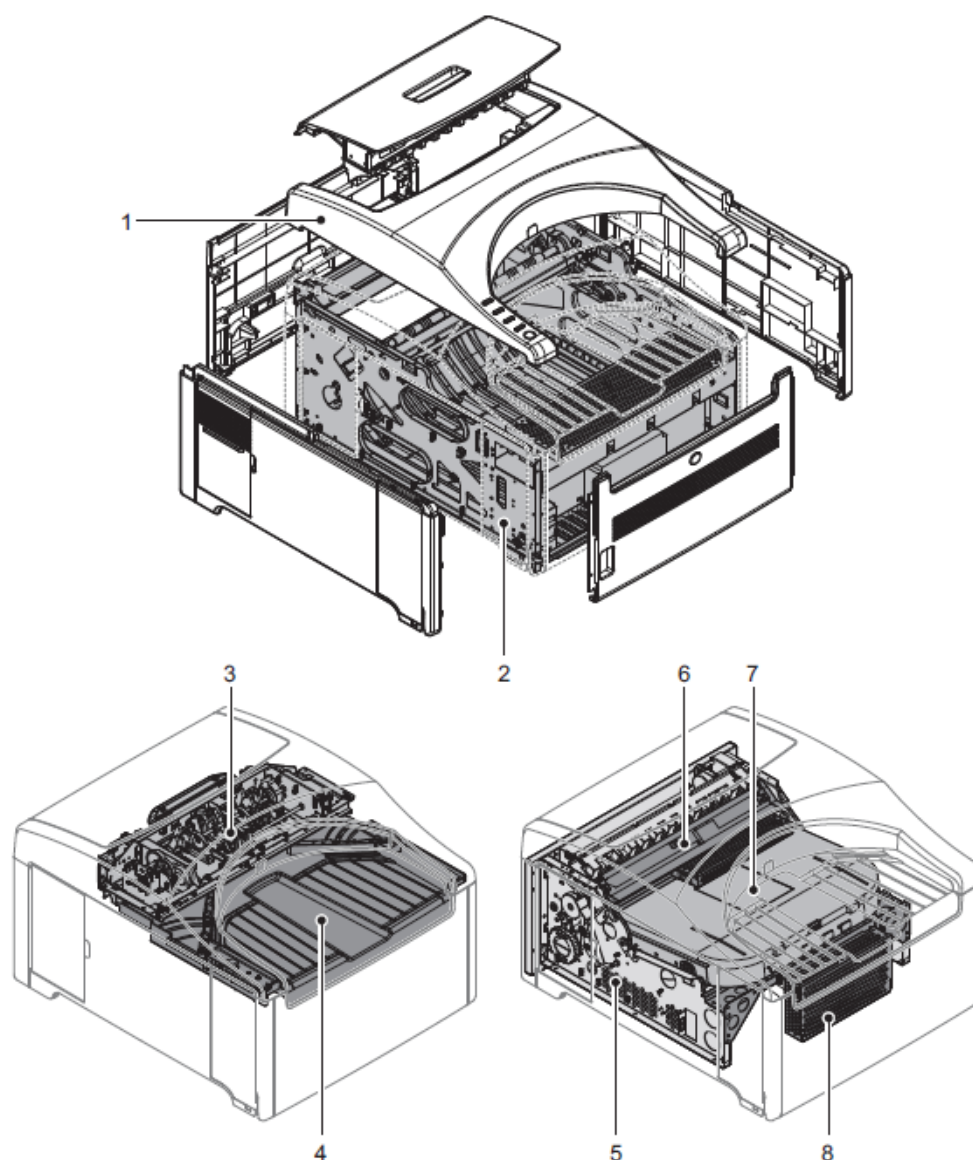


The 4 LEDs that indicate this device's current status are called "Status Lamps."

Name	Color	Function
Ready Lamp	Green	Indicates when a cassette is loaded or ejected. • Lit: The cassette can be loaded. • Blinking: The process has finished, and the cassette can be ejected.
Busy Lamp	Blue	Indicates the process status. • Lit: Process in progress. • Blinking: Preparing to start process (inspection questions during image reading) • Unlit: The process has stopped.
Error Lamp	Yellow	Indicates error and warning status. • Lit: Warning status alert, or connection with ImagePilot not established. • Blinking: Error status alert
Erase Lamp	Orange	Indicates Erase Mode status. • Lit: Image erase in progress, or the Erase switch is being pressed. • Blinking: Image erase mode is selected, and device is in standby mode for cassette loading (5 seconds).

2.2 Configuration

A description of how the device is configured.



No.	Name	Function
1	External Unit	Protects the device's interior parts from light, vibration, and other outside influences. It is made up of a upper panel, side panels, and a door for accessing interior parts of the device.
2	Frame	Make up the device's frame. Control parts, an inlet for power supply, and power sources are also included.
3	Upper Insertion Unit	Has a mechanism to release the loaded cassette's lock, and to eject or return the plate to normal. Display and operational parts are also included.
4	Lower Insertion Unit	
5	Sub-scan Unit	Feeds the plate (with sub scan).
6	Eraser Unit	Erases any image information remaining on the plate.
7	Exposure Unit	Scan excited laser light for the plate.
8	Light Collection Unit	Gathers phosphor light generated from the plate and converts it into electric signals.

2.3 Main Specifications

2.3.1 Device Specifications

Item	Content	
External dimensions	510 (W) × 610 (D) × 355 (H) mm	
Mass	Approximately 28 kg (excluding cassette and plate)	
Noise level (Excluding noise originating from user operation (such as cassette loading).)	During operation	Average = 53 dB or less Maximum = 61 dB or less
	During standby	Average = 25 dB or less
Maximum power consumption	100 VA or less (100 V, 1 A or less)	
Power source	Japan	100 V ± 10% (50/60 Hz)
	Other areas	115/120 V ± 10% (60 Hz) 110/220/230/240 V ± 10% (50/60 Hz) 200 V ± 10% (50 Hz)
Hygrothermal environment	Transport packing status	-20 to 60 °C / 90% RH or less (no condensation) 0.7 to 1.0 atmospheric pressure (= 709 to 1013 (hPa))
	During operation	10 to 30 °C / 15 to 80% RH (no condensation)
	When not operating	-10 to 40 °C / 90% RH or less (no condensation)
External connection	USB 2.0 (Konica Minolta independent communications specifications)	
Service life	7 years (10 cassettes per day × 25 days per month × 12 months per year × 7 years = 21,000 cassettes) IMPORTANT User should consider service life 6 years. (A 6-year service life is also mentioned in the operation manual.)	
Accessories	- Power cable (length: 3 m) - USB Cable (length: 5 m, USB 2.0 High Speed compatible) - Operation manual - Opening tools for cassette	
Parts requiring periodic change	None	

Compatible Cassettes & Plates

Item	Content	
Readable cassette types	RC-300 cassettes with isolated open/close structures and equivalent	
	Japan	14" × 17", 14" × 14", 11" × 14", 10" × 12", 8" × 10", 15cm × 30 cm
	Other areas	14" × 17", 14" × 14", 11" × 14", 10" × 12", 8" × 10", 24cm × 30 cm, 18cm × 24 cm, 15cm × 30 cm
Readable plate types	Regius Plate FP-1S or equivalent	

IMPORTANT

Cassettes/plates that can be used with this device are device exclusive. Rigid type cassettes used with the REGIUS MODEL 110, etc. cannot be used for this device.

2.3.2 Image Reading & Erasing

Processing Ability

Item	Content		
Reading resolution	Normal read mode (175 μm) High-definition read mode (87.5 μm) According to ImagePilot licensing. There are no resolution settings for this unit.		
Cycle time	REGIUS Σ	Japan	Up to 75 seconds per cassette (14" × 14" / normal read mode)
		Other areas	Up to 80 seconds per cassette (14" × 17" / normal read mode)
	REGIUS ΣII	Japan	Up to 54 seconds per cassette (14" × 14" / normal read mode)
		Other areas	Up to 60 seconds per cassette (14" × 17" / normal read mode)
	The fastest time from the cassette being loaded and locked, to the end of the reading process when the cassette is removed. The fastest erase speed for standard dose exposure. Cassette replacement time = 0 seconds.		

Item	Content		
Processing Ability	REGIUS Σ	Japan	About 48 cassettes per hour (14" × 14" / normal read mode)
		Other areas	About 45 cassettes per hour (14" × 17" / normal read mode)
	REGIUS Σ II	Japan	About 66 cassettes per hour (14" × 14" / normal read mode)
		Other areas	About 60 cassettes per hour (14" × 17" / normal read mode)
	When continuous cassette loading, for the number of cassettes going through the process per hour.		
Digital gradation level	12-bit (4096 gradation)		

Read image size (Japan)

Cassette type	Read capacity (mm)		Read pixel count			
			Normal read mode		High-definition read mode	
	X way	Y way	X way	Y way	X way	Y way
14" × 17"	365.75	442.05	2090	2526	4180	5052
14" × 14"	365.75	365.75	2090	2090	4180	4180
11" × 14"	289.10	365.75	1652	2090	3304	4180
10" × 12"	264.25	315.35	1510	1802	3020	3604
8" × 10"	213.50	264.25	1220	1510	2440	3020
15 cm × 30 cm	310.45	160.30	1774	916	3548	1832

Read image size (Other areas)

Cassette type	Read capacity (mm)		Read pixel count			
			Normal read mode		High-definition read mode	
	X way	Y way	X way	Y way	X way	Y way
14" × 17"	365.75	442.05	2090	2526	4180	5052
14" × 14"	365.75	365.75	2090	2090	4180	4180
11" × 14"	289.10	365.75	1652	2090	3304	4180
10" × 12"	264.25	315.35	1510	1802	3020	3604
8" × 10"	213.50	264.25	1220	1510	2440	3020
24 cm × 30 cm	250.25	310.10	1430	1772	2860	3544
18 cm × 24 cm	190.40	250.25	1088	1430	2176	2860
15 cm × 30 cm	310.45	160.30	1774	916	3548	1832

Exposure Unit

Item	Content
Method	Laser scanning by polygon mirror
Laser wavelength	659 ± 5 nm

Light Collection Unit

Item	Content
Method	Light collecting mirror + reed-shaped light guide + angled excitation light cut filter + rounded photomultiplier tube (PMT)
Excitation light elimination	An angled excitation light cut filter is placed in the light guide window to eliminate excitation light.

Eraser Unit

Item	Content
Method	Erasing done by white LED.

Sub-scan Unit

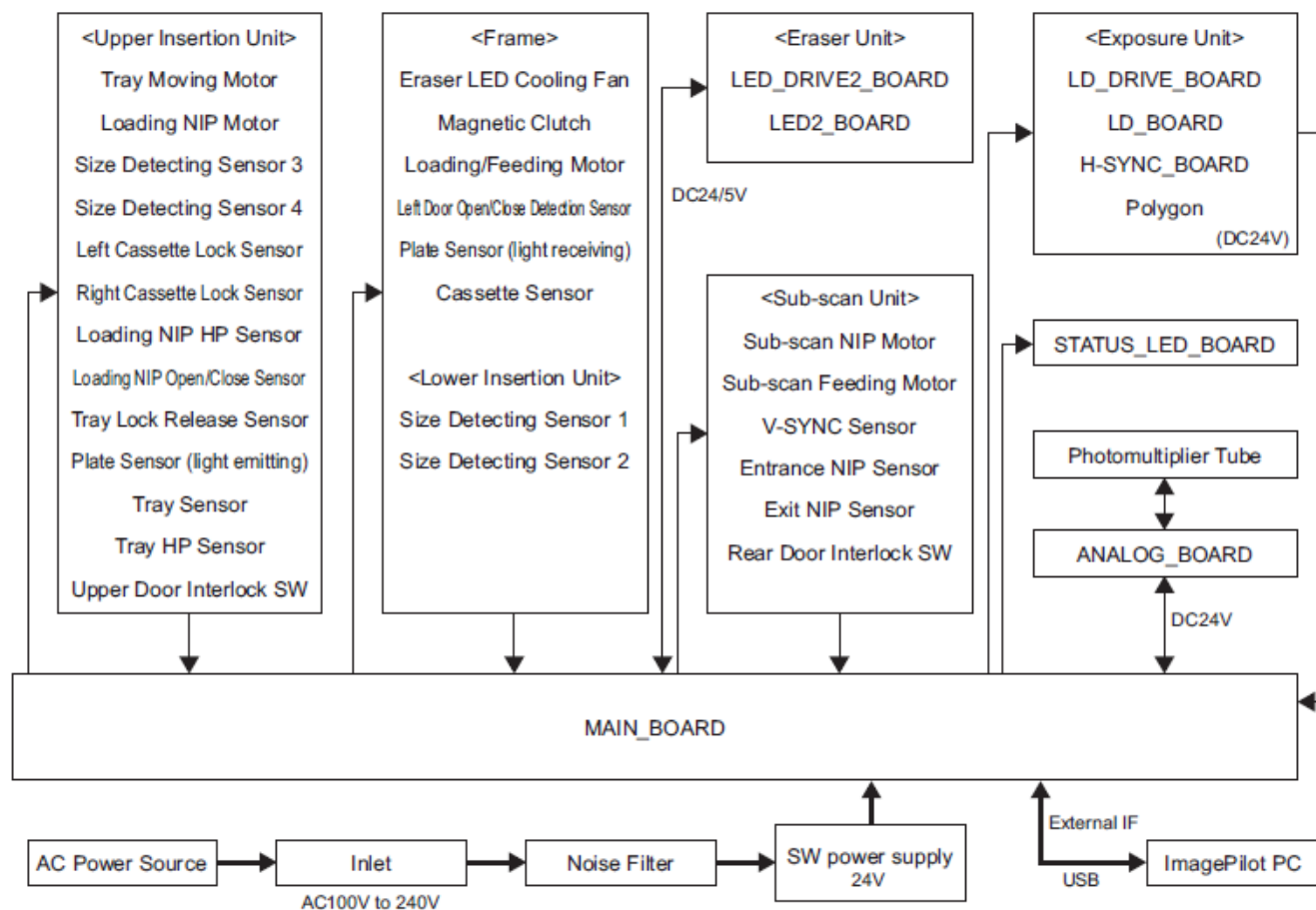
Item	Content
Method	Exposure unit fixed plate feed method Sub-scan (read/erase) of feeded photostimulable phosphor plate by optical/light collection unit.
Driving method	The sheet-type plate roller feed method using a 3-phase stepping motor.
Operation	- When reading: reading is outward, erasing is out and back. - When erasing: reading is outward (max signal value confirmation), erasing is out and back.

2.3.3 Control Part Functions

The main control part functions are as follows.

Function name	Content
Image data generation function	This function generates image data via laser scanning.
Image data revision/calculation function	- Moire elimination calculation - Gain/offset revision calculation - Shading irregularity correction, polygon revision - Erase speed calculation
Communication function	- Command receiving function - This function transmits read image data to a console PC. - This function provides a device status notification.
Control function	- This function is triggered by cassette loading, and performs a series of controls including mechanical control, optical control, signal processing control, communication control, etc. - This function performs the controls needed to execute commands received via USB 2.0 interface communication. - This function saves the base board serial number to device memory which should recognize the device's uniqueness.
Calibration function (during maintenance/shipment settings)	Mainly the following functions are performed. - Gain/offset function - Irregularity calibration function - Sensitivity calibration function - Image position/size adjustment function

2.4 Block Diagram

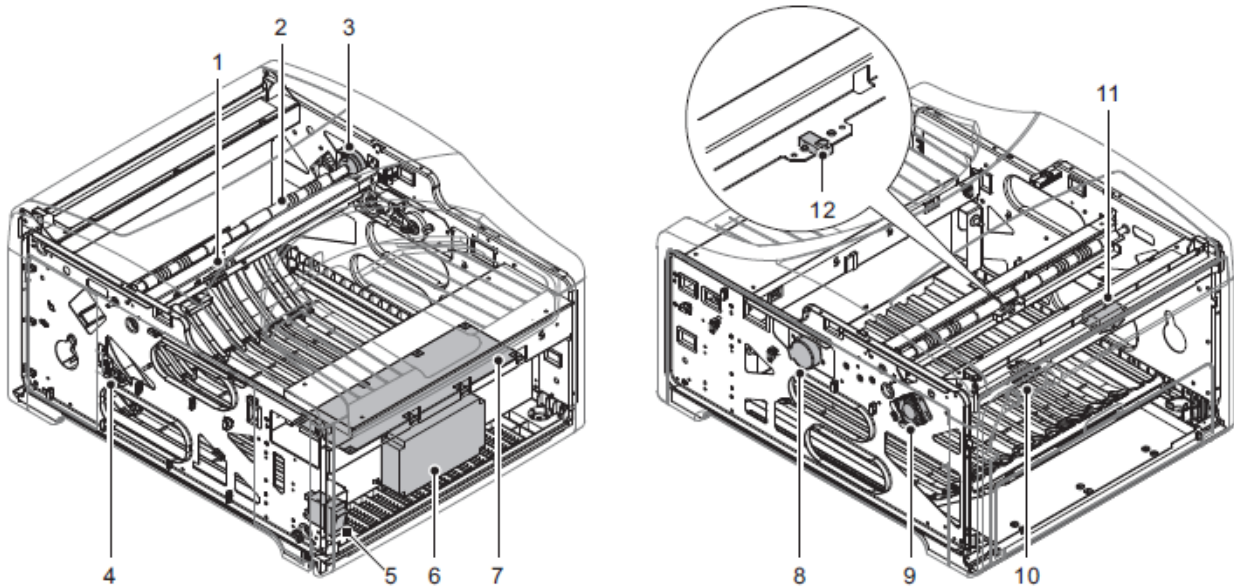


Excluding what is listed in the diagram, motors are supplied 24V, and other devices are supplied 5V of electricity from the main board.

2.5 Main Part Locations

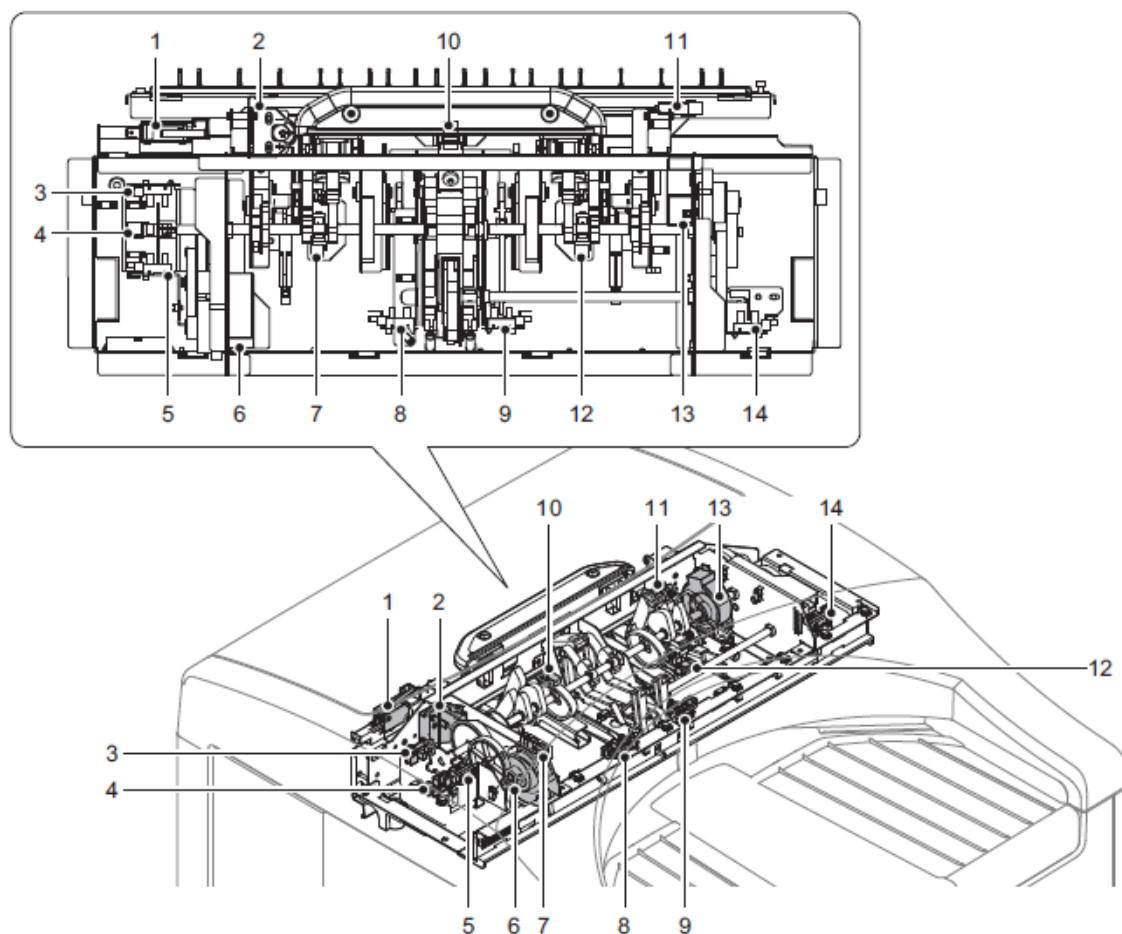
This is an explanation of where each device's main parts are positioned and how they function.

2.5.1 Frame



No.	Name	Function	Reference for how to change
1	Cassette Sensor	Turned ON when the cassette is loaded.	"5.3.9 Changing the cassette sensor (Page 5-34)"
2	Feeding Roller	The loading/feeding motor operates to remove or insert the plate from/into the cassette.	"5.4.5 Changing the magnetic clutch (Page 5-45)"
3	Magnetic Clutch	Regulates power sent to feeding roller.	"5.4.5 Changing the magnetic clutch (Page 5-45)"
4	Left Door Open/Close Detection Sensor	Turned ON when the left door is open.	-
5	Power supply circuit breaker	Used to turn the device's power ON/OFF.	"5.3.6 Changing the power supply circuit breaker (Page 5-31)"
6	SW power supply	-	"5.3.5 Changing the SW power supply (Page 5-30)"
7	MAIN_BOARD	A base board that controls the device.	"5.3.4 Changing the main board (Page 5-28)"
8	Insertion feed motor	Removes or inserts the plate from/into the cassette.	"5.4.1 Changing the loading/feeding motor (Page 5-40)"
9	Eraser LED Cooling Fan	Takes in outside air to keep the eraser unit LED cool.	"5.4.4 Changing the Eraser LED cooling fan (Page 5-44)"
10	Inlet-NF wiring Assy	-	"5.3.7 Changing the Inlet-NF wiring Assy (Page 5-32)"
11	Noise filter	-	-
12	Plate Sensor (light receiving)	Works together with the plate sensor (light emitting) of the upper insertion unit to detect whether a plate exists or not. Turns ON when a plate exists.	"5.3.10 Changing the plate sensor (light receiving) (Page 5-34)"

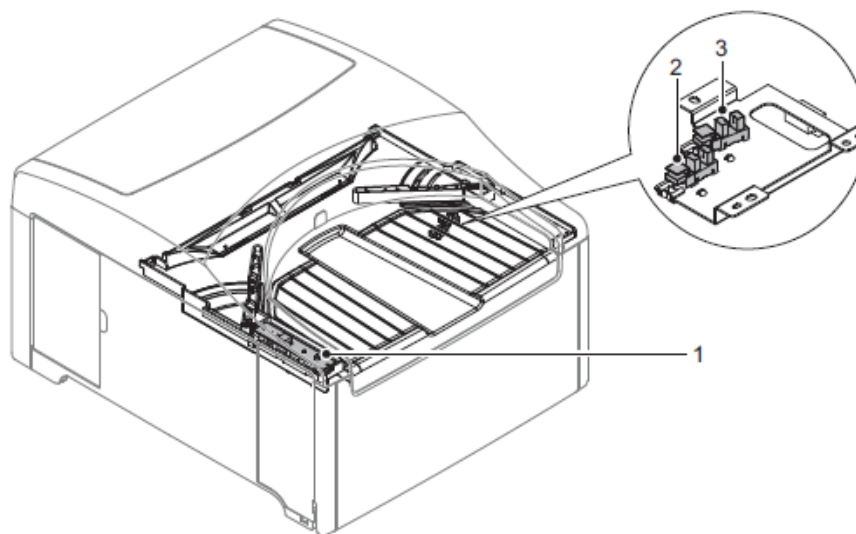
2.5.2 Upper Insertion Unit



No.	Name	Function	Reference for how to change
1	Switch (interlock 1)	Turned OFF when the upper door is open. This is not a safety interlock, but when turned OFF, power is disconnected from the following. • Mechanical control power for motors, fans, etc. • Eraser LED, power for high-voltage generation • LD power	-
2	Adjusting cam, adjusting board	Used for oblique adjustments. Refer to "6.1 Oblique adjustments (Page 6-2)".	Not changed as a single unit.
3	Loading NIP Open/Close Sensor	Detects NIP status, the tray's lock status, etc. when the plate is moving in and out of the cassette. Refer to "4.4.3 Insertion site sensor status (Page 4-6)".	-
4	Tray Lock Release Sensor		
5	Loading NIP HP Sensor		
6	Insertion nip motor	Performs NIP operation, or moves the tray's abutment plate up and down when the image plate is moving in and out of the cassette.	Not changed as a single unit.
7	Left Cassette Lock Sensor	Turned ON when the cassette is locked on the left side.	Not changed as a single unit.
8	Size Detection Sensor 4	Determines cassette size in a group of 4 sensors together with size detection sensors 1 and 2 of the lower insertion unit. Refer to "4.4.2 Cassette size & sensor status (Page 4-5)".	Not changed as a single unit.
9	Size Detection Sensor 3		Not changed as a single unit.

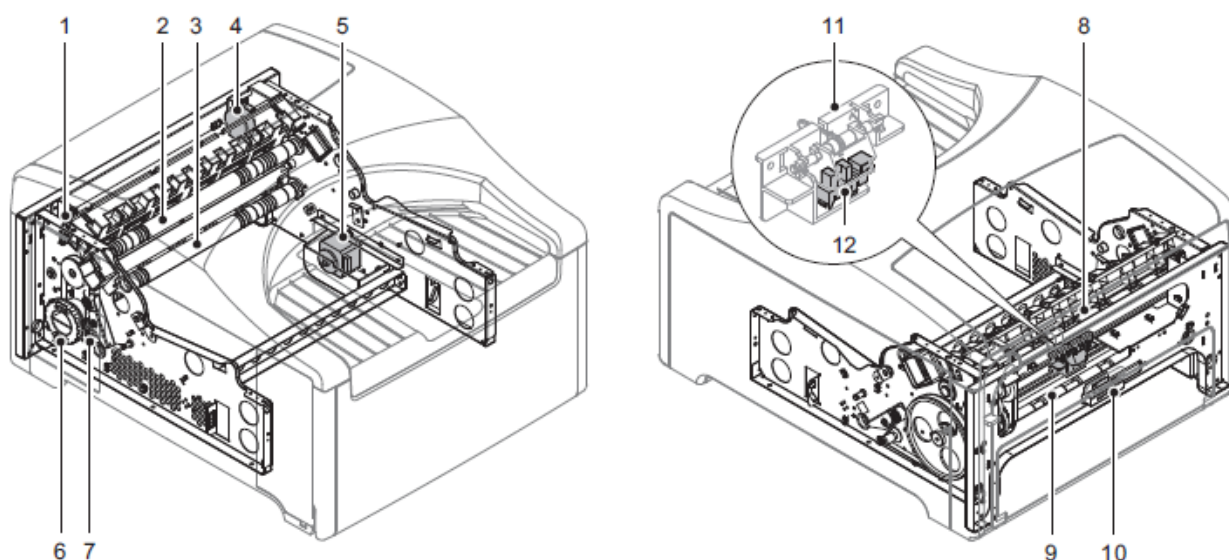
No.	Name	Function	Reference for how to change
10	Plate Sensor (light emitting)	Works together with the frame's plate sensor (light receiving) to detect whether a plate exists or not.	"5.3.11 Changing the plate sensor (light emitting) (Page 5-35)"
11	Tray Sensor	Turned ON when the tray is outside of the cassette.	Not changed as a single unit.
12	Right Cassette Lock Sensor	Turned ON when the cassette is locked on the right side.	Not changed as a single unit.
13	Cassette tray move motor	Moves the tray in and out, and corrects cassette tilt.	Not changed as a single unit.
14	Tray HP Sensor	Turned ON when the notches used for moving the tray in and out are in their home position.	-

2.5.3 Lower Insertion Unit



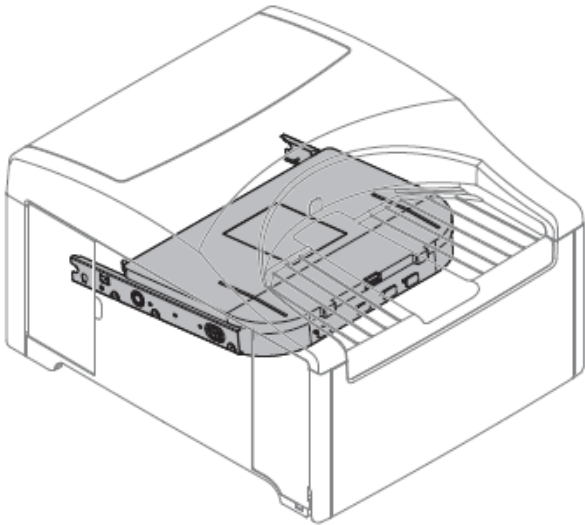
No.	Name	Function	Reference for how to change
1	Status LED board	A base board for status display (status lamps) and operations (Erase switch).	"5.3.1 Changing the status LED board (Page 5-25)"
2	Size Detection Sensor 1	Determines cassette size in a group of 4 sensors together with size detection sensors 3 and 4 of the upper insertion unit. Refer to "4.4.2 Cassette size & sensor status (Page 4-5)".	"5.3.8 Changing size detection sensors 1 and 2 (Page 5-33)"
3	Size Detection Sensor 2		

2.5.4 Sub-scan Unit



No.	Name	Function	Reference for how to change
1	Entrance NIP Sensor	Turned ON when Drive Roller U is in NIP status.	"5.3.13 Changing the entrance NIP sensor (Page 5-38)"
2	Drive Roller U	Feeds the plate via a Sub-scan Feeding Motor.	"5.4.9 Changing the drive rollers (Page 5-49)"
3	Drive Roller D		
4	Sub-scan NIP Motor	Performs NIP for Drive Roller U and Drive Roller D.	"5.4.2 Changing the sub-scan NIP motor and sub-scan drive Assy (Page 5-41)"
5	Sub-scan Feeding Motor	Feeds the plate.	"5.4.3 Changing the sub-scan feeding motor (Page 5-43)"
6	Handle (Unjamming dial)	Used to rotate Drive Rollers U and D when a plate jam occurs.	-
7	Exit NIP Sensor	Turned ON when Drive Roller D is in NIP status.	"5.3.14 Changing the exit NIP sensor (Page 5-39)"
8	Driven entrance assembly - driven roller	Operates via Sub-scan NIP motor to perform NIP for Drive Roller U and Drive Roller D.	-
9	Driven exit assembly - driven roller		-
10	Switch (interlock 2)	Turned OFF when the rear door is open. This is not a safety interlock, but when turned OFF, power is disconnected from the following. • Mechanical control power for motors, fans, etc. • Eraser LED, power for high-voltage generation • LD power	-
11	V-SYNC assembly	Detects when passage is available for the plate in a reading area. Sensor is ON when a plate exists.	"5.3.12 Changing the V-SYNC sensor and V-SYNC assembly (Page 5-37)"
12	V-SYNC Sensor		

2.5.5 Exposure Unit

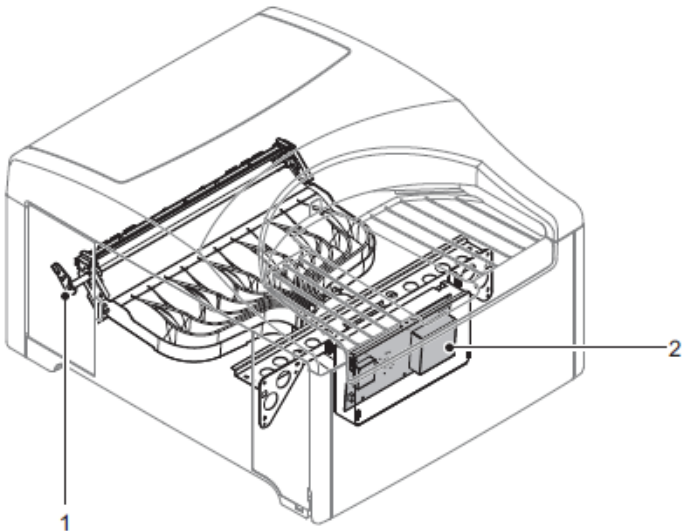


Parts within the exposure unit cannot be changed individually. If a problem occurs within the exposure unit, the entire exposure unit must be changed.

IMPORTANT

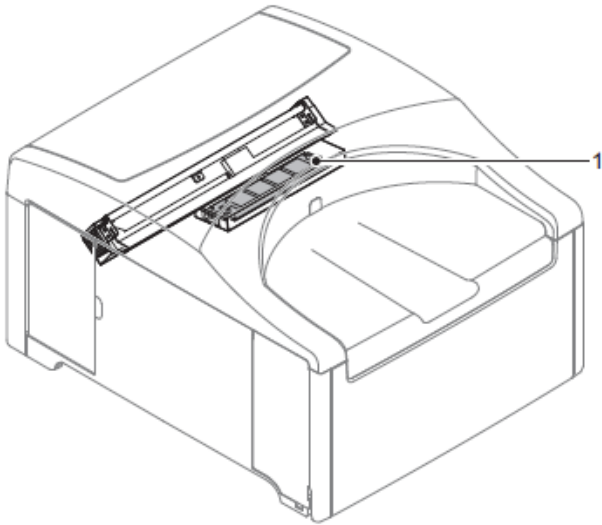
Do not disassemble the exposure unit.

2.5.6 Light Collection Unit



No.	Name	Function	Reference for how to change
1	Handle (Light Collection Unit Cleaning Handle)	Activates a cleaning mechanism to keep the light guide and laser scanning line of the light collection unit free of dust.	"5.2.10 Removing & attaching the Sub-scan unit (Page 5-12)"
2	Analog board	A base board that takes in analog signals from the photomultiplier tube and converts them.	Not changed as a single unit.

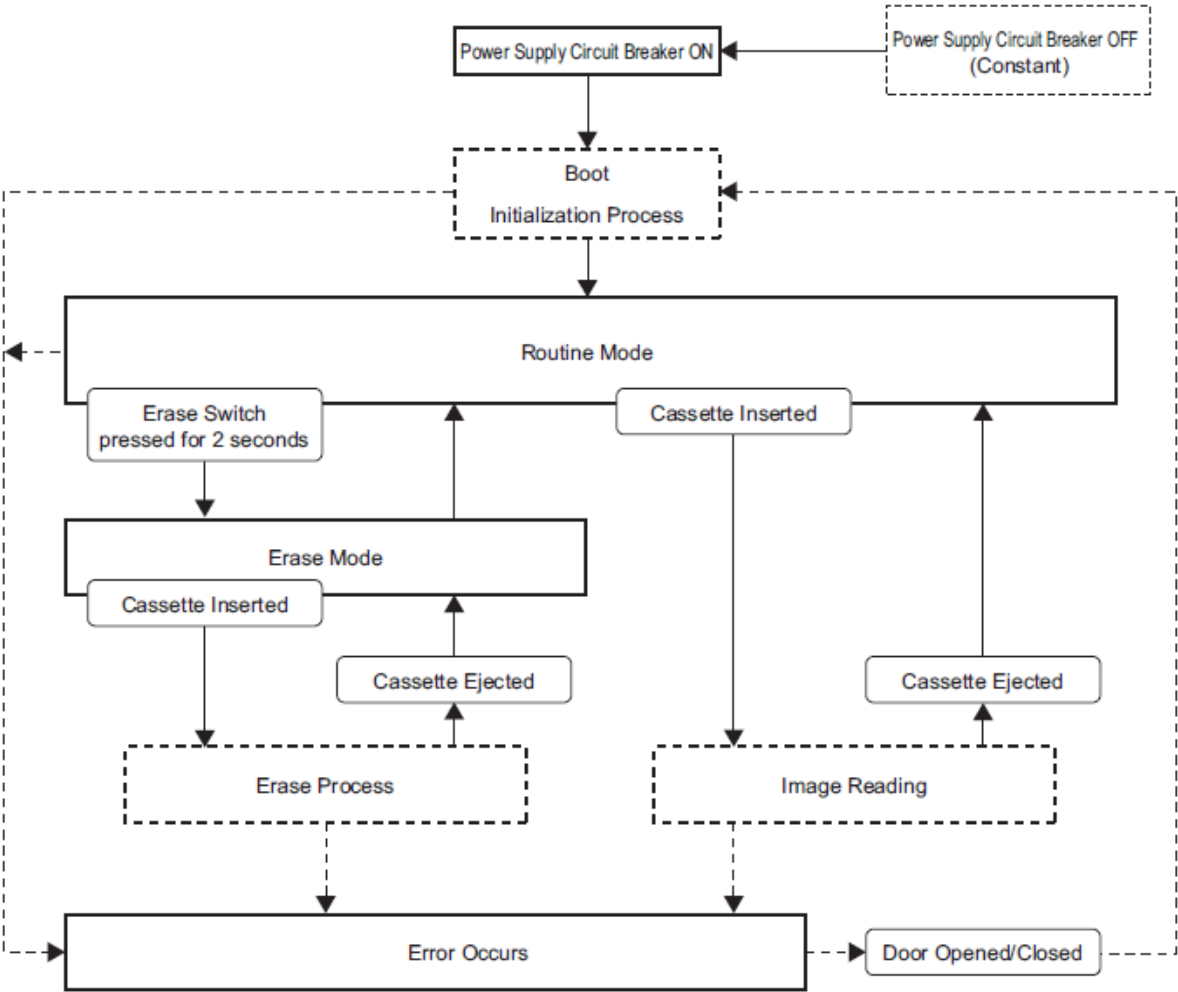
2.5.7 Eraser Unit



No.	Name	Function	Reference for how to change
1	LED_DRIVE2_BOARD	A base board that controls the Eraser LED.	"5.3.2 Changing the LED drive2 board (Page 5-25)"

2.6 Behavior Explained

2.6.1 State Transition Diagram

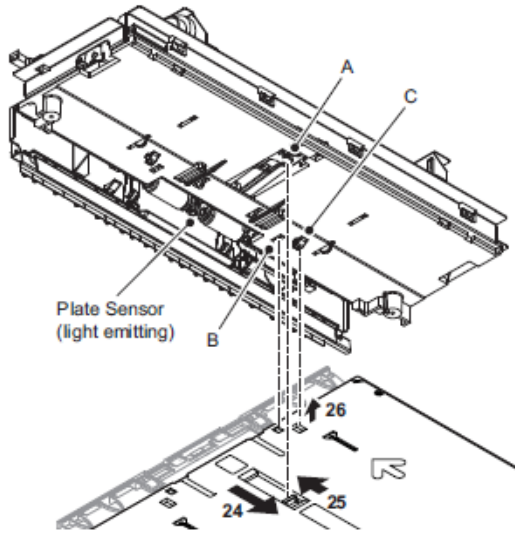


No sleep mode as with REGIUS MODELS 110/210 is available. (No operation switch.)

2.6.2 Normal Operation (Image Reading)

Operation/control	No.	Feed (letters in parenthesis correspond to the diagram)	Exposure / Optical / Eraser	Diagram (Numbers here correspond with the "No." column)
Initialization	1	Notch (A) for tray removal retracts. • Drive source: Tray moving motor (F)		Upper Insertion Unit
	2	The tray's lock release Notch (B) retracts. • Drive source: Loading NIP motor (G)		
User loads cassette	3	Notch (C) locks the cassette. • No drive source	Polygon starts to rotate for cassette detection.	
	4	Notch (D) releases tray lock (bolt mechanism). (For large cassettes only) • No drive source Part (E) is pushed into operation by the cassette.		
Plate is ejected	5	Notch (A) is lowered, and connects to the cassette's tray ejection notch (cassette open/close lever). • Drive source: Loading NIP motor (G)		
	6	Notch (A) moves in the tray's return direction and corrects the cassette's position. • Drive source: Tray moving motor (F)		
	7	Notch (B) is lowered, and the tray lock is released. • Drive source: Loading NIP motor (G)		
	8	Notch (A) ejects the tray, and bumps it against the abutment plate. • Drive source: Tray moving motor (F)		
	9	Magnetic Clutch (H) is turned ON. (Feeding roller connects to drive source.)		
	10	The NIP roller is lowered, and moves into NIP status with the feeding roller. • Drive source: Loading NIP motor (G)		
	11	The abutment plate is raised with the tray attached, and the image plate is removed from the tray. • Drive source: Loading NIP motor (G)		
	12	Notch (A) slightly returns the tray (to avoid tray and roller contact). • Drive source: Tray moving motor (F)		
Plate is sent (feed)	13	Loading/Feeding Motor (I) starts the plate into send feed. At the same time, Sub-scan Feeding Motor (J) starts up. • Drive source: Loading/Feeding Motor (I)	Seven seconds later, the polygon lock is confirmed, the laser emits low level light, and PMT high voltage is turned ON. (PMT voltage level rises until V-SYNC detection)	

Operation/ control	No.	Feed (letters in parenthesis correspond to the diagram)	Exposure / Optical / Eraser	Diagram (Numbers here correspond with the "No." column)
(V-SYNC detection)	14	Loading/Feeding Motor (I) stops. Magnetic Clutch (H) is turned OFF. The feeding roller is released from NIP status. (Otherwise the plate will scrape against the roller.)	When V-SYNC detection has 42 line counts (H-SYNC) remaining, the laser emits strong light, and image reading begins.	
	15	Drive Roller D goes into NIP status. (The timing is the same regardless of cassette size.) Drive source: Sub-scan NIP Motor (K)		
	16	Drive Roller U is released from NIP status. (Timing differs according to cassette size.) Drive source: Sub-scan NIP Motor (K)		
	17	Send feed stops at 10 mm from V-SYNC detection.		
Plate is returned (feed)	18	Sub-scan Feeding Motor (J) begins the plate's return feed. At the same time, an operation begins in which Drive Roller U moves into NIP status. Drive source: Sub-scan Feeding Motor (J)	The Eraser LED lights up.	
(V-SYNC detection)	19	From V-SYNC detection to 45 mm, feed speed gets faster, then slows down.		
(Plate detected)	20	When the plate sensor detects a plate (at the front end of its return direction), the Magnetic Clutch turns ON, and the feeding roller begins its return operation. (Timing differs according to plate size.)		
	21	The NIP roller is lowered, and moves into NIP status with the feeding roller. Drive source: Loading NIP motor (G)		
(No plate detected)	22	When the plate sensor does not detect a plate (at the far end of its return direction), the plate is determined to be stored in the cassette, and the tray's abutment plate is lowered. At the same time, Notch (B) is enabled to release the tray lock to take in tray storage. Drive source: Loading NIP motor (G)	The Eraser LED shuts down.	
	23	The Magnetic Clutch is turned OFF.		

Operation/ control	No.	Feed (letters in parenthesis correspond to the diagram)	Exposure / Optical / Eraser	Diagram (Numbers here correspond with the "No." column)
Plate storage	24	Notch (A) returns the tray to the cassette. (Max send: Abutment) • Drive source: Tray moving motor (F)		
	25	Notch (A) moves slightly in the extraction direction. (To relieve the tray of push stress.) • Drive source: Tray moving motor (F)		
	26	Notch (C) releases the cassette lock. • Drive source: Loading NIP motor (G)		
User ejects the cassette (and initialization begins).	27	Moves into standby mode to prepare for the next image reading.		

2.6.3 Initialization Operation

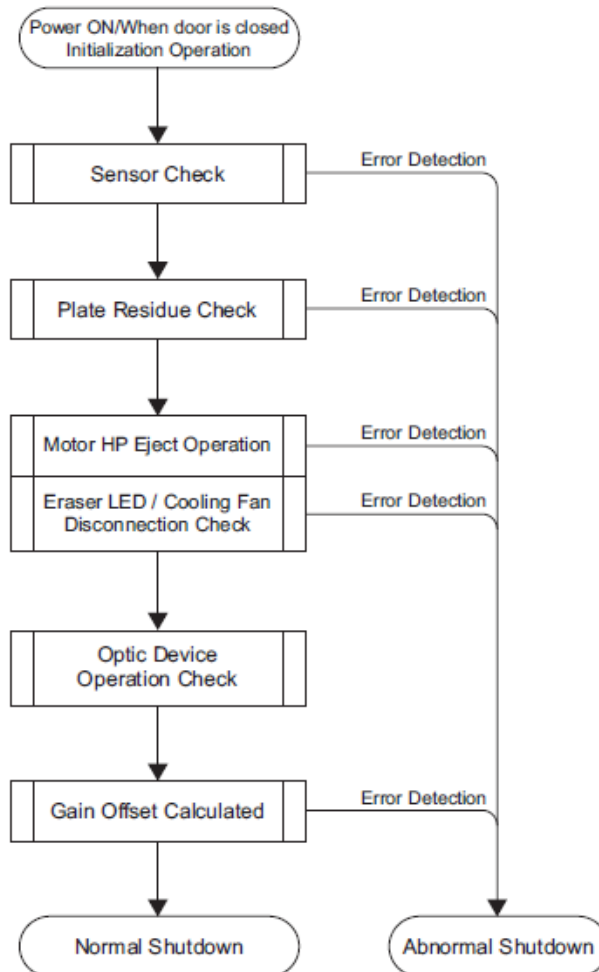
By switching the power supply circuit breaker to ON, device operation will begin, starting with the initialization process.

After starting up, firmware initialization content is as follows.

- 1) CPU Initialization Process
- 2) OS Initialization Process
- 3) FPGA Configuration & Initialization
- 4) Generation of all resources, such as each firmware task, etc.
- 5) Kernel runs

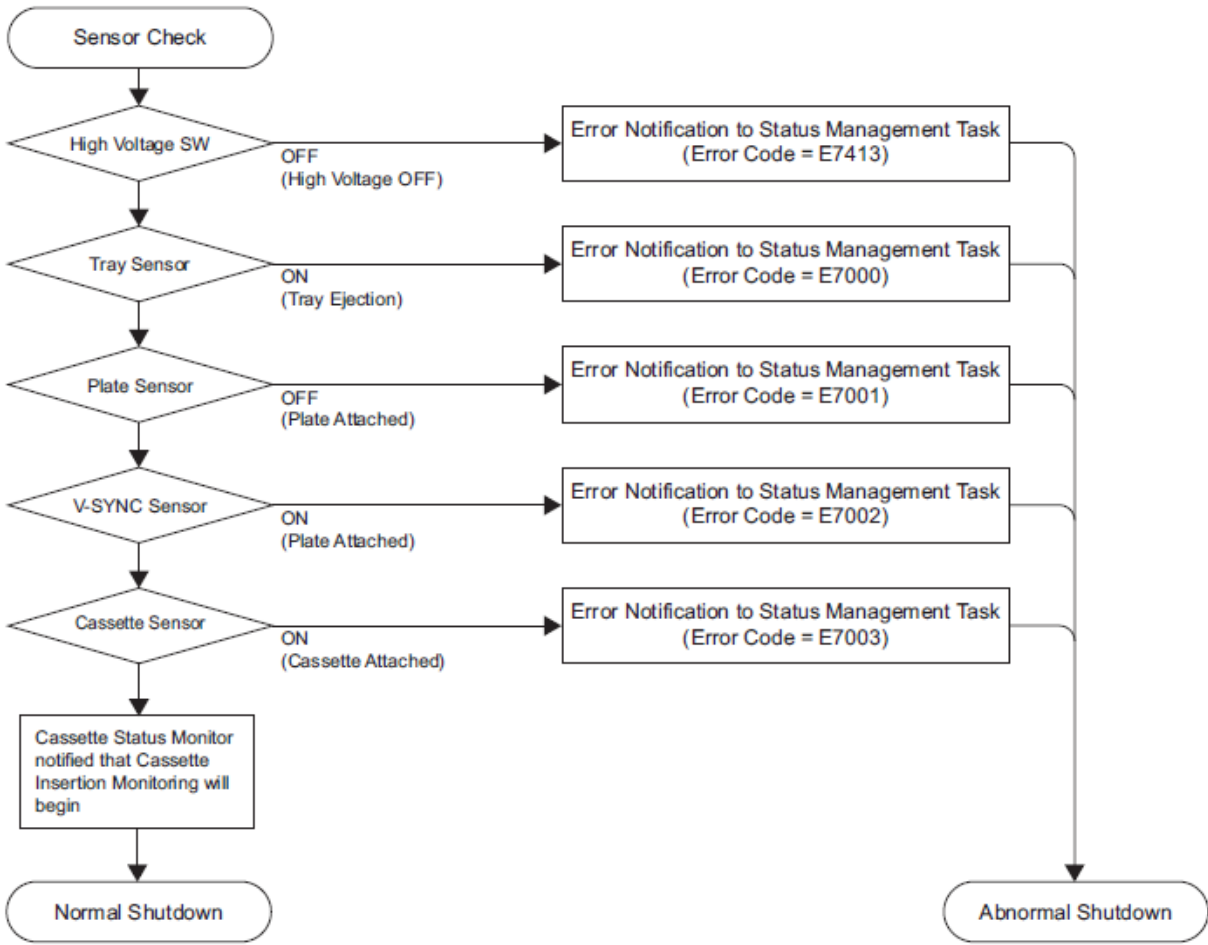
Each task operation will begin from this point, and the device will start up. USB communication initialization, mechanical and optical initialization etc. will also be performed simultaneously.

Mechanical Initialization Operation

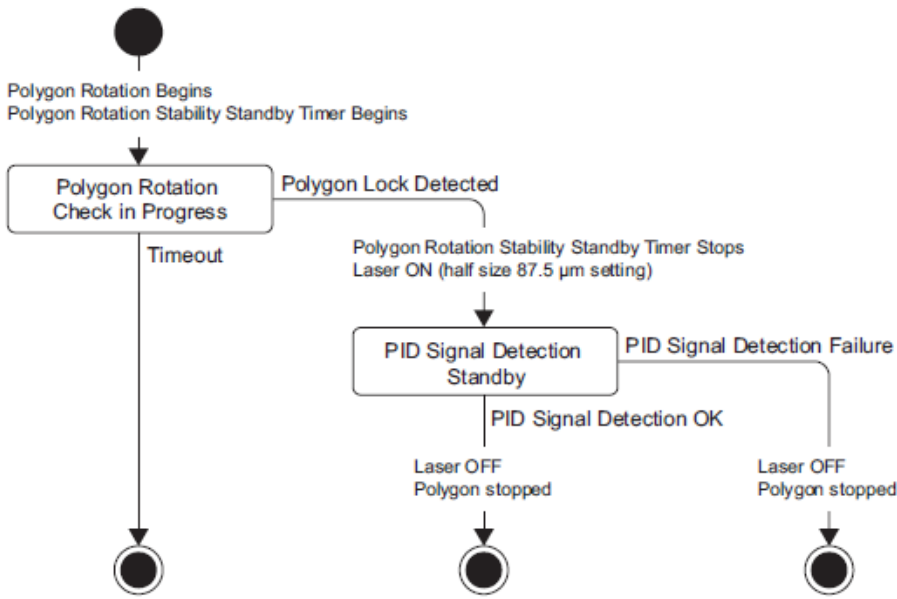


Summary
Sensor Check All sensors are checked, then checks are made of whether or not there is a cassette in the Cassette Insertion Slot, and whether or not a plate is inside the device. If one is detected, error notification is provided. For details, refer to "Sensor Check Flow (Page 2-20)".
Plate Residue Check The Sub-scan Feeding Motor is rotated for a certain amount of time, and if the V-SYNC sensor is turned ON (plate detected) at that time, the following operation is performed. 1. After the Sub-scan Feeding Motor has stopped, it will turn a determined number of revolutions, and the plate is sent to an area where it can easily be ejected from the upper Door (unjamming send process). 2. An error message is displayed.
Motor HP Eject Operation The Tray Moving Motor, Loading NIP Motor, and Sub-scan NIP Motor HP eject operation is performed. During each operation, changes to the sensors attached to each motor are confirmed to make sure that no errors have occurred in the sensors. Tray Moving Motor: Tray HP Sensor Loading NIP Motor: Lock Release Sensor Loading NIP HP Sensor Loading NIP Open/Close Sensor Sub-scan NIP Motor: Entrance NIP Sensor Exit NIP Sensor
Eraser LED/Cooling Fan Disconnection Check The Eraser LED and Eraser LED Cooling Fan are turned ON for a certain time, and if an Eraser LED error signal or Erase Fan error signal is detected during that time, error notification is given. For details, refer to "Moving into Eraser LED/Cooling Fan Disconnection Check Status (Page 2-21)".
Optic Device Operation Check Polygon is rotated, and confirmation is made that polygon lock signal detection can be performed. Laser is activated to emit light, while HSYNC updates and PID (Polygon-ID) detection are confirmed. If HSYNC updates and PID detection cannot be confirmed, error notification is provided. For details, refer to "Moving into Optical Device Operation Check Status (Page 2-20)".
Gain Offset Calculated Pixel values are entered with two reference current source types, and the gain offset is calculated from each average. • Motor HP Eject Operation and Eraser LED/Cooling Fan Disconnection Check will be performed in parallel.

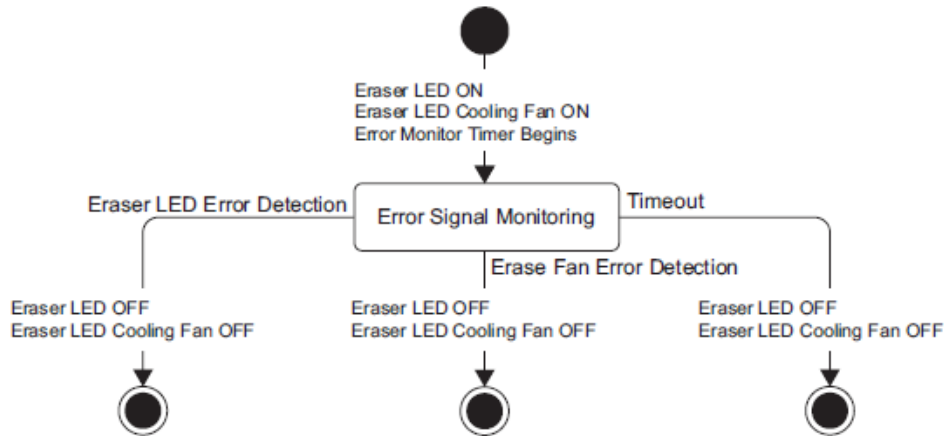
Sensor Check Flow



Moving into Optical Device Operation Check Status



Moving into Eraser LED/Cooling Fan Disconnection Check Status



2.6.4 Erase Operation

When the device's status is READY, holding down the Erase switch for at least 2 seconds will put it into Erase Mode status. Loading a cassette while the Erase Lamp is blinking will start the Erase operation. The Erase operation adjusts to the plate status to perform at the optimal speed.

2.7 Cassette/Plate Handling

Carefully adhere to the following instructions when handling a cassette or plate.

IMPORTANT

Only cassettes and plates exclusively produced for this device can be used with it. Rigid cassettes that are used with REGIUS MODEL 110 etc. cannot be used.

Maintaining image quality

- Always make sure the plate's reading surface has not been scratched or contaminated with foreign objects.
- If you find any foreign objects on the plate's reading surface, immediately clean it off.
- When handling a plate, wear the proper type of gloves according to film handling.
- Keep the plate away from sunlight.
- If the plate is exposed to fluorescent light during cleaning, etc. make sure to complete the task as quickly as possible and immediately load the plate into the cassette when finished.
- Do not let the cassette get scratched or dirty.

Transport & storage

- Follow the specifications below in regards to temperature and humidity.
 - Temperature: 10 to 30 °C
 - Humidity: 15 to 80% RH (with no condensation)
- Do not expose it to X-rays, γ rays or any other sort of radiation.
- A plate which has been unsealed must be loaded and stored in the cassette.
- Do not drop or bump the cassette.
- Stand the cassette in a rack for storage during operation.

Things to confirm before insertion

To prevent trouble during feed of the plate, insert the cassette only after confirming the following points.

- Make sure the cassette has not been altered in form or damaged. If the cassette is found to have a problem, do not insert it into the device.
- Confirm that no memo clips, paper, or other such objects are caught inside.
- Confirm that the cassette is locked.

Disposal

- Refer to "1.3.6 Disposal Cautions (Page 1-5)"

2.8 Tools, Instruments, Jigs, etc. Needed for Service

The following tools, instruments, jigs are needed to service this device.

Jig / Tool	Comment
Standard Tools • Phillips-head screwdriver (#1 bit): for cassette • Phillips-head screwdriver (#2 bit): for device • Flathead screwdriver • Hex wrench • Wrench • Nippers • Tray	The tray is used to store loose screws and parts.
A static electricity countermeasure type wristband (for grounding)	
Laser safety glasses	Capable of protecting against 659 nm laser wavelength
Tester	
Gloves	
Portable vacuum cleaner	Used to clean the plate's feeding path.
Cloth	Used to clean the device's external parts after servicing.
Plastic sheet	Use to cover light collection unit when removing them to keep dust out.
Cloth, cardboard, etc.	
Maintenance PC	
USB Cable	Used to connect a maintenance PC to the device.
A 14" × 17" size cassette and plate	Used when performing oblique light adjustment. (If not possessed by the user)

Supplies

Jig / Tool	Comment
Alcohol	
Zip ties	
Snap bands	

Chapter 3

Troubleshooting

An explanation of the various methods for dealing with device-related troubles.

3.1	Work Flow	3-2
3.2	Recovery Operation	3-2
3.3	How to Retrieve Logs.....	3-3
3.4	Error Message Content & Countermeasures.....	3-4
3.5	Countermeasures for dealing with problems not displayed as error messages.....	3-70

3.1 Work Flow

The following flow provides countermeasures to various issues that may occur with the device.

Problem	Countermeasure Flow
An error message is displayed on the ImagePilot for this problem.	1. If the plate is still inside the device, remove it. 2. Restart the device. - Refer to "3.2 Recovery Operation (Page 3-2)". 3. Check if the error occurs again. 4. Refer to "3.4 Error Message Content & Countermeasures (Page 3-4)" and perform the appropriate countermeasure.
	↓ If unable to perform the countermeasure. Contact our service window with a detailed explanation of the problem (refer to "Contact Information").
An error message is not displayed on the ImagePilot for this problem.	Refer to "3.5 Countermeasures for dealing with problems not displayed as error messages (Page 3-70)" and perform the appropriate countermeasure. If the plate is still inside the device, remove it before doing anything else.
	↓ If unable to perform the countermeasure. Contact our service window with a detailed explanation of the problem (refer to "Contact Information").

Contact Information

When contacting our service window regarding a problem, provide the following information with as much detail as possible.

Item	Content
Log, etc.	• When trouble occurs with the device, extract a device log etc. from the ImagePilot. (refer to "3.3 How to Retrieve Logs") • If a defective image is created, compile the following log from the ImagePilot. - App Log - System Log - Reader Log - JM Log
Facility connection format	
Recovery information	• Did the device recover with the door opened/closed? • Did the breaker have to be switched OFF/ON?
System Information	
Image Data	If a defective image is created, make a backup of the defective image from the ImagePilot.
Other Details	

3.2 Recovery Operation

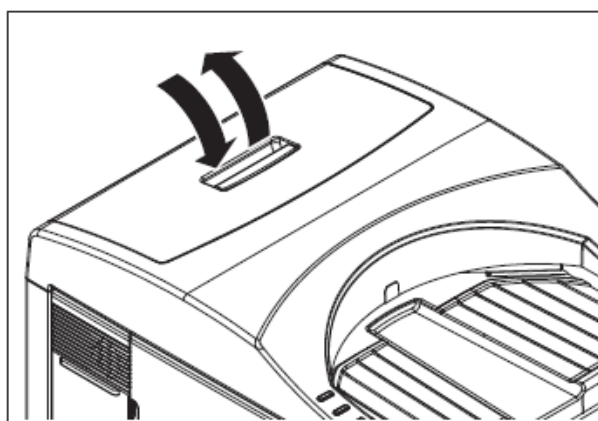
When an error occurs, restart this device and check if the error occurs again. The following is an explanation on how to restart.

⚠ CAUTION

If the plate is still inside the device, remove it before performing a recovery operation.

For details on how to eject a plate, refer to the operation manual.

1. Open and close the upper door.



Opening and closing the upper Door triggers the device's initialization process.

💡 HINT

Opening/closing any other door on the device will also work. The initialization process will be triggered in the same fashion.

If a sensor, etc. has gone bad, the initialization process will be halted, and the error will occur again.

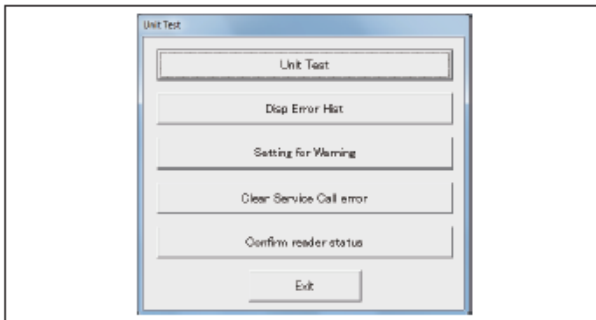
Dealing with a Service Call Error

If a V-SYNC error occurs during the image read process, or a basic operation failure occurs in the base board (RAM/ROM cannot be accessed), the reader will be in Service Call Error status. Opening/closing the door or turning the power OFF/ON will not correct the error in this case.

If a Service Call Error (Error code 7118, 7125, 7318, etc.) occurs, repair/replace the part that caused the error, then have a service engineer perform the following steps for recovery.

1. **Start up the Reader Settings Service Tool on an ImagePilot connected to this device.**
The [Service Tool] screen will be displayed for reader settings.
2. **Select [Reader Settings], then click the [Settings] button for [Unit test].**

The [Unit test] screen is displayed.



3. Click the [Clear Service Call error] button.
The error will be cleared.
4. Turn the device power OFF and ON.

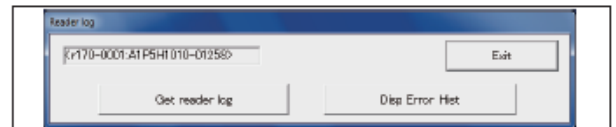
3.3 How to Retrieve Logs

This section describes how to retrieve logs.

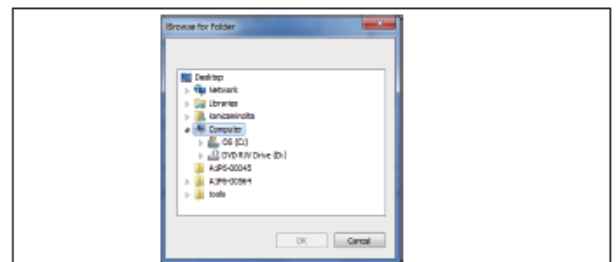
IMPORTANT

Retrieve logs before powering ON/OFF.

1. Launch the reader setting service tool using Image Pilot with this device connected.
The [Service Tool] screen for reader setting is displayed.
2. Select [Reader Setting] and click on the [Setting] button of [Reader log].
The [Reader log] screen is displayed.



3. Click the [Get reader log] button.
The behavior depends on the Image Pilot version.
 - For V1.50/1.51
The log is stored in the following location:
C:\¥Konicaminolta¥kim¥Server¥ReaderSetting¥Prog¥
Pegasus¥ ~ Board Number ~ ¥Log¥
Board number example r170-0001_A1P5H1010-00050
 - V1.52 or higher
Select the save destination folder from the [Browse for Folder] screen, and click the [OK] button.



The log is saved in the selected folder.

3.5 Countermeasures for dealing with problems not displayed as error messages

3.5.1 Device Sound

Examine the area where an abnormal noise was heard, then confirm the following.

- Check gear tooth surface (if a gear tooth is chipped, an abnormal noise will occur)
- Check for dents in the feeding guide
- If near the upper door, check the upper door's positioning. It may be hitting against the guide to produce a noise.
- Check for foreign objects or chipped areas
- Check the connectors
- Change the MAIN_BOARD and check for noises

3.5.2 If power cannot be turned ON

Confirm the following:

1. Check the power cable
2. Check the SW power supply's input voltage (100 to 200 V)
If it's no good, check the inlet and noise filter to pinpoint the abnormal area.
3. Check the SW power supply's output voltage (24 V)
If it's no good, change the SW power supply.
4. Check wiring from the SW power supply to the MAIN_BOARD
If 24 V of power is coming to the MAIN_BOARD, change the MAIN_BOARD.

3.5.3 Plate related

- When dirty
Clean the plate. (Refer to the Operation Manual for cleaning methods.)
- When bent
Change the plate.

3.5.4 Defective image

If an image is defective, take the appropriate steps to deal with it.

Problem	Countermeasure
Image or erase remnants	If the Eraser LED warning indication is on, replace the Eraser LED. (Refer to "5.3.3 Changing the Eraser Unit Assy (Eraser LED) (Page 5-27)") Leave the following content to the customer. • After unjamming the plate, have them perform the image erase operation. (The image erase operation might have not been performed when the error occurred.) • Have them perform the plate image erase operation every morning.
The image size displayed on the ImagePilot will differ.	Details: A 14 × 17 sized image was read, but only the top part is shown (14 × 14 size), and the bottom portion is a white image. Make sure a plate of a different size is not loaded in the cassette.
Vertical lines (sharp)	1. Use the cleaning mechanism to remove dust from the light collection unit. 2. Remove the light collection unit, and make sure no dust or foreign objects are obstructing the laser scanning path. Make sure the cleaning mechanism is not interfering with the laser scanning path. 3. Clean the exposure unit's window to make sure it is free of dust. 4. Change the exposure unit. (Refer to "5.2.12 Removing & attaching the exposure unit (Page 5-20)")
Vertical lines (faded)	1. Clean the light collection unit and light guide surface to make sure they are free of dust. 2. Clean the exposure unit's window to make sure it is free of dust.
Horizontal lines	1. Check the installation environment. If the environment is as described below, horizontal lines will occur. - The table is unsteady. - The table rocks when someone moves. - The unit is touching the table in some area other than its rubber feet. A cable is crushed between it. HINT Small vibrations directly from the table are being absorbed by the 4 rubber feet. (Vibration can only be absorbed to a limit) 2. Clean as needed so that no dust, sticky residue, or dents are on the guide board or roller. Make sure no sticky residue is on the plate. 3. Change the Sub-scan unit. (Refer to "5.2.10 Removing & attaching the Sub-scan unit (Page 5-12)") 4. Change the exposure unit. (Refer to "5.2.12 Removing & attaching the exposure unit (Page 5-20)") 5. Change the light collection unit. (Refer to "5.2.13 Removing & attaching light collection unit (Page 5-21)")
White spots (left/right)	Refer to "6.1 Oblique adjustments (Page 6-2)" and perform plate oblique adjustment.

Chapter 4

Using a Service Tool to Confirm Behavior

An explanation of how to confirm device operation.

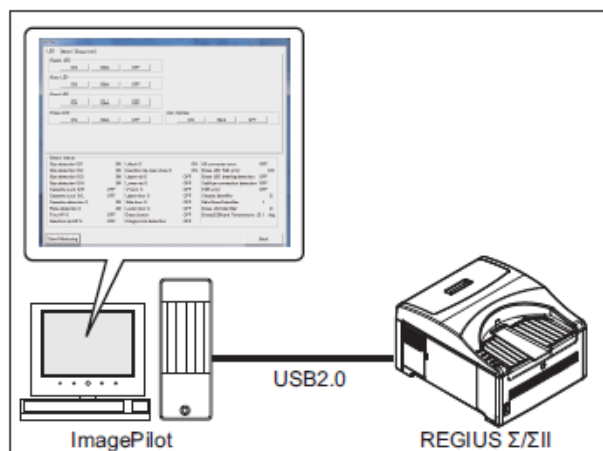
IMPORTANT

The ImagePilot service tool will only start up with the ImagePilot server. Moreover, when confirming operation of this device connected from the client's side, run service tool and user tool operations from the ImagePilot server side.

4.1	Behavior Confirmation Summary	4-2
4.2	[Unit Test] (Check) Screen Display.....	4-2
4.2.1	The [Unit test] Screen & Function Summary	4-2
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4.4.2	Cassette size & sensor status.....	4-5
4.4.3	Insertion site sensor status.....	4-6

4.1 Behavior Confirmation Summary

To confirm the operation of each device within the device, as well as check sensor status, an ImagePilot service tool called the Unit Test function (REGIUS $\Sigma/\Sigma II$ exclusive) can be used.



When displaying the [Unit test] (Check) screen, the device will move into maintenance mode.

The following actions can be performed at the [Unit test] (Check) screen.

- Independent operation of each device (motors, LEDs, etc.)
- Display sensor status
- Setting reader parameters

This chapter provides a summary of the functions available at the [Unit test] (Check) screen, steps to take in order to confirm single device operation, and an explanation of how to confirm sensor status.

For details not covered in this chapter, including detailed screen explanations, refer to "8.1 Service Tool Screen (Unit test) (Page 8-2)".

4.2 [Unit Test] (Check) Screen Display

Below is an explanation of how to display the [Unit Test] screen needed to confirm device operation, as well as a summary of available functions.

4.2.1 The [Unit test] Screen & Function Summary

Displaying the "Unit test" Screen

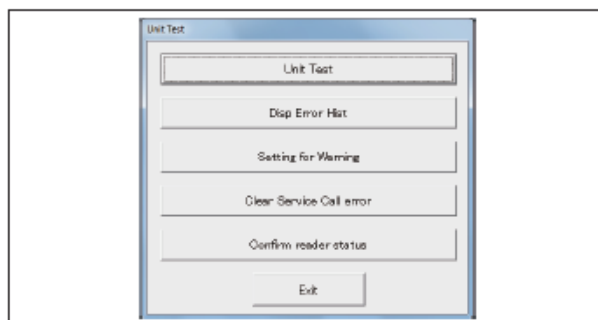
1. Start up the Reader Settings Service Tool on an ImagePilot connected to this device.

The "Service Tool" screen will be displayed for reader settings.

For details on the "Service Tool" screen, refer to the ImagePilot Installation & Service Manual.

2. Select "Reader Settings", then click the "Settings" button for "Unit Test".

The [Unit test] screen is displayed.



A Summary of [Unit test] Screen Functions

The "Unit test" screen includes 6 buttons, each capable of activating a function as explained below.

Button	Content
[Unit Test] button	Allows you to perform an independent operation check for each device in the device. (A function explained in this chapter.)
[Disp Error Hist] button	Allows you to check the error history.
[Setting for Warning] button	Allows you to toggle the "Study is not registered." and image re-erase request dialogs ON/OFF.
[Clear Service Call error] button	Used for a Service Call Error status.
[Confirm reader status] button	Allows you to check the following reader adjustment results. • Unit settings • Irregularity • Sensitivity • Mechanical counter • Base board number
[Exit] button	Allows you to exit the "Unit test" screen.

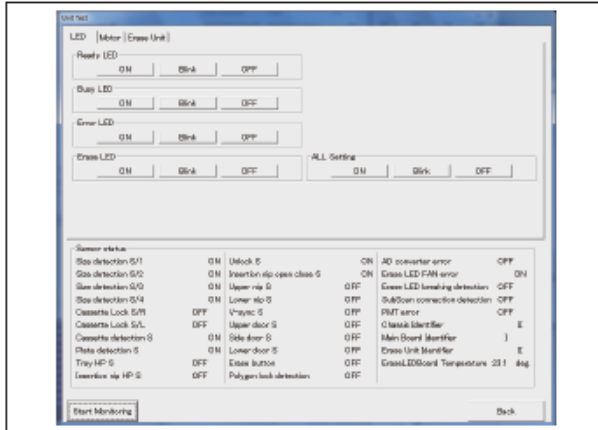
For screens displayed during execution of these functions, refer to "8.1 Service Tool Screen (Unit test) (Page 8-2)".

4.2.2 The [Unit Test] Screen & Function Summary

Displaying the [Unit Test] screen.

1. Click the [Unit test] button at the "Unit test" screen (Page 4-2).

The [Unit Test] screen is displayed.



Summary of functions at the [Unit Test] screen

The [Unit Test] screen consists of 3 panels, each of which can be accessed by clicking a tab.

Each panel is explained below.

Panel	Content
[LED]	Enables you to check the operation for each of the LEDs used as status lamps.
[Motor]	Allows you to check the operation for each motor.
[Erase Unit]	Allows you to check operation for the Eraser LED and its cooling fan.

In addition, the "Sensor status" area at the bottom of the screen allows you to display the status of every sensor, even if you switch to another panel.

When you are finished at the "Unit Test" screen, clicking the "Back" button on the lower right of the screen will return you to the "Unit test" screen.

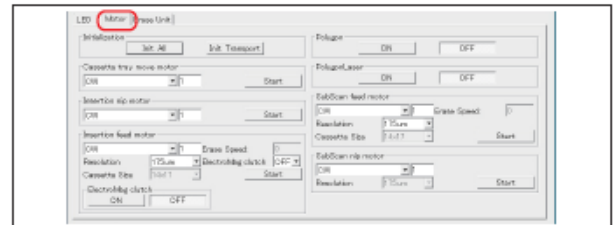
- For details on how to confirm device operation, refer to "4.3 Confirming behavior of a single device (Page 4-3)".
- For details on how to confirm sensor status, refer to "4.4 Sensor status Check (Page 4-5)".
- For screen details, refer to "8.1 Service Tool Screen (Unit test) (Page 8-2)".

4.3 Confirming behavior of a single device

The loading/feeding motor will be used to illustrate the steps involved in confirming the operation of a single device in the device.

1. Click the "Motor" tab at the "Unit Test" screen.

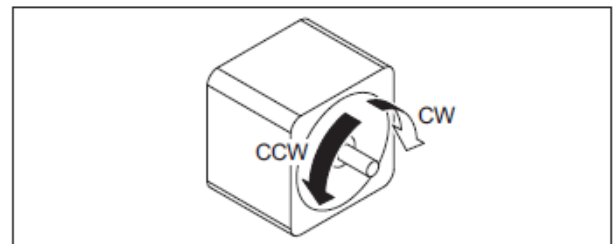
The "Motor" panel is displayed.



2. Select which operation you would like performed at the Operation Selection column.

IMPORTANT

- If you only want to confirm motor operation, select [CW] (Clock Wise) or [CCW] (Counter Clock Wise).

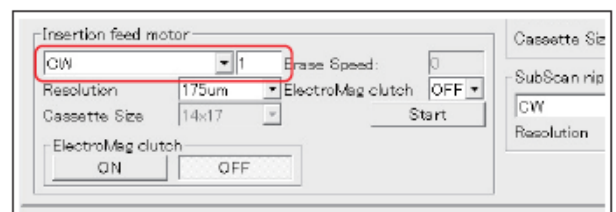


- If using a cassette, refer to " Unit Test: Cassette Feeding Flow (Page 4-4)", and follow the process described for operation.

- When selecting [CW] or [CCW] for operation, enter the number of rotations you want the motor to perform in the Set Rotations column.

IMPORTANT

The [CW] of the Cassette tray move motor does not work if the Tray HP Sensor is turned off. [CW] also stops if the Tray HP Sensor is turned off while it is working.



4. Select all other operation settings.

Item	Content
[Erase Speed]	Set the erase speed. (When selecting the "Transport Plate (Return)" operation.)
[Resolution]	Select a resolution to be used during the image read operation. (When selecting the "Transport Plate (Send)" operation.)
[ElectroMag clutch]	Select whether or not the magnetic clutch is operated. (When selecting the "Transport Plate (Send)" or "Transport Plate (Return)" operation.) When selecting OFF, the loading/feeding operation will be performed with the magnetic clutch as it was before operation. This will not change regardless of whether the magnetic clutch is ON or OFF. Refer to " • Setting combinations for the loading/feeding motor's magnetic clutch ".
[Cassette Size]	Select the cassette size. (When selecting the "Transport Plate (Return)" operation.)

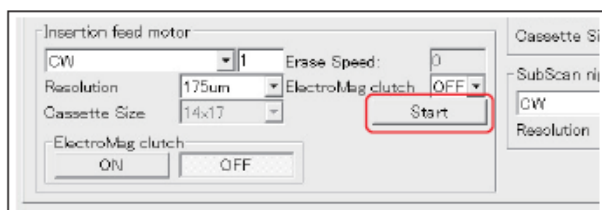
- Setting combinations for the loading/feeding motor's magnetic clutch

Magnetic Clutch	Button is ON	Button is OFF
Pulldown set to OFF	Magnetic clutch operates via ON. After operating, turns OFF.	Magnetic clutch operates via ON. After operating, turns OFF.
Pulldown set to OFF	Magnetic clutch operates via ON. After operating, stays ON.	Magnetic clutch operates via OFF. After operating, stays OFF.

HINT

Leaving the magnetic clutch ON for a long period of time will not cause the reader any errors. Also, initialization will occur when exiting the "Unit Test" screen, so no problems will occur.

5. Click the [Start] button.



The loading/feeding motor will operate according to the selected content.

Unit Test: Cassette Feeding Flow

IMPORTANT

Other than [CW] and [CCW] for motor operations performed via [Unit Test], only the following order of operation be guaranteed to work. Also, a cassette is used for this operation. (15 × 30 cm cassettes cannot be used.)

No.	Motor	Operation	Magnetic Clutch (Pulldown)
1		Full initialization	
2		Cassette insertion	
3	Loading NIP motor	Cassette position control	
4	Tray moving motor	Cassette position control 1	
5	Tray moving motor	Cassette position control 2	
6	Loading NIP motor	Tray lock release	
7	Tray moving motor	Tray pullout	
8	Loading NIP motor	Roller NIP + Tray raised	
9	SubScan nip motor	Initialize	
10	Magnetic clutch (button)	OFF	
11	Loading/Feeding motor	Plate feeding (send)	ON
12	Insertion nip motor	Scanning	
13	Sub-scan NIP motor	Plate feeding (send) 1	
14	Sub-scan feeding motor	Plate feeding (send)	
15	Sub-scan feeding motor	Plate feeding (return)	
16	Insertion nip motor	Erasing	
17	Magnetic clutch (button)	ON	
18	Loading/feeding motor	Plate feeding (return)	OFF
19	Loading NIP motor	Tray lowered + roller NIP	
20	Magnetic clutch (button)	OFF	
21	Tray moving motor	Tray return 1	
22	Tray moving motor	Tray return 2	
23	Loading NIP motor	Cassette lock release	
24		Cassette removal	

4.4 Sensor status Check

Perform the following steps to confirm sensor status.

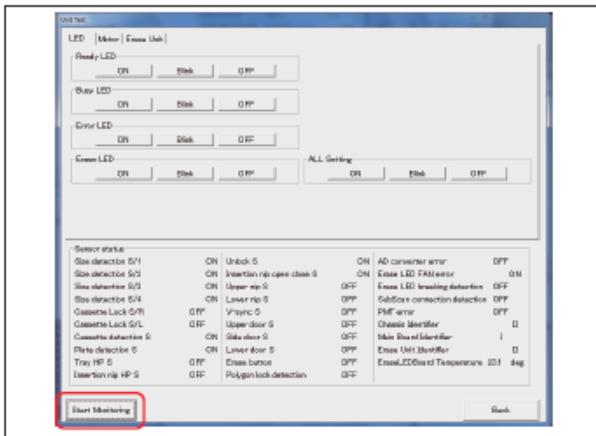
Also, the relationship between sensor status and device status when calculating the device status with multiple sensors is explained here.

- "4.4.2 Cassette size & sensor status (Page 4-5)"
- "4.4.3 Insertion site sensor status (Page 4-6)"

4.4.1 Sensor status Confirmation Process

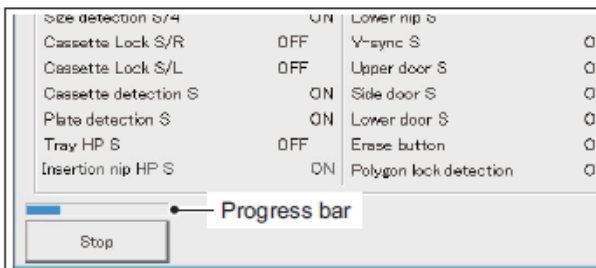
Perform the following steps to confirm sensor status.

1. Click the "Start Monitoring" button at the "Unit Test" screen.



The sensor status is then monitored, and the "Start Monitoring" button changes to a "Stop" button.

Once monitoring begins, an updated sensor status is displayed on the [Sensor status] screen about once every 3 seconds. Also, a progress bar is displayed above the "Stop" button.



2. Confirm the sensor status.

IMPORTANT

Other single device operation commands cannot be sent while the sensors are being monitored. For example, sensors cannot be monitored while operating a motor.

If you want to run a different single device operation command, you'll have to stop sensor monitoring first.

3. After confirming sensor operations, click the "Stop" button.

Sensor status monitoring will come to a halt.

4.4.2 Cassette size & sensor status

Cassette size calculation is performed based on the status of 4 sensors.

Cassette size	Sensor status			
	S1	S2	S3	S4
14" × 17"	ON	ON	ON	ON
14" × 14"	ON	ON	ON	
15 cm × 30 cm		ON	ON	ON
11" × 14"		ON		ON
10" × 12"	ON		ON	ON
24 cm × 30 cm	ON			ON
8" × 10"				ON
18 cm × 24 cm		ON	ON	
No cassette				

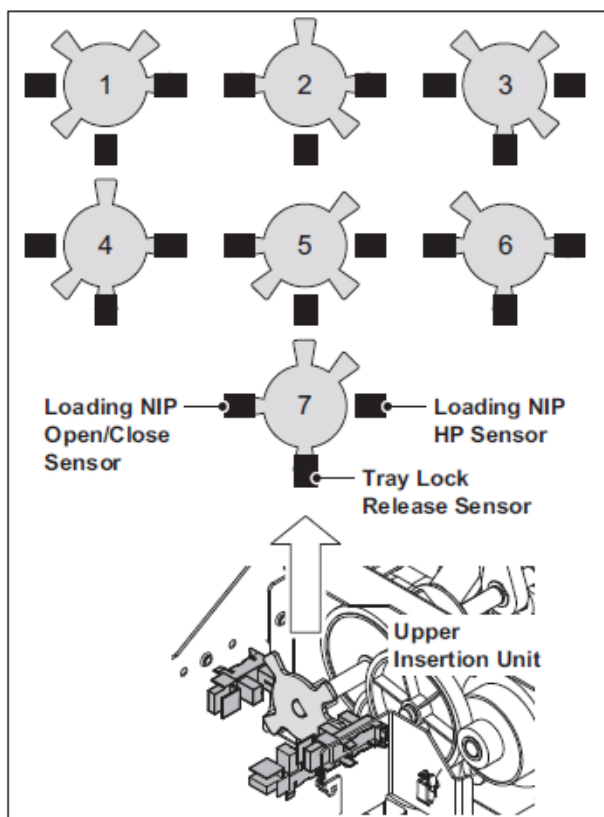
- S1: Size detection sensor 1 status
- S2: Size detection sensor 2 status
- S3: Size detection sensor 3 status
- S4: Size detection sensor 4 status

4.4.3 Insertion site sensor status

The status of NIP insertion by the loading NIP motor, and the status of 3 sensors located to the left of the upper insertion unit, are defined in the following chart.

No.	NIP insertion status	Sensor status			Rotation angle
		NIP open/close	Lock release	HP	
1	Home position			ON	0°
2	Tray ejecting notch lowered	ON		ON	45°
3	Tray lock released		ON		85°
4	NIP insertion status		ON	ON	135°
5	Tray raised	ON			180°
6	Inserted NIP released	ON	ON	ON	225°
7	Cassette lock released	ON	ON		-45°

- Rotation angle: The rotation angle from home position.



Chapter 5

Disassembly & Assembly

An explanation of the steps needed to disassemble and assemble the device.

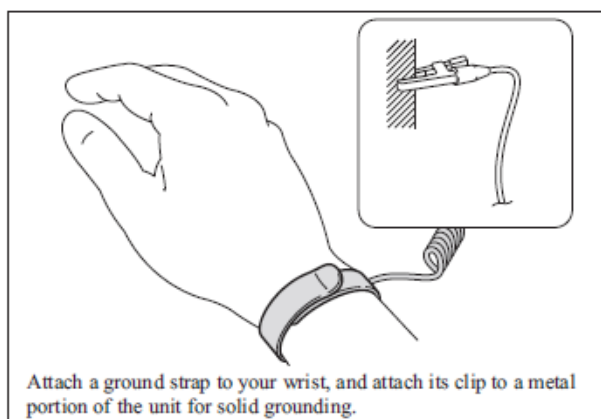
5.1	Before Disassembly	5-2
5.2	Fundamental Steps	5-3
5.3	Changing electrical parts.....	5-25
5.4	Changing mechanical parts	5-40
5.5	Changing cassette parts	5-51

5.1 Before Disassembly

5.1.1 Disassembly/Assembly Cautions

⚠ CAUTION

- Before disassembling/assembling this device, be sure to switch OFF the power supply circuit breaker and disconnect the power cable from the device. Otherwise, your fingers may get caught in sliding or rotating parts inside of the device due to accidental operation.
- Please use a static electricity resistant wristband when handling electronic components inside the device. Touching a charged base board or other electronic component may damage or destroy it.



- Do not touch the inside of this device with bare hands. Dirt or oils from the skin may cause the device to rust, resulting in defective sliding or rotating parts.
- When assembling this device, make sure all wiring is returned to its proper location, and confirm that no wires are lodged in any attachments, etc. If wires are lodged or caught in any attachments, it may cause the device to malfunction.
- Do NOT loosen any screws that are painted white or yellow. If any white screws are loosened, device recovery may be impossible to perform at the service site.

5.1.2 Checking & adjustments needed after assembly

When devices or parts have been reattached or changed, adjustments and confirmation must be done as described in the following chart.

📌 IMPORTANT

After making vertical/horizontal magnification adjustments, read starting point adjustments, or oblique adjustments, make sure to correct irregularity. Correction of irregularity must be done for both 175 μm and 87.5 μm resolutions.

Device/part	Confirmation/adjustment
All units and parts	- Confirm that all status lamps are lit right after turning the power ON. - Confirm that the Ready Lamp is lit after initialization. - When performing an image read, confirm that the image is displayed on the ImagePilot.
Upper Insertion Unit	Oblique confirmation/adjustment For details, refer to "6.1 Oblique adjustments (Page 6-2)".
Exposure Unit	- Confirm/adjust horizontal magnification - Confirm/adjust read starting point - Perform sensitivity calibration - Correct irregularity Refer to the ImagePilot Installation & Service Manual.
Light Collection Unit	- Perform sensitivity calibration - Correct irregularity Refer to the ImagePilot Installation & Service Manual.
Eraser Unit/Assy (Eraser LED)	When changing parts: - Confirm that the Eraser LED lights up Use the service tool's [Unit Test] (Check) function to make the Eraser LED light up, then confirm visually that it has indeed lit up. - If it flickers or does not light up at all, there is a defective connection. For a detailed explanation of the [Unit Test](Check) function, refer to "4.2 [Unit Test] (Check) Screen Display (Page 4-2)". - Clear the Erase Counter Use the service tool's [Processing Counter] function to perform this task. Refer to the ImagePilot Installation & Service Manual.

Device/part	Confirmation/adjustment
Sub-scan Unit	- Confirm image If there are no problems, no adjustment is necessary. If a problem is found, perform the following confirmation/adjustment. - Confirm/adjust vertical and horizontal magnification - Confirm/adjust read starting point - Perform sensitivity calibration - Correct irregularity Refer to the ImagePilot Installation & Service Manual. - Oblique confirmation/adjustment For details, refer to "6.1 Oblique adjustments (Page 6-2)".
SW power supply	Confirm voltage Details can be found in the change process descriptive text ("5.3.5 Changing the SW power supply (Page 5-30)").
Main board	Make a backup of base board data before changing. After changing, the backup data must be restored via the new base board. Details can be found in the change process descriptive text ("5.3.4 Changing the main board (Page 5-28)").

5.1.3 Problems that often occur after assembly

When forgetting to connect the connectors, or they are not securely attached, the following problems will occur. Make sure to thoroughly check that all connectors are fastened securely.

Trouble	Connectors to check when a problem occurs
The Status Lamp will not light up.	Status LED board SCN1
An error occurs during initialization.	Left side of device (Sub-scan unit) - JP36-2(-A) Right side of device (Sub-scan unit) - JJ37 - JP7-2
After the image reading process is complete and the cassette is removed, the Ready Lamp continues to blink. (No problem before the image reading process)	Right side of device (lower insertion unit) JP16-2

5.2 Fundamental Steps

When disassembling and reassembling this device, certain fundamental steps must be taken, including a method of removing and attaching the external panel, which are fully explained here.

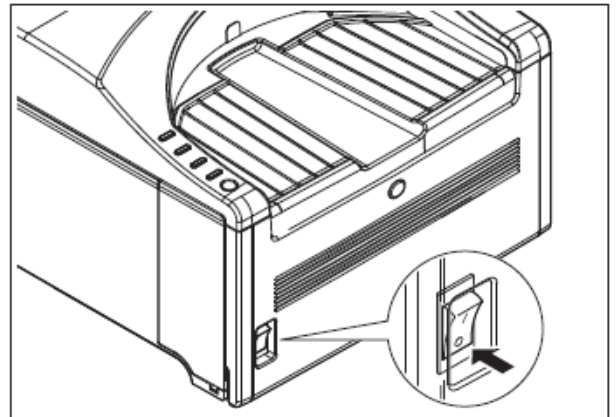
5.2.1 Power OFF/ON

The device's power OFF/ON method is explained below.

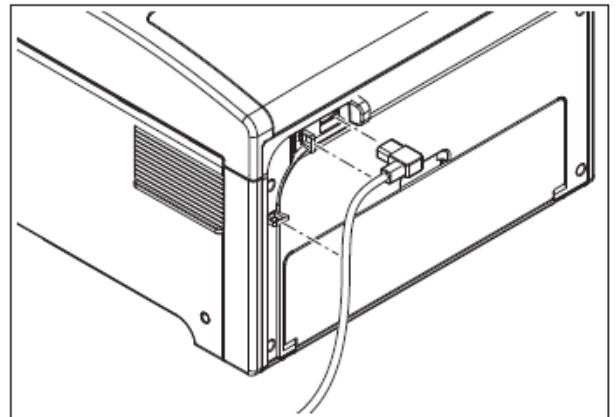
Power OFF

Turn the power supply circuit breaker OFF, then remove the power cable and USB cable.

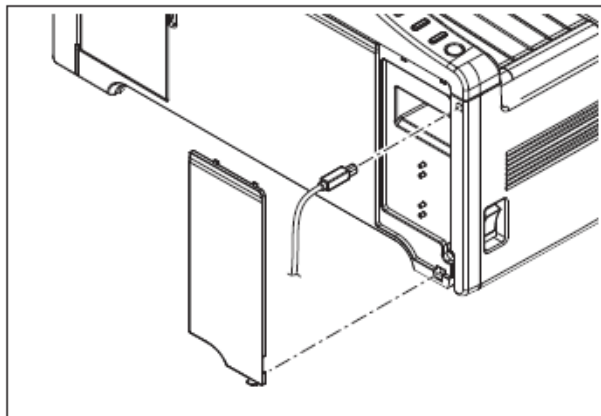
1. Make sure the Busy Lamp is unlit.
2. Turn OFF the power supply circuit breaker.



3. Unplug the power cable from its outlet and disconnect it from the device.



4. Disconnect the USB cable from the device.



This concludes the power OFF sequence.

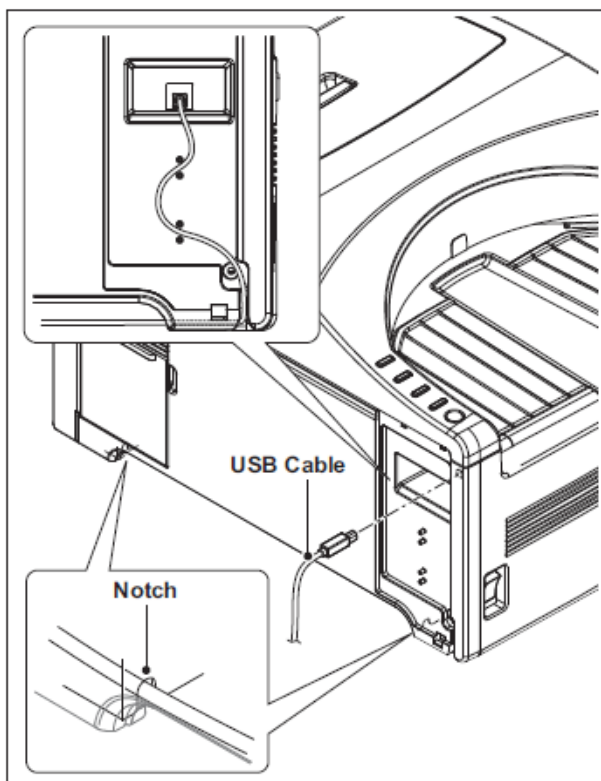
Power ON

Connect the USB cable and power cable to the device, then switch the device power ON.

1. Connect the USB cable to the device.



When wiring the cable to the back of the device, it can be threaded through the notches at the bottom.



2. Connect the power cable to this device, then plug it into an outlet.



To prevent the power cable from coming loose, make sure to thread it through the locking wire saddles (× 2).

3. Switch ON the power supply circuit breaker.

4. Check the status lamps.

- After turning the power ON, all lamps should be lit for approximately 5 seconds during initialization. If any of the lamps are not lit, they may be in need of repair.
- Once initialization has ended, and image reading can be performed, the Ready Lamp should be lit.

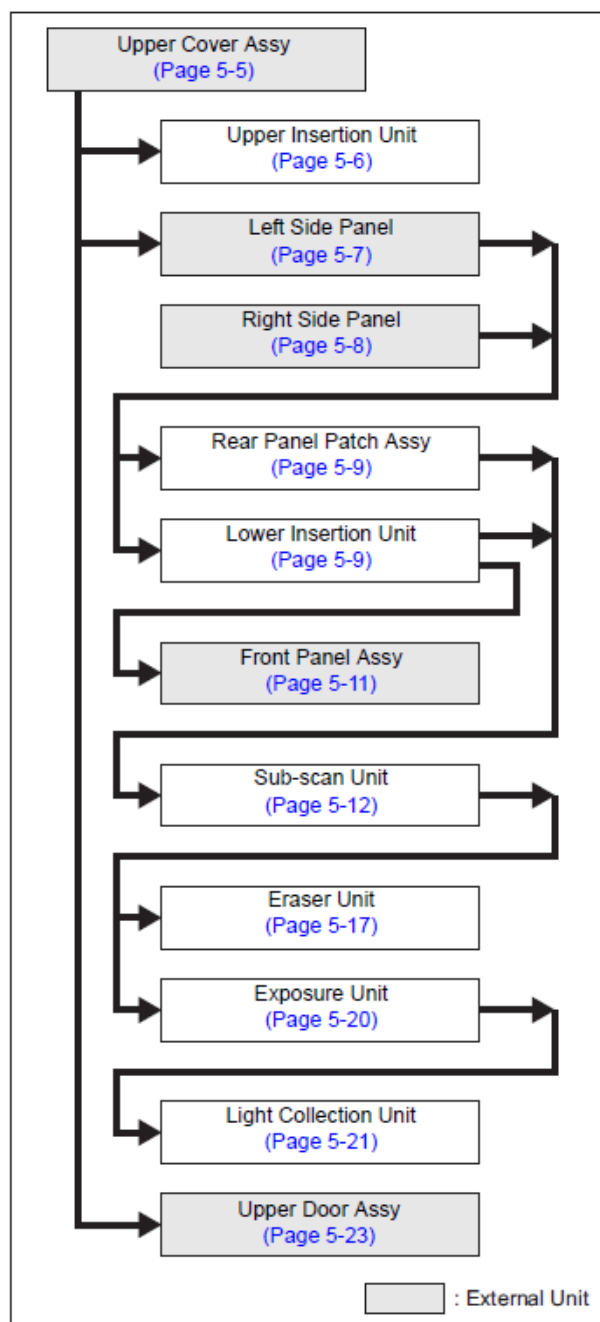


If this device is not connected to the ImagePilot, or the ImagePilot has not started up, the Error Lamp will be lit.

This concludes the power ON sequence.

5.2.2 Removing the external panel & parts

External panel and part removal should be performed as follows.



5.2.3 Removing & attaching the upper cover assy

The following is a description of how to remove and attach the upper cover assy.

⚠ CAUTION

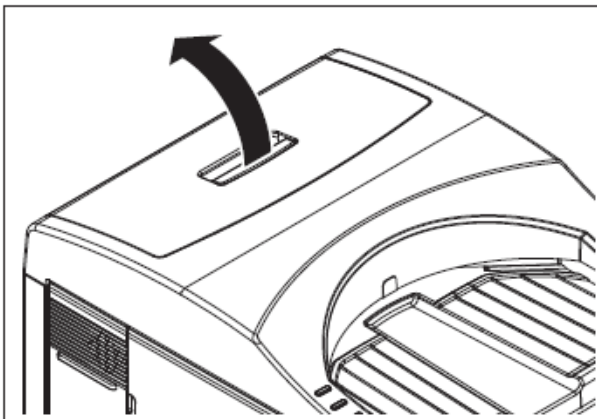
Before disassembling/assembling this device, be sure to switch OFF the power supply circuit breaker and disconnect the power cable from the device.

Things to prepare

Standard tools are required to perform the following steps.

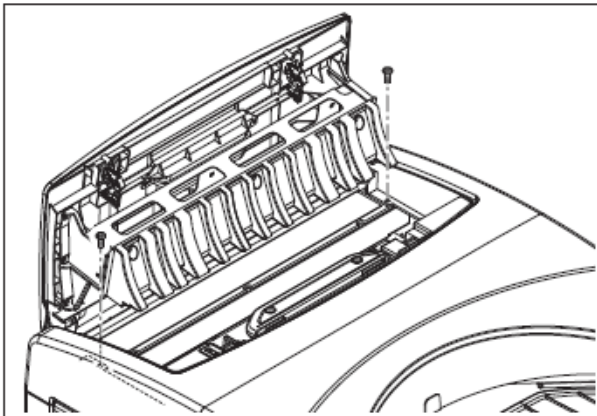
The removal process

1. Open the upper Door.

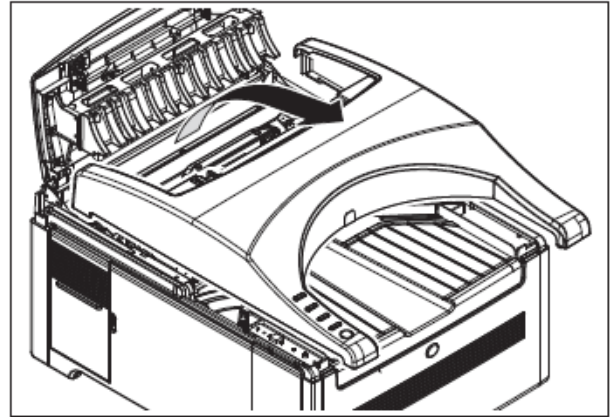


2. Remove the screws from the upper cover assy.

- Screw × 2



3. Remove the upper cover assy.



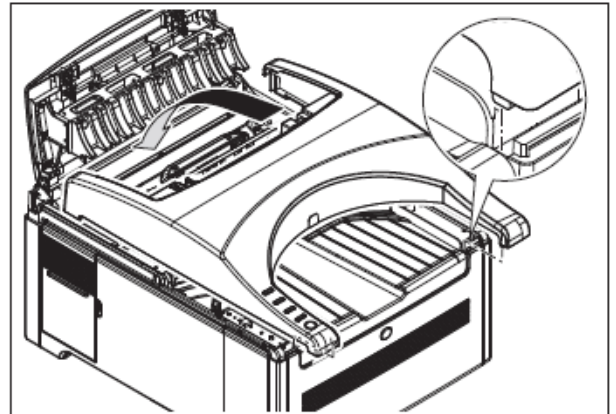
This concludes the upper cover assy removal process.

The attachment process

1. Open the upper door.

2. Attach the upper cover assy.

First attach the 2 protruding parts in front, then slide the back part down and in until it fits.



3. Attach the screws to the upper cover assy.

- Screw × 2

This concludes the upper cover assy attachment process.

5.2.4 Removing & attaching the upper insertion unit

The following is a description of how to remove and attach the upper insertion unit.

⚠ CAUTION

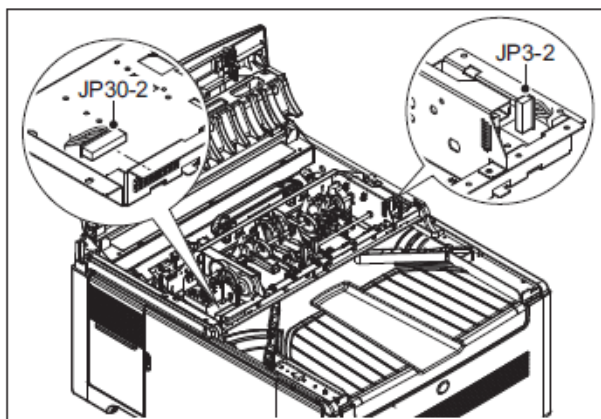
Before disassembling/assembling this device, be sure to switch OFF the power supply circuit breaker and disconnect the power cable from the device.

Things to prepare

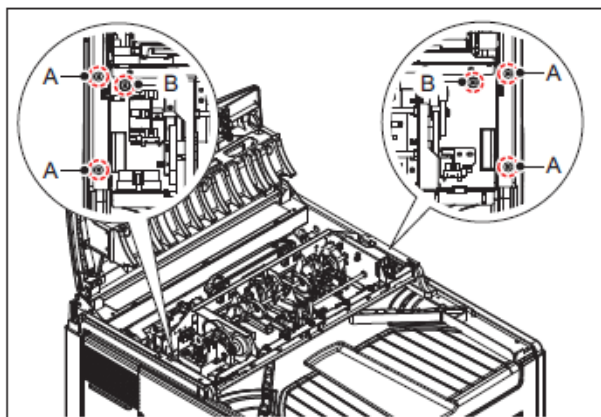
Standard tools are required to perform the following steps.

The removal process

1. Refer to "5.2.3 Removing & attaching the upper cover Assy (Page 5-5)", and remove the upper cover Assy.
2. Remove the cables (connectors) as shown in the diagram.
 - JP30-2
 - JP3-2



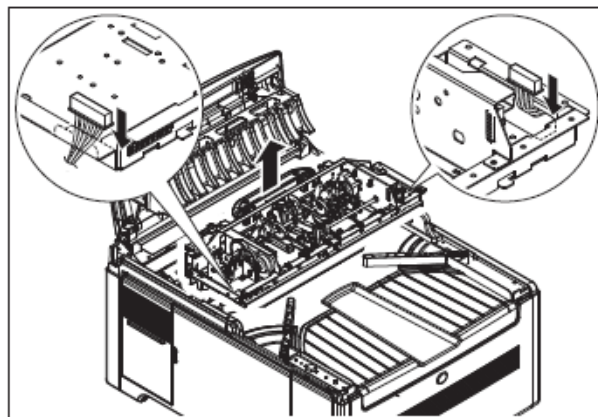
3. Remove the screws from the upper insertion unit.
 - A: TP hexagonal Phillips-head screw × 4 (M3 × 6)
 - B: TP hexagonal Phillips-head semi screws × 2 (M3 × 8)



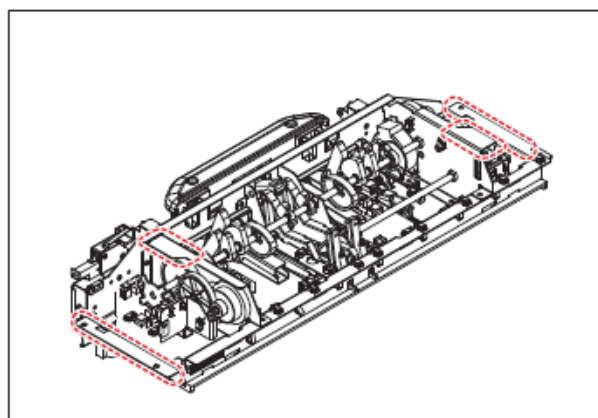
4. Insert the cables removed in step 2 into the square holes from the top.

Lift up the upper insertion unit a bit from each side, then insert cables into the square holes.

5. Remove the upper insertion unit.

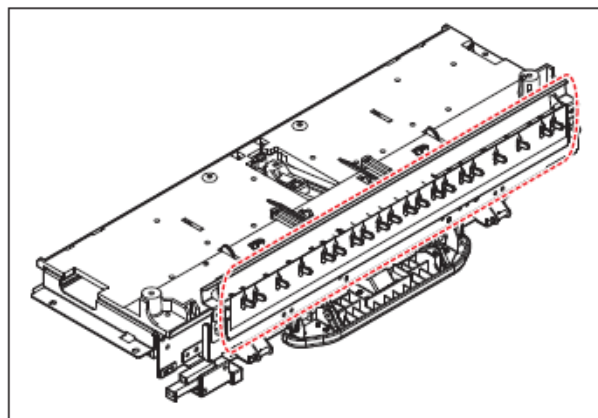


Handle the upper insertion unit from the sections circled in the illustration below (normally from the 2 sections on the inside).



⚠ IMPORTANT

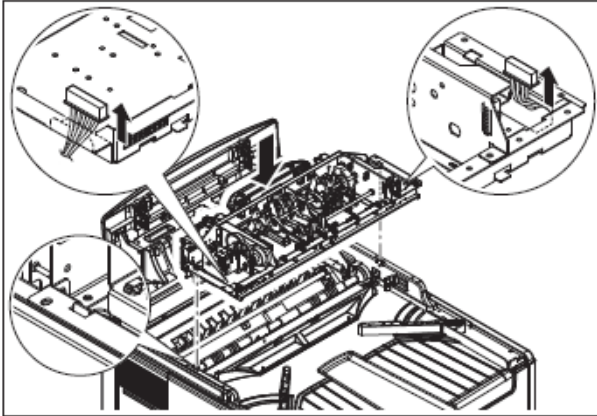
When setting down the removed upper insertion unit, place it upside down to avoid scratching the antistatic sheet Assy (circled in the illustration below).



This concludes the upper insertion unit removal process.

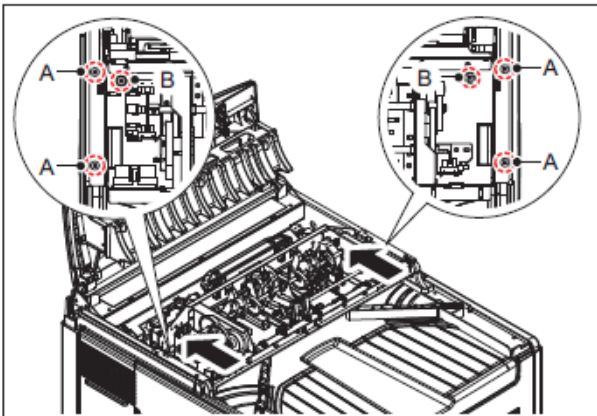
The attachment process

1. Pull the connectors out of the square holes they were placed in while removing the upper insertion unit, and put the upper insertion unit on top of the frame.



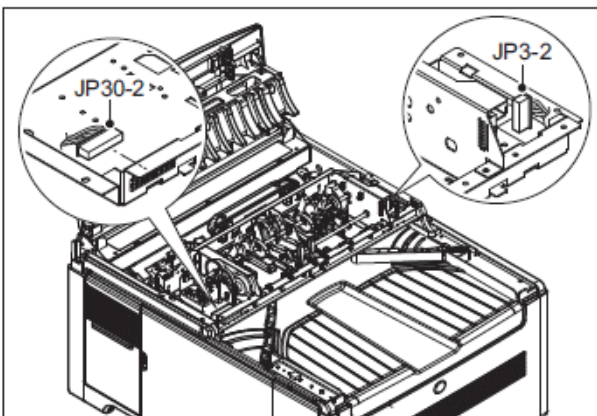
2. Press the upper insertion unit in towards the rear, and tighten it down with screws.

- A: TP hexagonal Phillips-head screw × 4 (M3 × 6)
- B: TP hexagonal Phillips-head semi screws × 2 (M3 × 8)



3. Connect the cables.

- JP30-2
- JP3-2
(After connecting the connectors, thread the cable through the 2 wire saddle points.)



4. Refer to "5.2.3 Removing & attaching the upper cover assy (Page 5-5)" and attach the upper cover assy.

This concludes the upper insertion unit attachment process.

Things to check/adjust when re-attaching the same upper insertion unit.

Perform the following checks/adjustments when re-attaching the upper insertion unit.

- Checking/adjusting oblique lighting
For details, refer to "6.1 Oblique adjustments (Page 6-2)".

5.2.5 Removing & attaching the left side panel

The following is a description of how to remove and attach the left side panel.

⚠ CAUTION

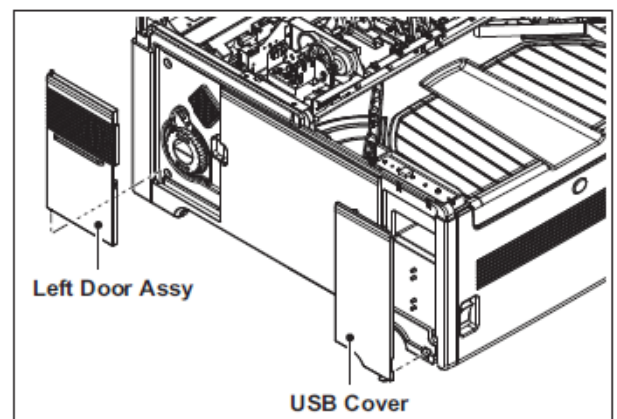
Before disassembling/assembling this device, be sure to switch OFF the power supply circuit breaker and disconnect the power cable from the device.

Things to prepare

Standard tools are required to perform the following steps.

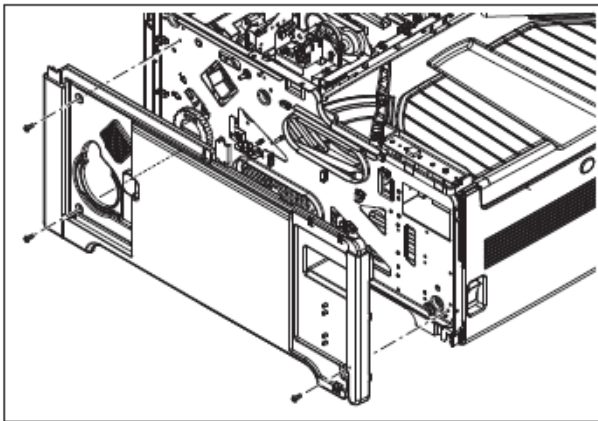
The removal process

1. Refer to "5.2.3 Removing & attaching the upper cover assy (Page 5-5)", and remove the upper cover assy.
2. Remove the left door and USB cover.



3. Remove the left side panel.

- Screws × 3



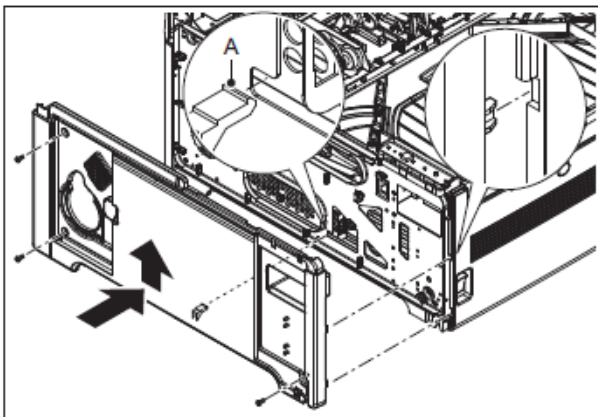
This concludes the left side panel removal process.

The attachment process

1. Attach the left side panel.

- Screws × 3

Attach while abutted in the upward direction.



IMPORTANT

Make sure Notch A shown above does not get caught on the frame, and that the panel bottom is not open.

2. Attach the left door Assy.

HINT

If no difficulties are experienced during this process, attach the USB cover after connecting the USB cable.

3. Refer to "5.2.3 Removing & attaching the upper cover Assy (Page 5-5)" and attach the upper cover Assy.

This concludes the left side panel attachment process.

5.2.6 Removing & attaching the right side panel

The following is a description of how to remove and attach the right side panel.

CAUTION

Before disassembling/assembling this device, be sure to switch OFF the power supply circuit breaker and disconnect the power cable from the device.

Things to prepare

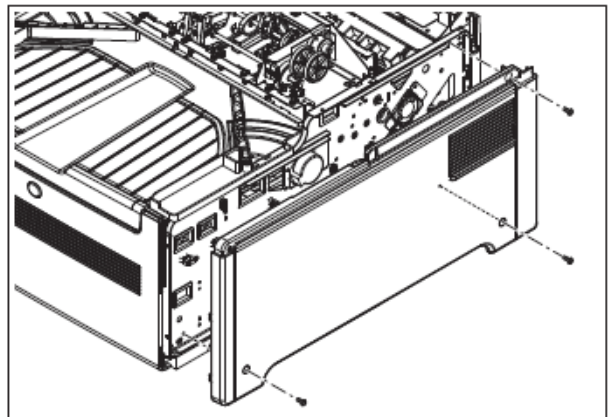
Standard tools are required to perform the following steps.

The removal process

1. Refer to "5.2.3 Removing & attaching the upper cover Assy (Page 5-5)", and remove the upper cover Assy.

2. Remove the right side panel.

- Screws × 3



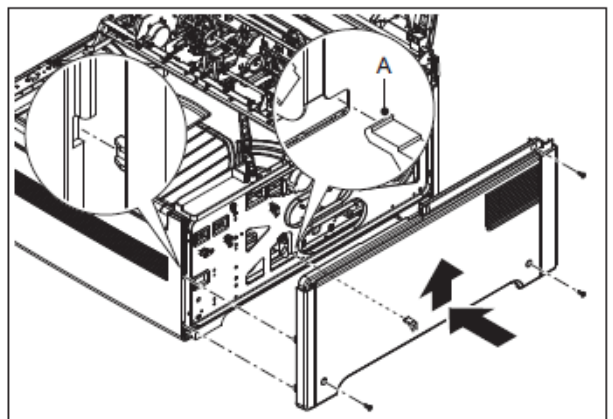
This concludes the right side panel removal process.

The attachment process

1. Attach the right side panel.

- Screws × 3

Attach while abutted in the upward direction.



IMPORTANT

Make sure Notch A shown above does not get caught on the frame, and that the panel bottom is not open.

2. Refer to "5.2.3 Removing & attaching the upper cover assy (Page 5-5)" and attach the upper cover assy.

This concludes the right side panel attachment process.

5.2.7 Removing & attaching the rear panel patch assy.

The following is a description of how to remove and attach the rear panel patch assy.

CAUTION

Before disassembling/assembling this device, be sure to switch OFF the power supply circuit breaker and disconnect the power cable from the device.

Things to prepare

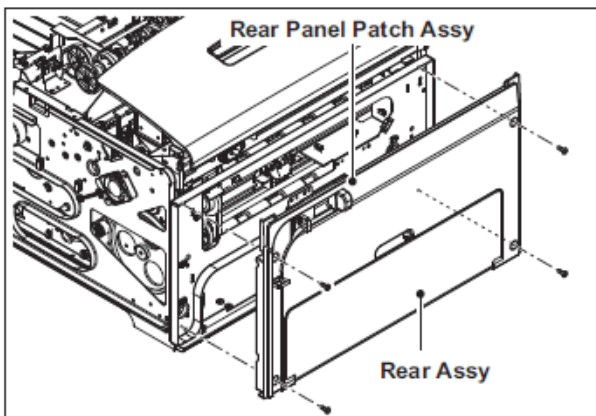
Standard tools are required to perform the following steps.

The removal process

1. Refer to "5.2.3 Removing & attaching the upper cover assy (Page 5-5)", and remove the upper cover assy.
2. Refer to "5.2.5 Removing & attaching the left side panel (Page 5-7)" and remove the left side panel.
3. Refer to "5.2.6 Removing & attaching the right side panel (Page 5-8)" and remove the right side panel.
4. Remove the rear panel patch assy together with the rear assy.
 - Screws × 4

IMPORTANT

While supporting the rear assy so that it doesn't fall off, remove the rear panel patch assy.



This concludes the rear panel patch assy removal process.

The attachment process

1. Attach the rear panel patch assy together with the rear assy.

- Screws × 4

IMPORTANT

While supporting the rear assy so that it doesn't fall off, attach the rear panel patch assy.

2. Refer to "5.2.6 Removing & attaching the right side panel (Page 5-8)" and attach the right side panel.
3. Refer to "5.2.5 Removing & attaching the left side panel (Page 5-7)" and attach the left side panel.
4. Refer to "5.2.3 Removing & attaching the upper cover assy (Page 5-5)" and attach the upper cover assy.

This concludes the rear panel patch assy attachment process.

5.2.8 Removing & attaching the lower insertion unit

The following is a description of how to remove and attach the lower insertion unit.

CAUTION

- Before disassembling/assembling this device, be sure to switch OFF the power supply circuit breaker and disconnect the power cable from the device.
- In order to avoid potential base board damage while discharging it, refer to "5.1.1 Disassembly/Assembly Cautions (Page 5-2)" and wear a grounding strap during the process.

Things to prepare

Standard tools are required to perform the following steps.

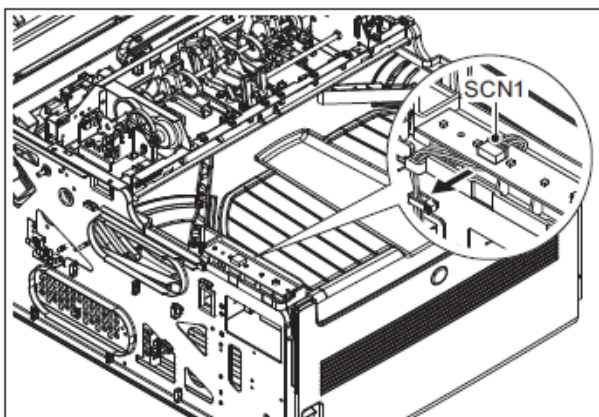
The removal process

1. Refer to "5.2.3 Removing & attaching the upper cover assy (Page 5-5)", and remove the upper cover assy.
2. Refer to "5.2.5 Removing & attaching the left side panel (Page 5-7)" and remove the left side panel.
3. Refer to "5.2.6 Removing & attaching the right side panel (Page 5-8)" and remove the right side panel.

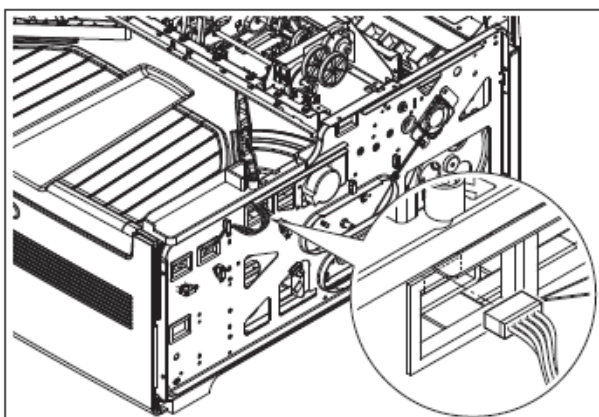
4. Remove cables connected to the status LED board.

- SCN1

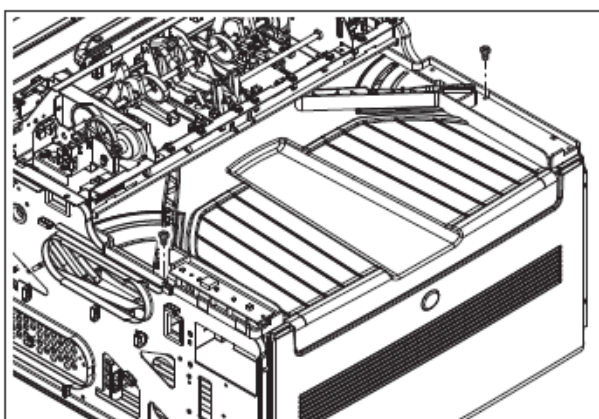
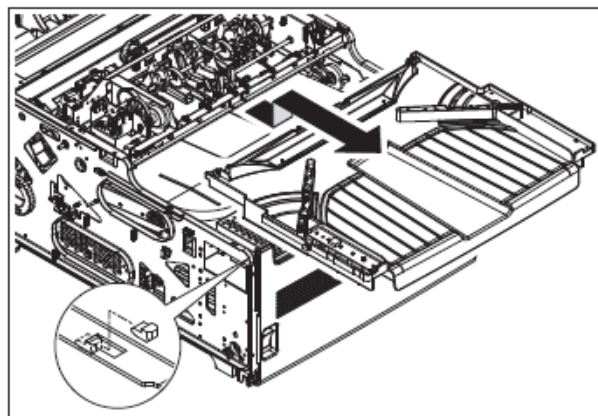
These cables travel underneath the status LED board, so make sure to pull them out.

**5. Remove connectors from the right side of the device as illustrated.**

- JP16-2

**6. Remove the screws from the lower insertion unit.**

- TP hexagonal Phillips-head screws 2 (M3 × 8)

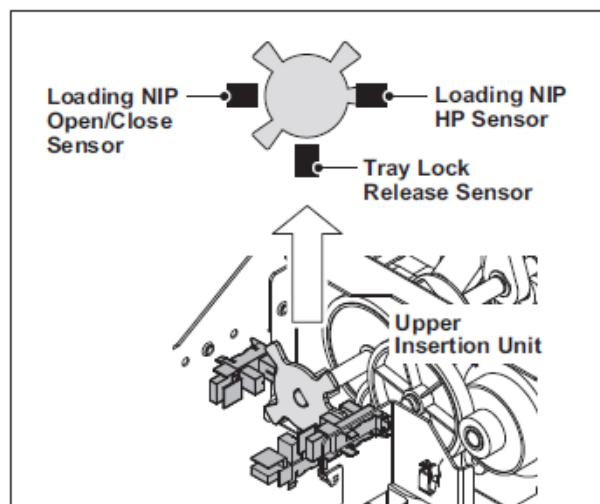
**7. Remove the lower insertion unit.**

This concludes the lower insertion unit removal process.

The attachment process

1. Check the upper insertion unit's cam part.

If the situation is not as shown below, rotate the cam part so that the detector is angled according to the illustration.

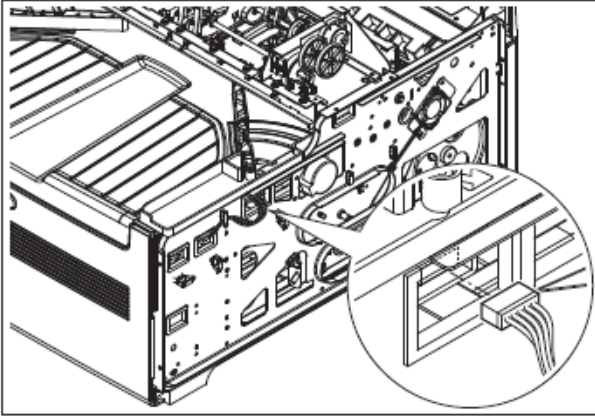
**IMPORTANT**

When the situation is the same as illustrated, the upper insertion unit's notches are in the escape position. If the notches are not in the escape position, contact may occur when attaching the lower insertion unit.

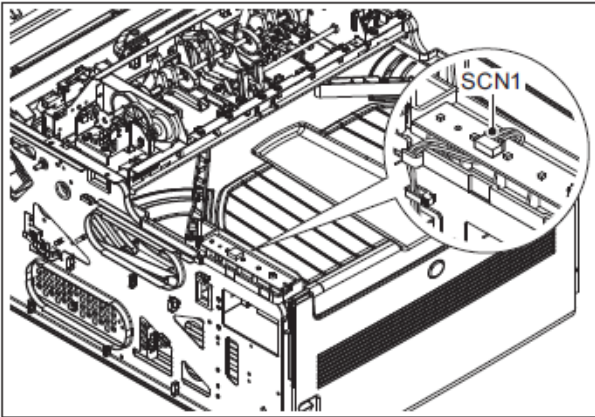
2. Attach the lower insertion unit to the frame.**3. Secure the lower insertion unit with screws.**

- TP hexagonal Phillips-head screws 2 (M3 × 8)

4. Connect cables to the lower insertion unit from the right side of the device.
- JP16-2



5. Connect cables to the status LED board.
- SCN1
- Pass the cables underneath the status LED board to connect them.



6. Refer to "5.2.6 Removing & attaching the right side panel (Page 5-8)" and attach the right side panel.
7. Refer to "5.2.5 Removing & attaching the left side panel (Page 5-7)" and attach the left side panel.
8. Refer to "5.2.3 Removing & attaching the upper cover Assy (Page 5-5)" and attach the upper cover Assy.

This concludes the lower insertion unit attachment process.

5.2.9 Removing/attaching the front panel Assy

The following is a description of how to remove and attach the front panel Assy.

⚠ CAUTION

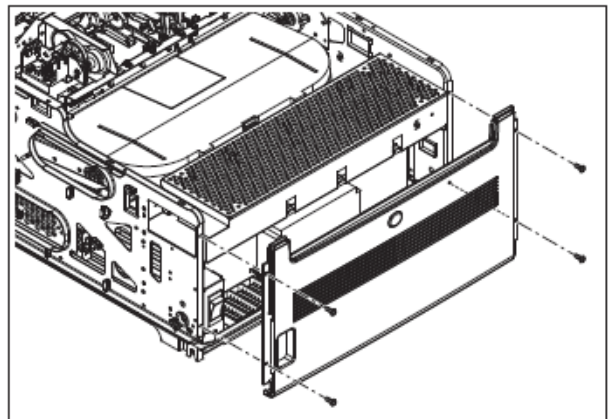
- Before disassembling/assembling this device, be sure to switch OFF the power supply circuit breaker and disconnect the power cable from the device.
- In order to avoid potential base board damage while discharging it, refer to "5.1.1 Disassembly/Assembly Cautions (Page 5-2)" and wear a grounding strap during the process.

Things to prepare

Standard tools are required to perform the following steps.

The removal process

1. Refer to "5.2.3 Removing & attaching the upper cover Assy (Page 5-5)", and remove the upper cover Assy.
2. Refer to "5.2.5 Removing & attaching the left side panel (Page 5-7)" and remove the left side panel.
3. Refer to "5.2.6 Removing & attaching the right side panel (Page 5-8)" and remove the right side panel.
4. Refer to "5.2.8 Removing & attaching the lower insertion unit (Page 5-9)", and remove the lower insertion unit.
5. Remove the front panel Assy.
 - Screws x 4



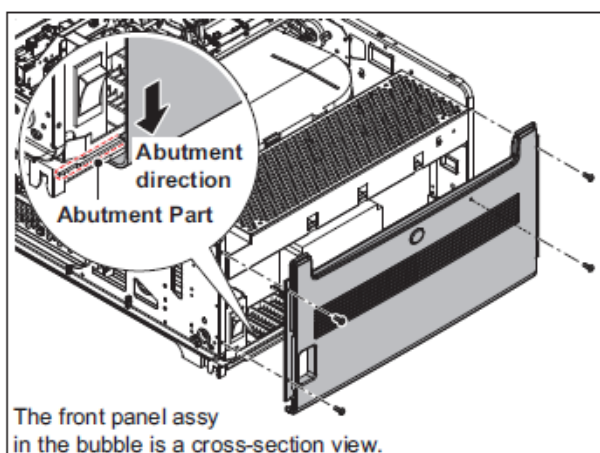
This concludes the front panel Assy removal process.

The attachment process

1. Attach the front panel assy.

- Screws × 4

Attach while abutted as shown by the illustration.



2. Refer to ["5.2.8 Removing & attaching the lower insertion unit \(Page 5-9\)"](#) and attach the lower insertion unit.
3. Refer to ["5.2.6 Removing & attaching the right side panel \(Page 5-8\)"](#) and attach the right side panel.
4. Refer to ["5.2.5 Removing & attaching the left side panel \(Page 5-7\)"](#) and attach the left side panel.
5. Refer to ["5.2.3 Removing & attaching the upper cover assy \(Page 5-5\)"](#) and attach the upper cover assy.

This concludes the front panel assy attachment process.

5.2.10 Removing & attaching the Sub-scan unit

The following is a description of how to remove and attach the Sub-scan unit.

(In this explanation, the term "sub-scan unit" refers to the exposure unit, light collection unit, and eraser unit together as a single assembly.)

⚠ CAUTION

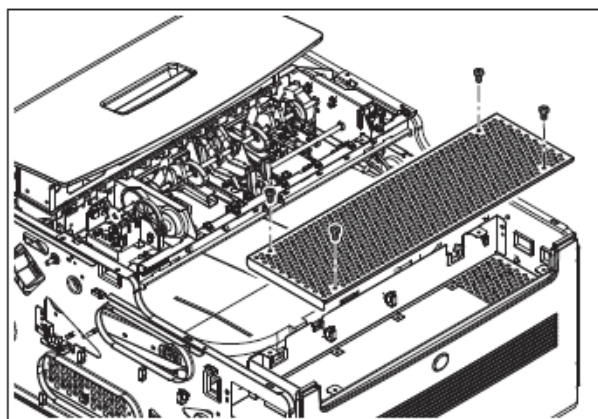
- Before disassembling/assembling this device, be sure to switch OFF the power supply circuit breaker and disconnect the power cable from the device.
- In order to avoid potential base board damage while discharging it, refer to ["5.1.1 Disassembly/Assembly Cautions \(Page 5-2\)"](#) and wear a grounding strap during the process.

Things to prepare

Standard tools are required to perform the following steps.

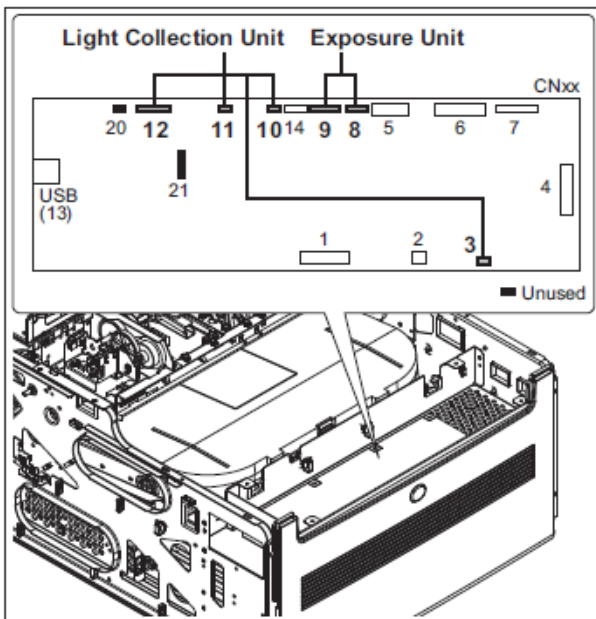
The removal process

1. Refer to ["5.2.3 Removing & attaching the upper cover assy \(Page 5-5\)"](#), and remove the upper cover assy.
2. Refer to ["5.2.5 Removing & attaching the left side panel \(Page 5-7\)"](#) and remove the left side panel.
3. Refer to ["5.2.6 Removing & attaching the right side panel \(Page 5-8\)"](#) and remove the right side panel.
4. Refer to ["5.2.7 Removing & attaching the rear panel patch assy. \(Page 5-9\)"](#) and remove the rear panel patch assy.
5. Refer to ["5.2.8 Removing & attaching the lower insertion unit \(Page 5-9\)"](#), and remove the lower insertion unit.
6. Remove the frame's anterior cover.
 - TP hexagonal Phillips-head screw × 4 (M3 × 6)



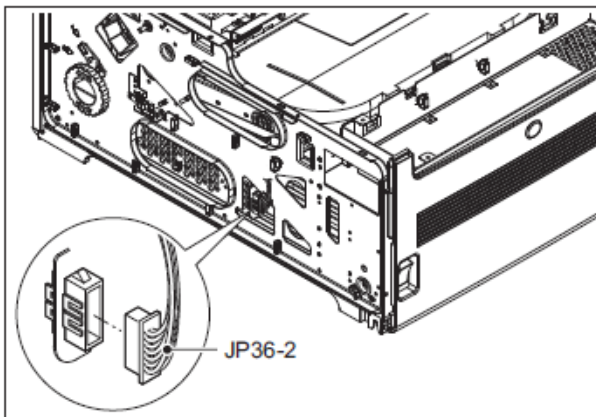
7. Remove cables connected to the main board from the exposure unit and light collection unit.

- Exposure unit: CN8, CN9
- Light Collection Unit: CN3, CN10, CN11, CN12
(Together with the main board connector names.)



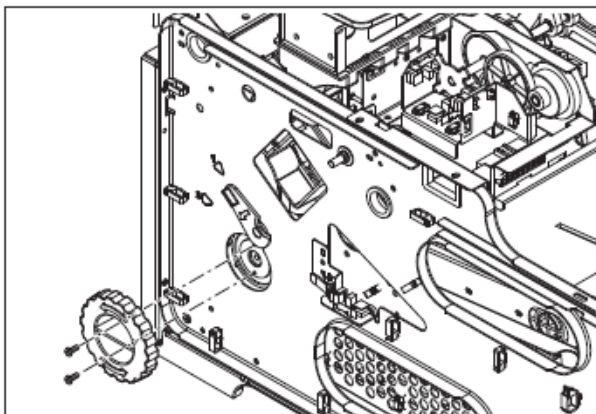
8. Remove connectors from the left side of the device as illustrated.

- JP36-2



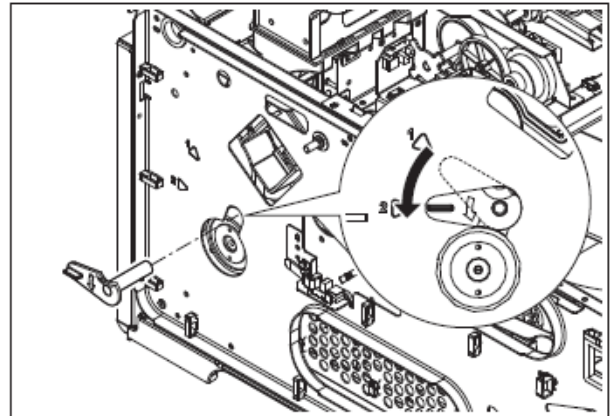
9. Remove the handle (unjamming dial).

- TP hexagonal Phillips-head screw × 2 (M3 × 6)



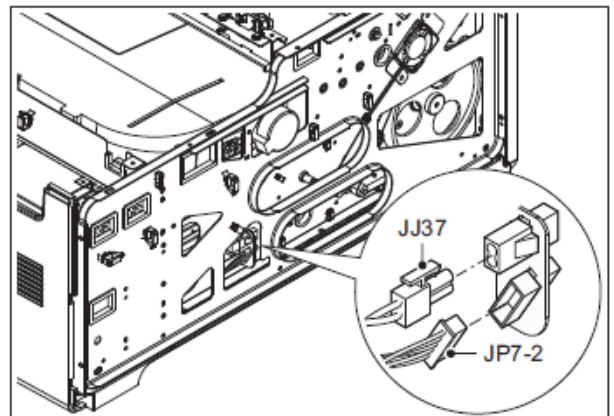
10. Remove the handle (light collection unit cleaning handle).

Take it out while turned in the #2 position.



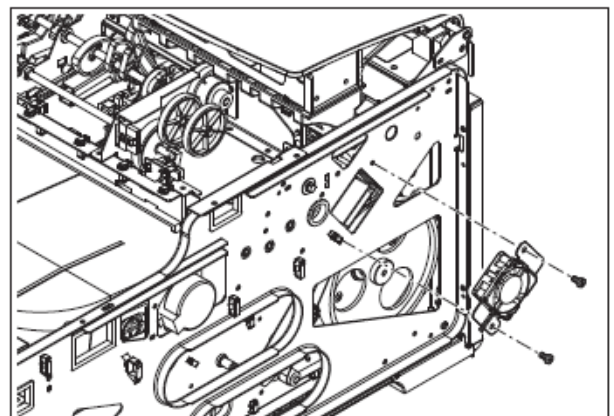
11. Remove connectors from the right side of the device as illustrated.

- JJ37
- JP7-2

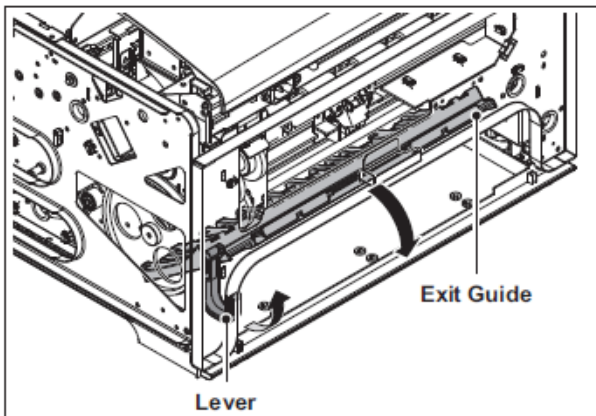


12. Remove the fan assembly (Eraser LED cooling fan).

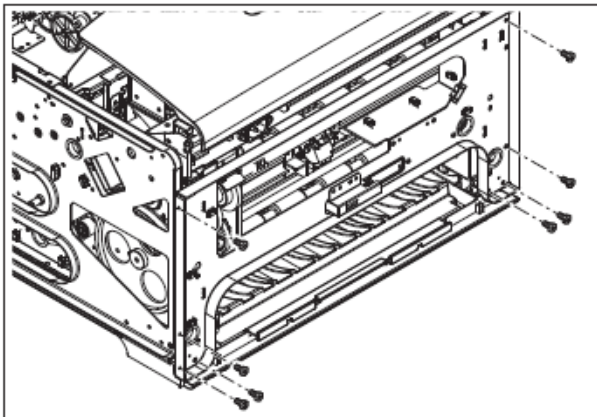
- TP hexagonal Phillips-head screw × 2 (M3 × 6)



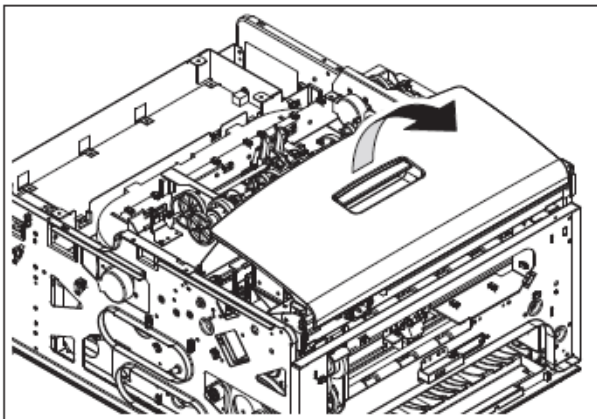
13. Pull the device's rear lever and slide down the exit guide.



14. Remove the screws from the sub-scan unit.
- TP hexagonal Phillips-head screw × 8 (M3 × 6)



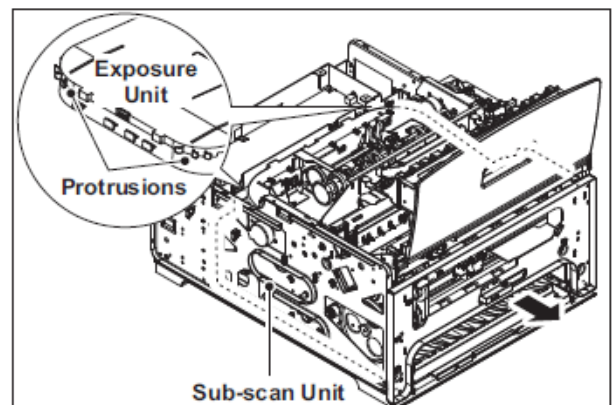
15. Open the upper door.



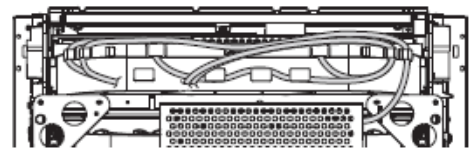
16. Pull the sub-scan unit about 5cm out of the device, then coil the cables removed from the main board around the exposure unit protrusions.

IMPORTANT

When removing the sub-scan unit, be careful not to get the cables caught on the frame, upper insertion unit, etc. Space between the exposure unit and upper insertion unit is extremely small, so be sure not to pull out the sub-scan unit while cables remain on top of the exposure unit.

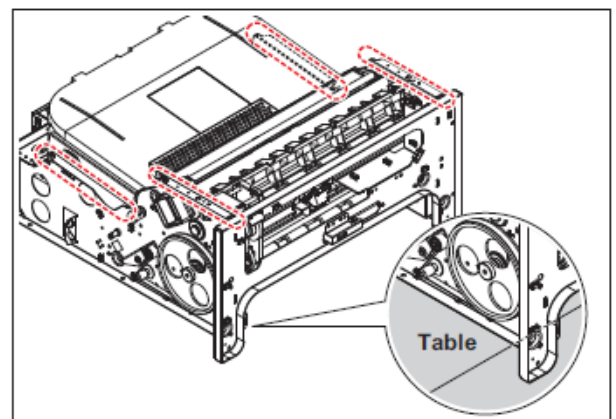


Wrapping Reference



17. Pull out and remove the sub-scan unit.

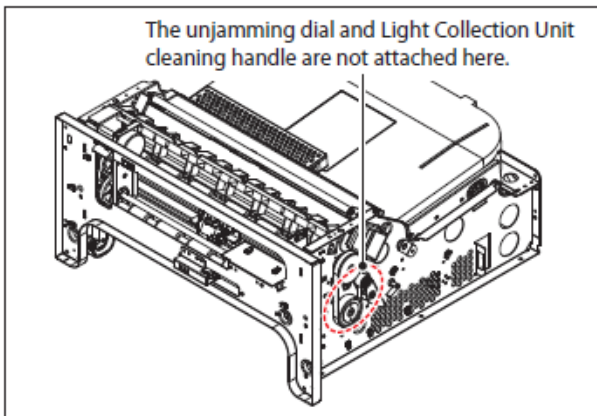
Handle the sub-scan unit from the sections circled in the illustration below. When setting it down, place it on the table so that the protruding parts hang off the edge, enabling it to sit flat as shown in the bubble.



This concludes the sub-scan unit removal process.

The attachment process

1. Make sure all parts removed from the sub-scan unit are back in place.



HINT

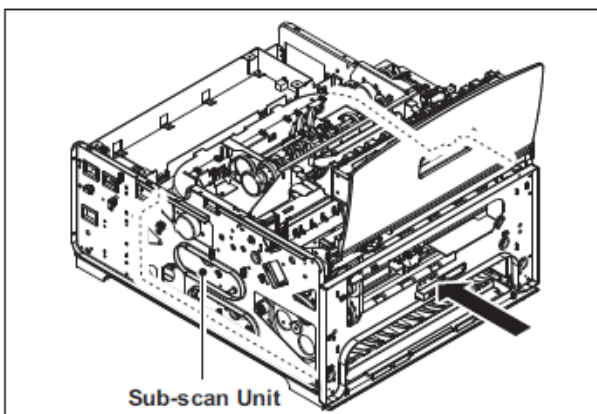
Attachments you might forget:

- Attach exposure unit screws
(Exposure unit: 6 places, Fixed pin Assy: 2 places)
- Reconnect cables disconnected when removing and attaching base board assembly (4 places)

2. As done during removal, coil the exposure unit cables, etc. around the exposure unit protrusions.
3. When inserting the sub-scan unit into the device, unravel the coiled cables part way through and place them on top of the exposure unit.

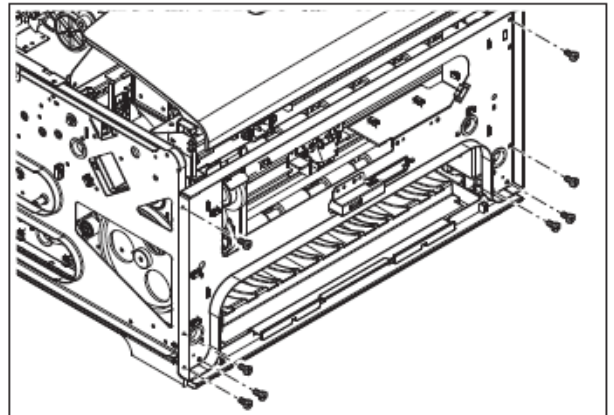
IMPORTANT

- When inserting the sub-scan unit, make sure it slides into the guides along the bottom of the device. Make sure it is not loaded at a slant, or that it does not hit the loaded part when lifted.
- Depending on how the cables were coiled, they might get smashed when putting the sub-scan unit back into the device. In some cases you may not be able to uncoil the cables, so make sure to check the cable status while inserting the sub-scan unit into the device, then uncoil them from the protrusions part way through.

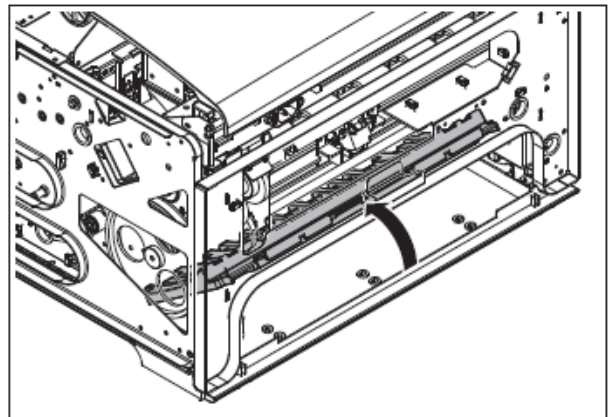


4. Close the upper door.
5. Secure the sub-scan unit with screws.

- TP hexagonal Phillips-head screw × 8 (M3 × 6)

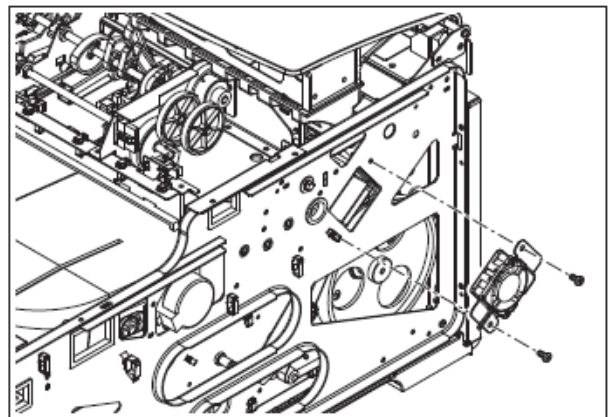


6. Raise the exit guide upright.



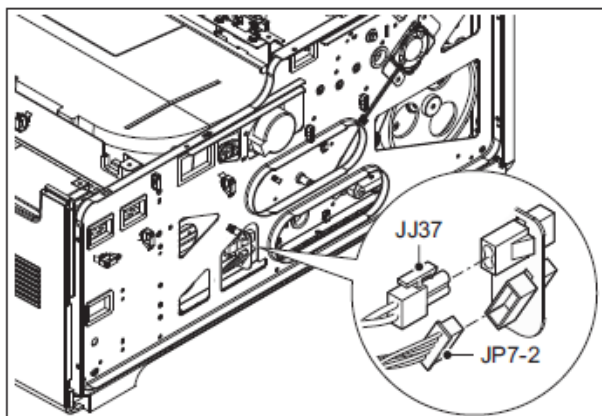
7. Attach the fan assembly (Eraser LED cooling fan).

- TP hexagonal Phillips-head screw × 2 (M3 × 6)



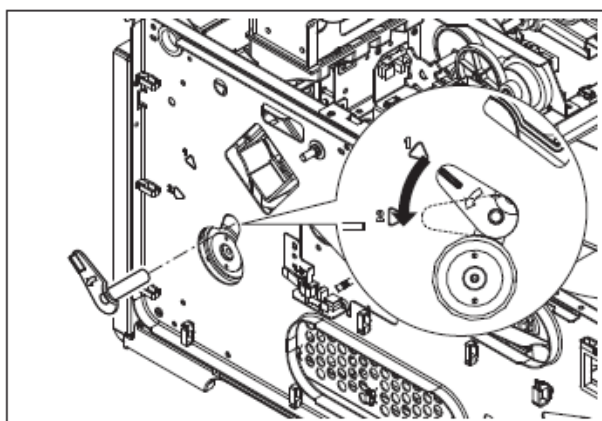
8. Connect cables on the right side of the device.

- JJ37
- JP7-2

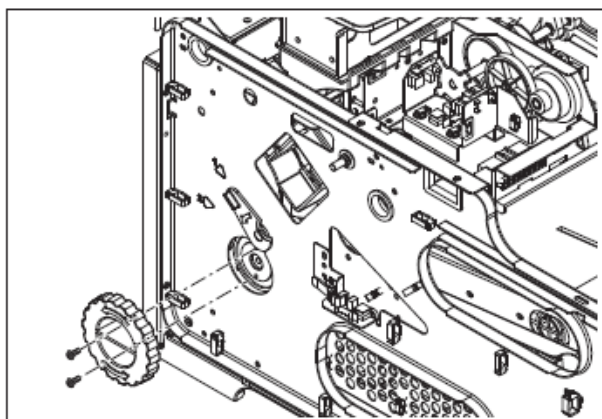
**9. Attach the handle (light collection unit cleaning handle).**

Insert aligned with mark #1. Once abutted, twist it to the mark #2 position and push it in until it stops.

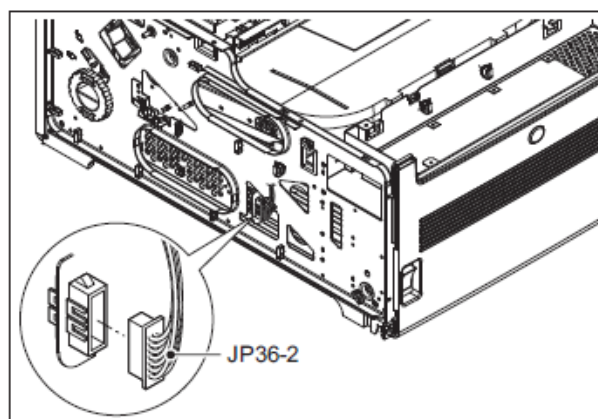
After pushing it in, remove your hand, and the handle will return to the mark #1 position.

**10. Attach the handle (unjamming dial).**

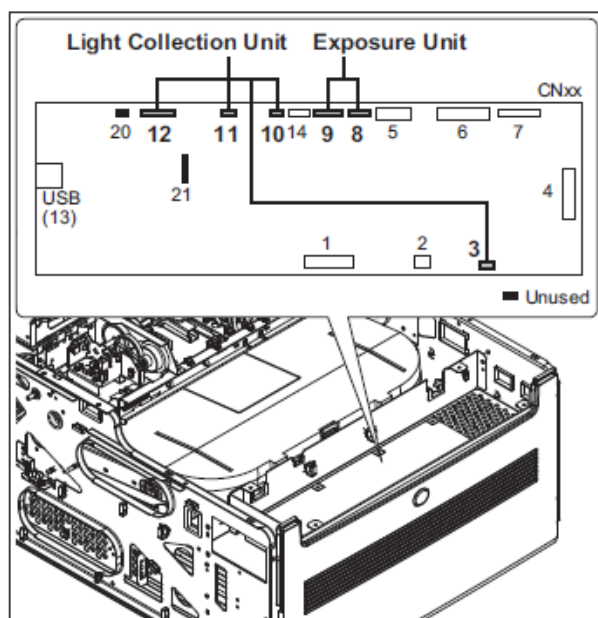
- TP hexagonal Phillips-head screw × 2 (M3 × 6)

**11. Connect cables on the left side of the device.**

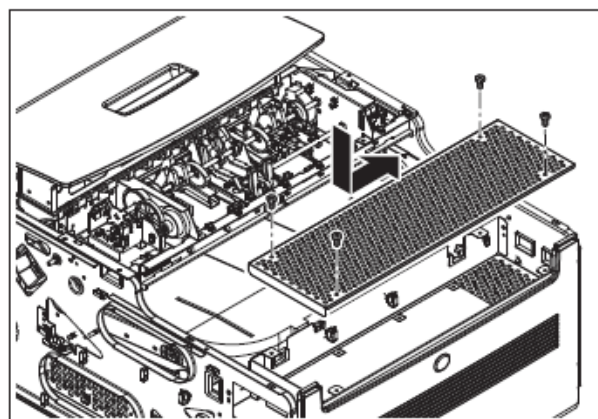
- JP36-2

**12. Connect exposure unit and light collection unit cables to the main board.**

- Exposure unit: CN8, CN9
- Light Collection Unit: CN3, CN10, CN11, CN12
(Together with the main board connector names.)

**13. Attach the frame's anterior cover.**

- TP hexagonal Phillips-head screw × 4 (M3 × 6)



14. Refer to "5.2.8 Removing & attaching the lower insertion unit (Page 5-9)" and attach the lower insertion unit.
15. Refer to "5.2.7 Removing & attaching the rear panel patch assy. (Page 5-9)" and attach the rear panel patch assy.
16. Refer to "5.2.6 Removing & attaching the right side panel (Page 5-8)" and attach the right side panel.
17. Refer to "5.2.5 Removing & attaching the left side panel (Page 5-7)" and attach the left side panel.
18. Refer to "5.2.3 Removing & attaching the upper cover assy (Page 5-5)" and attach the upper cover assy.

This concludes the sub-scan unit attachment process.

Things to check/adjust when re-attaching a sub-scan unit

When re-attaching a sub-scan unit, make sure to check the read image status as a final step. If the image has a problem, check/adjust the device as follows.

- Confirm/adjust vertical and horizontal magnification
- Confirm/adjust read starting point
- Perform sensitivity calibration
- Correct irregularity
Refer to the ImagePilot Installation & Repair Manual for details on the items above.
- Checking/adjusting oblique lighting
For details, refer to "6.1 Oblique adjustments (Page 6-2)".

5.2.11 Removing and attaching the eraser unit

The following is a description of how to remove and attach the eraser unit.

⚠ CAUTION

- Before disassembling/assembling this device, be sure to switch OFF the power supply circuit breaker and disconnect the power cable from the device.
- In order to avoid potential base board damage while discharging it, refer to "5.1.1 Disassembly/Assembly Cautions (Page 5-2)" and wear a grounding strap during the process.
- Do NOT touch the erase filter (the yellow part) of the eraser unit. Do not let dust or debris contaminate the parts while handling them.

Things to prepare

Standard tools are required to perform the following steps.

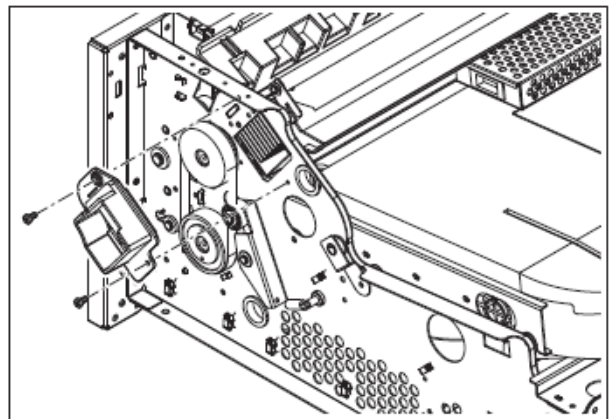
The removal process

1. Refer to "5.2.10 Removing & attaching the Sub-scan unit (Page 5-12)", and remove the sub-scan unit.

In this explanation, the term "sub-scan unit" refers to the exposure unit, light collection unit, and eraser unit together as a single assembly.

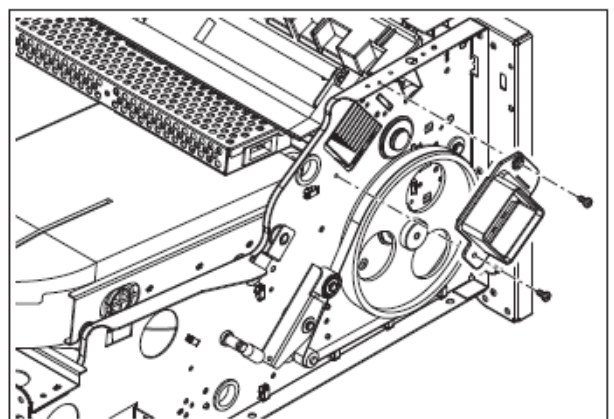
2. Remove the air intake duct from the left side of the sub-scan unit.

- TP hexagonal Phillips-head screw × 2 (M3 × 6)



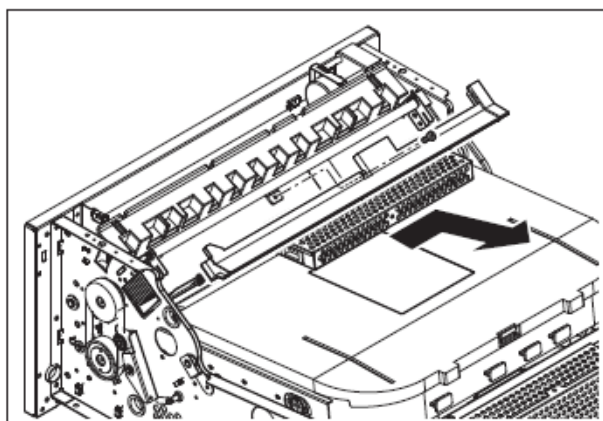
3. Remove the exhaust air duct from the right side of the sub-scan unit.

- TP hexagonal Phillips-head screw × 2 (M3 × 6)

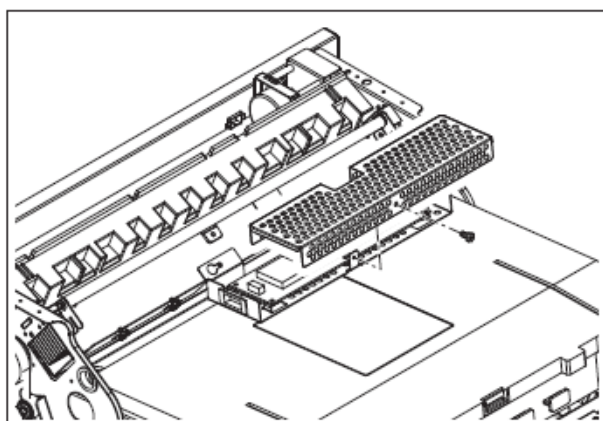


4. Remove the dustproof cover.

- TP hexagonal Phillips-head screw × 1 (M3 × 6)

**5. Remove the base board cover.**

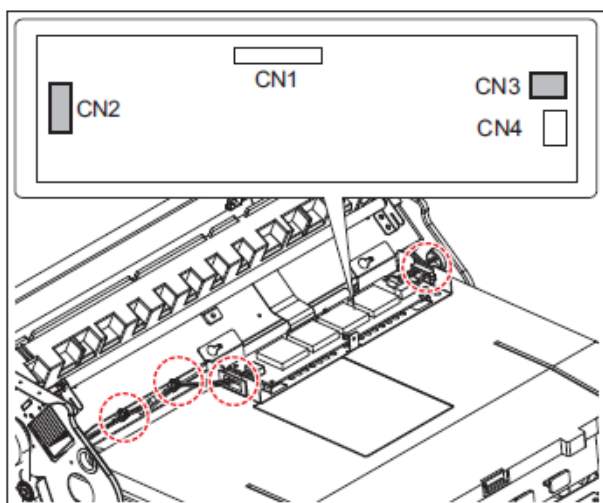
- TP hexagonal Phillips-head screw × 1 (M3 × 6)

**6. Remove 2 cables connected to the LED drive2 board.**

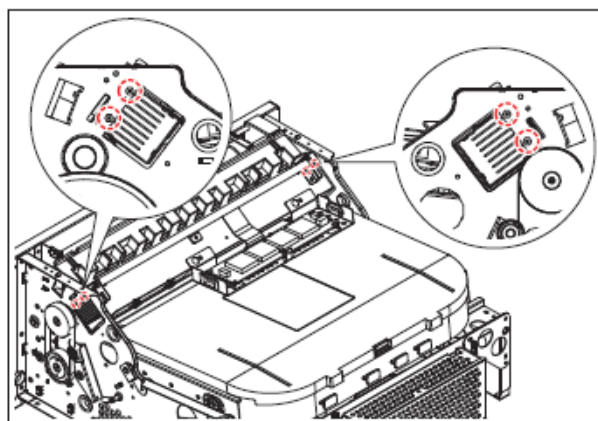
- CN2
- CN3

(Together with the LED drive2 board connector names.)

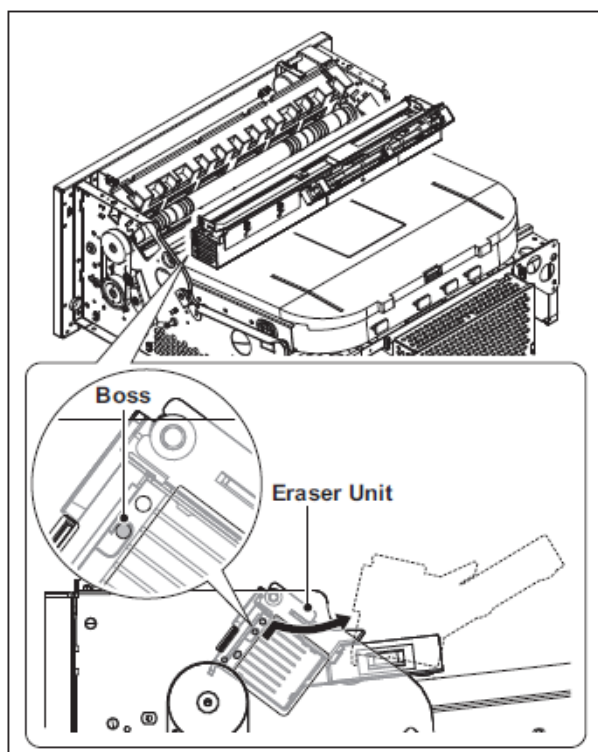
After removing each connector, unhook them from their clamps and wire saddles so that they are no longer secured to the eraser unit.

**7. Remove the screws from the eraser unit.**

- TP hexagonal Phillips-head screw × 4 (M3 × 6)

**8. Remove the eraser unit.**

At this point it will be hitched on the sub-scan unit's side plate boss, so slide it up a bit to remove it.

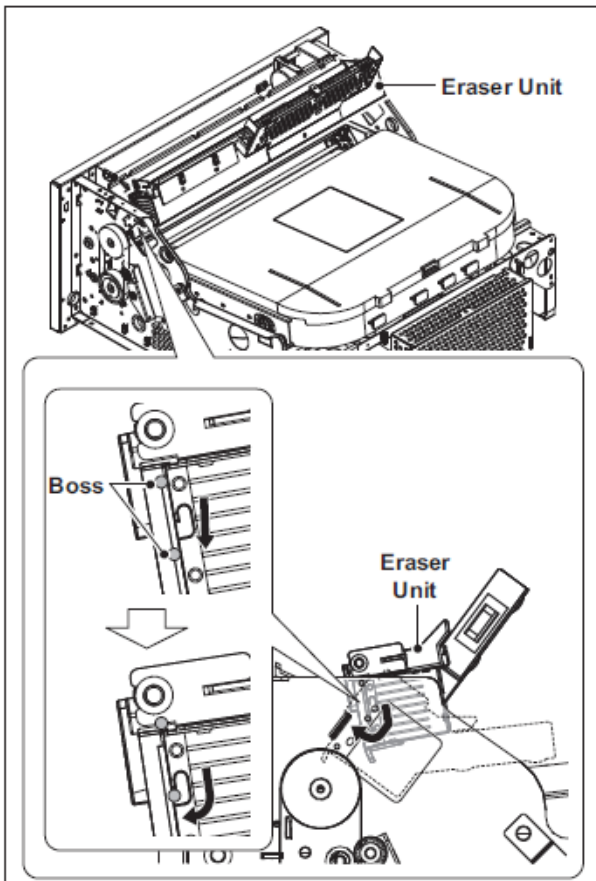


This concludes the eraser unit removal process.

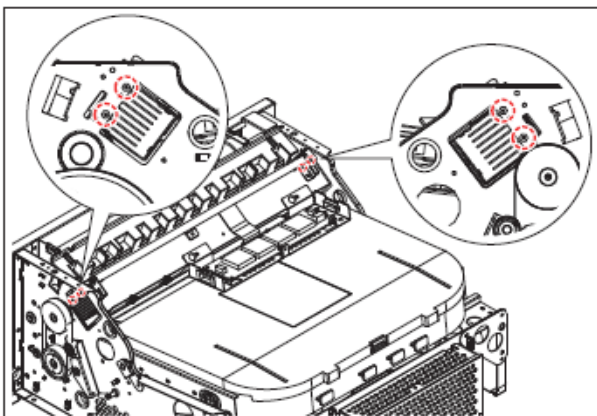
The attachment process

1. Hitch the eraser unit to the sub-scan unit's side plate boss.

There are 3 bosses on one side. Fit the eraser unit to the top 2 bosses and slide it down. Once the eraser unit's indentation enters the second boss from the top, it will slide down by its own weight.

**2. Secure the eraser unit with screws.**

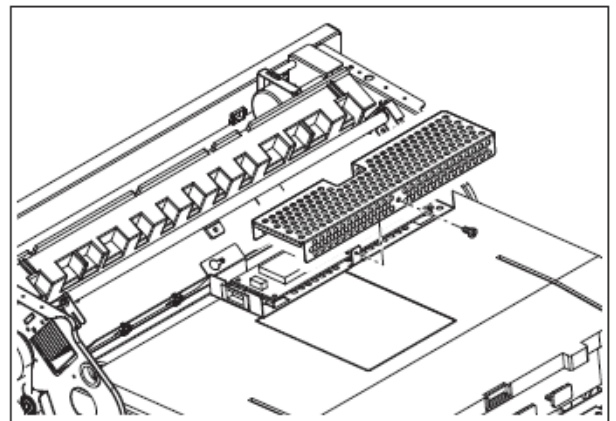
- TP hexagonal Phillips-head screw × 4 (M3 × 6)

**3. Connect 2 cables to the LED drive2 board.**

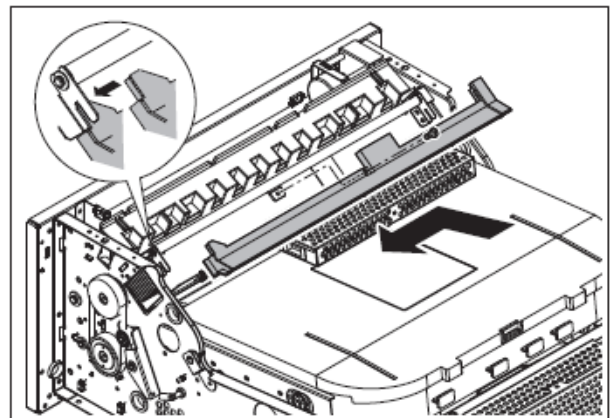
- CN2
 - CN3
- (Together with the LED drive2 board connector names.)

4. Attach the base board cover.

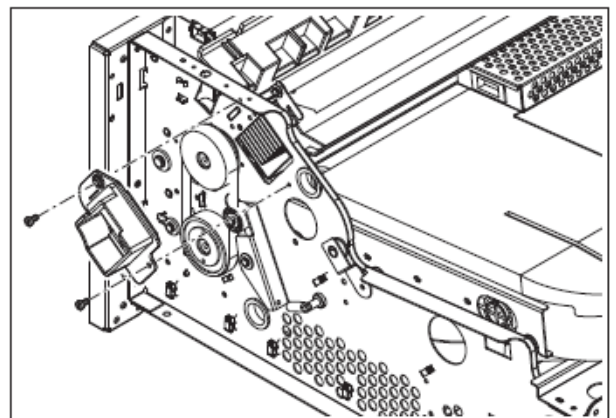
- TP hexagonal Phillips-head screw × 1 (M3 × 6)

**5. Attach the dustproof cover.**

- TP hexagonal Phillips-head screw × 1 (M3 × 6)

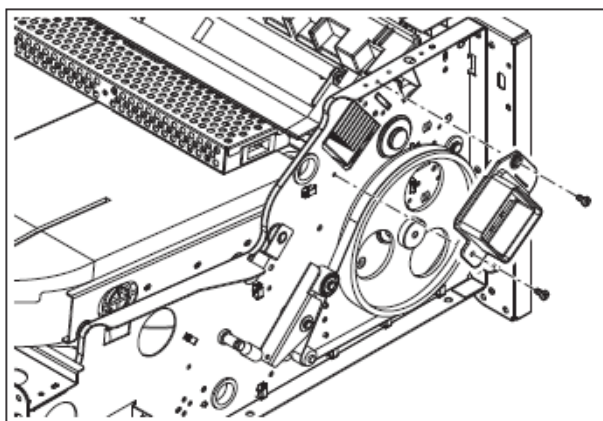
**6. Attach the air intake duct.**

- TP hexagonal Phillips-head screw × 2 (M3 × 6)



7. Attach the exhaust air duct.

- TP hexagonal Phillips-head screw × 2 (M3 × 6)



8. Refer to "5.2.10 Removing & attaching the Sub-scan unit (Page 5-12)" and attach the sub-scan unit.

This concludes the eraser unit attachment process.

Confirmation/settings needed when changing the Eraser Unit or Eraser Unit Assy

When the eraser unit or eraser unit assy have been changed, the following confirmation and settings must be performed.

- Confirm that the Eraser LED lights up
Use the service tool's [Unit test] to access the [Unit Test] (Check) screen, operate the Eraser LED, and confirm visually that it lights up.
- If the LED flickers or does not light up, cable connection may be defective.

For a detailed explanation of the [Unit Test] (Check) function, refer to "4.2 [Unit Test] (Check) Screen Display (Page 4-2)".

- Clear the Erase Counter
Use the service tool's [Processing Counter] function to perform this task.

Refer to the ImagePilot Installation & Service Manual for details.

HINT

When the Eraser LED has lit up more than 42,000 times, a warning message is displayed.

5.2.12 Removing & attaching the exposure unit

The following is a description of how to remove and attach the exposure unit.

CAUTION

- Before disassembling/assembling this device, be sure to switch OFF the power supply circuit breaker and disconnect the power cable from the device.
- Do not touch the exposure unit's dustproof glass (where laser light is emitted). Do not let dust or debris contaminate the parts while handling them.

Things to prepare

Standard tools are required to perform the following steps.

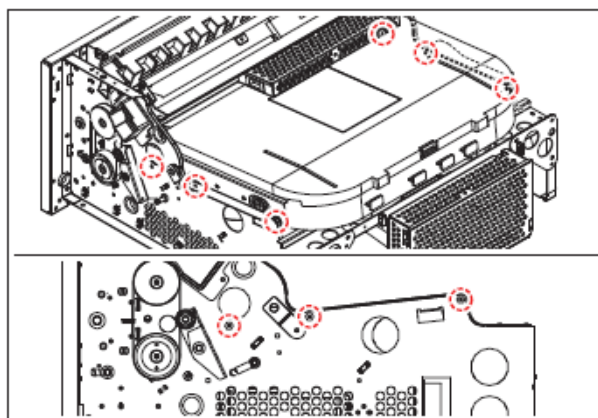
The removal process

1. Refer to "5.2.10 Removing & attaching the Sub-scan unit (Page 5-12)", and remove the sub-scan unit.

In this explanation, the term "sub-scan unit" refers to the exposure unit, light collection unit, and eraser unit together as a single assembly.

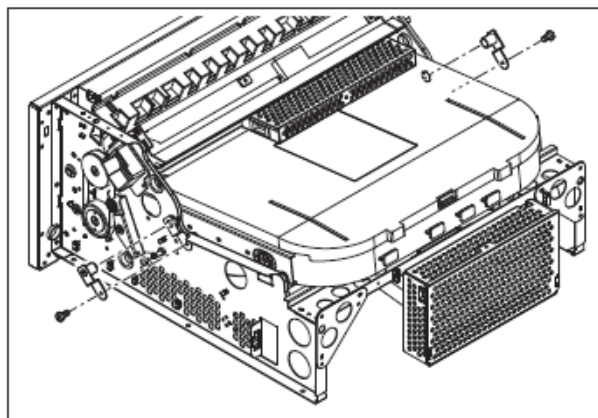
2. Remove the screws from the exposure unit.

- TP hexagonal Phillips-head screw × 6 (M3 × 6)

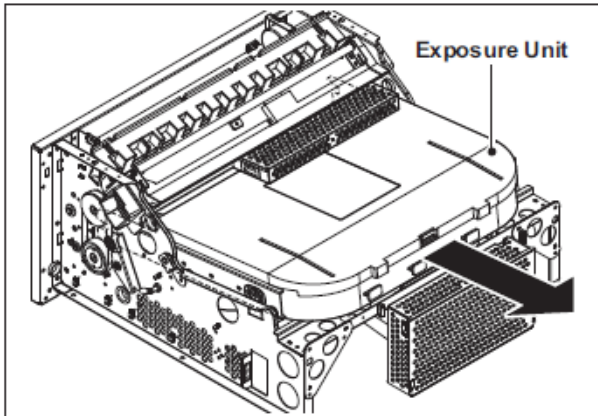


3. Remove the exposure unit's positioning pin Assys (one per side).

- TP hexagonal Phillips-head screw × 1 per side (M3 × 6)



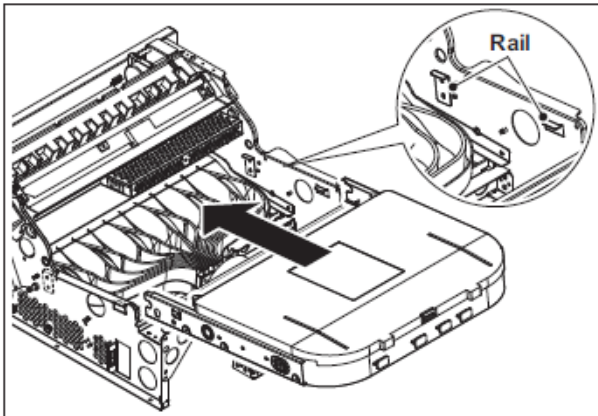
4. Remove the exposure unit.



This concludes the exposure unit removal process.

The attachment process

1. Place the exposure unit on top of the sub-scan unit's rail, then press it inward.



2. Attach the exposure unit's positioning pin Assys (one per side).
 - TP hexagonal Phillips-head screw $\times 1$ per side (M3 $\times$ 6)
3. Secure the exposure unit with screws.
 - TP hexagonal Phillips-head screw $\times 6$ (M3 $\times$ 6)
4. Refer to "5.2.10 Removing & attaching the Sub-scan unit (Page 5-12)" and attach the sub-scan unit.

This concludes the exposure unit attachment process.

Things to check/adjust when re-attaching an exposure unit

When the same exposure unit has been re-attached, adjustments and confirmation must be done as follows.

- Confirm/adjust horizontal magnification
- Confirm/adjust read starting point
- Perform sensitivity calibration
- Correct irregularity

Refer to the ImagePilot Installation & Service Manual for details.

5.2.13 Removing & attaching light collection unit

The following is a description of how to remove and attach the light collection unit.

⚠ CAUTION

- Before disassembling/assembling this device, be sure to switch OFF the power supply circuit breaker and disconnect the power cable from the device.
- Do not touch the light collecting mirror base (at the tip) or the light guide (transparent part). Do not let dust or debris contaminate the parts while handling them.

Things to prepare

Standard tools are required to perform the following steps.

The removal process

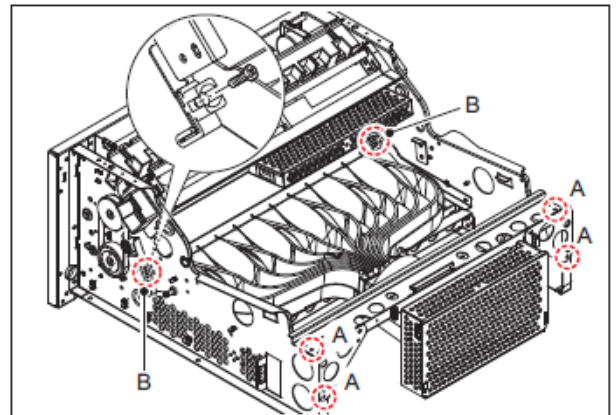
1. Refer to "5.2.10 Removing & attaching the Sub-scan unit (Page 5-12)", and remove the sub-scan unit.

In this explanation, the term "sub-scan unit" refers to the exposure unit, light collection unit, and eraser unit together as a single assembly.

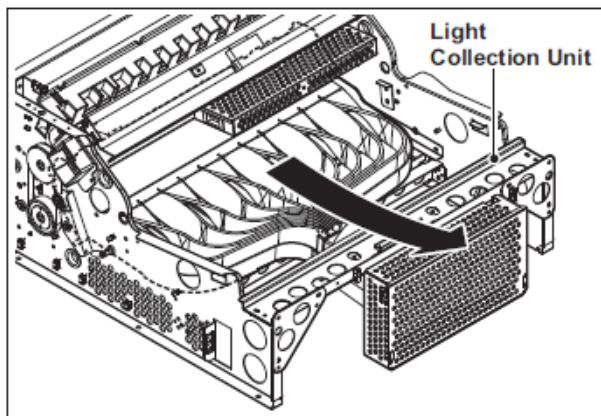
2. Refer to "5.2.12 Removing & attaching the exposure unit (Page 5-20)", and remove the exposure unit from the sub-scan unit.

3. Remove the screws from the light collection unit.

- A: TP hexagonal Phillips-head screw $\times 4$ (M3 $\times$ 6)
- B: TP hexagonal Phillips-head semi screws $\times 2$ (M3 $\times$ 10)



4. Remove the light collection unit.



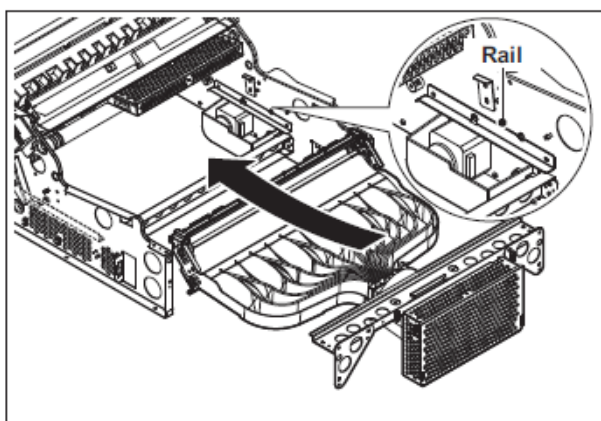
This concludes the light collection unit removal process.

The attachment process

IMPORTANT

When changing the light collection unit, the PMT support plate and light collection unit positions need to be adjusted. Refer to "The attachment process (when changing assemblies) (Page 5-22)".

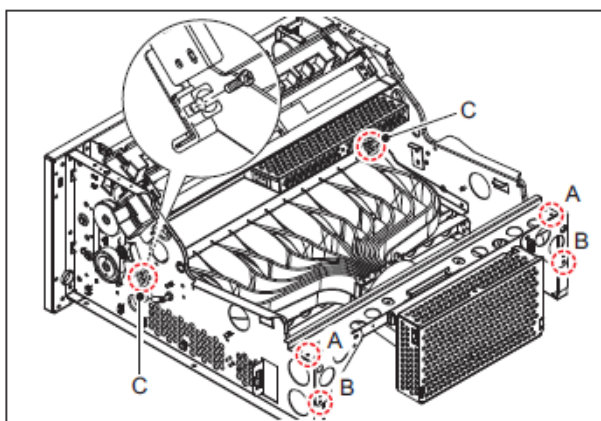
1. Place the light collection unit on top of the sub-scan unit's rail, then press them inward.



2. Secure the light collection unit with screws.

First attach the 2 screws marked "A" in the illustration.

- A, B: TP hexagonal Phillips-head screw × 4 (M3 × 6)
- C: TP hexagonal Phillips-head semi II screws × 2 (M3 × 10)



3. Refer to "5.2.12 Removing & attaching the exposure unit (Page 5-20)" and attach the exposure unit to the sub-scan unit.

4. Refer to "5.2.10 Removing & attaching the Sub-scan unit (Page 5-12)" and attach the sub-scan unit.

This concludes the light collection unit attachment process.

Things to check/adjust when re-attaching the same light collection unit.

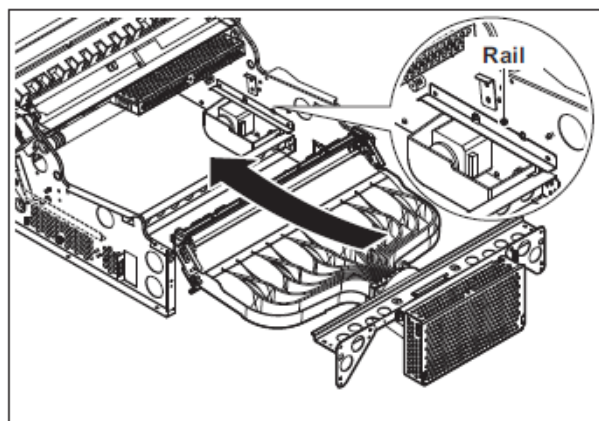
When the same light collection unit has been re-attached, adjustments and confirmation must be done as follows.

- Perform sensitivity calibration
- Correct irregularity

Refer to the ImagePilot Installation & Service Manual for details.

The attachment process (when changing assemblies)

1. Place the light collection unit on top of the sub-scan unit's rail, then press them inward.



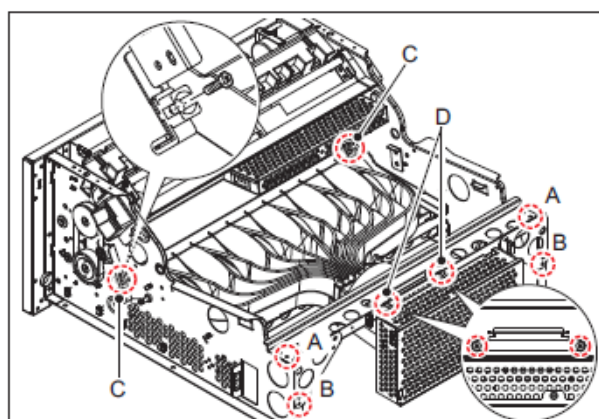
2. Loosen the screws on the PMT support plate.

- D: Screws × 2 (step 3 illustration)

3. Secure the light collection unit with screws.

Attach screws marked A through D in the illustration in the following order.

- 1) A: TP hexagonal Phillips-head screw × 2 (M3 × 6)
- 2) B: TP hexagonal Phillips-head screw × 2 (M3 × 6)
- 3) C: TP hexagonal Phillips-head semi II screws × 2 (M3 × 10)
- 4) D: Screws × 2



4. Refer to "5.2.12 Removing & attaching the exposure unit (Page 5-20)" and attach the exposure unit to the sub-scan unit.
5. Refer to "5.2.10 Removing & attaching the Sub-scan unit (Page 5-12)" and attach the sub-scan unit.

This concludes the light collection unit attachment process when changing them out.

Things to check/adjust when changing light collection unit.

When changing light collection unit, adjustments and confirmation must be done as follows.

- Perform sensitivity calibration
- Correct irregularity

Refer to the ImagePilot Installation & Service Manual for details.

5.2.14 Removing & attaching the upper door Assy

The following is a description of how to remove and attach the upper door Assy.

⚠ CAUTION

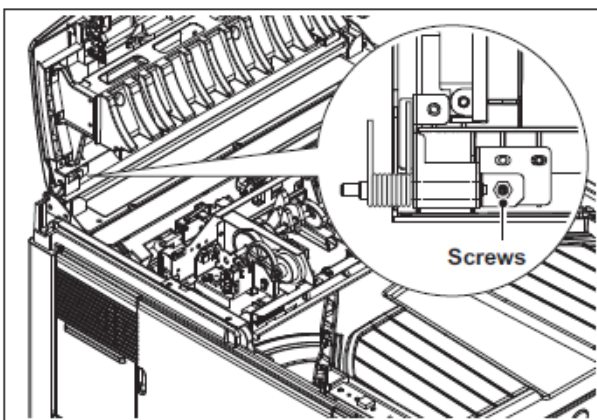
Before disassembling/assembling this device, be sure to switch OFF the power supply circuit breaker and disconnect the power cable from the device.

Things to prepare

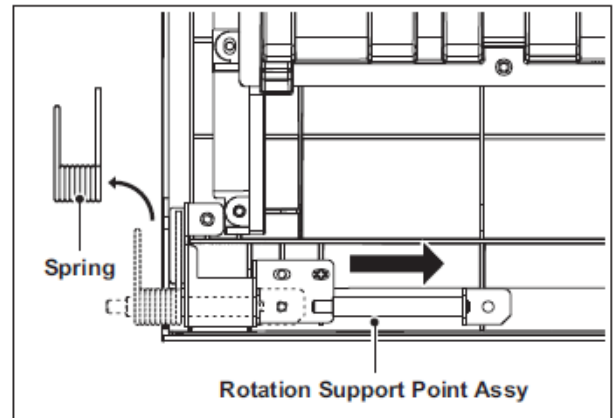
Standard tools are required to perform the following steps.

The removal process

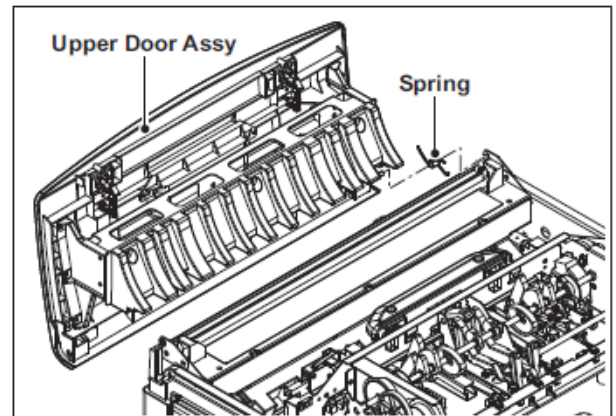
1. Refer to "5.2.3 Removing & attaching the upper cover Assy (Page 5-5)", and remove the upper cover Assy.
2. Remove the screw from the Rotation Support Point Assy on the left side.
 - TP hexagonal Phillips-head screw × 1 (M3 × 6)



3. Slide the Rotation Support Point Assy (left side) to the right and remove it.
Remove the spring at this point as well.



4. Slide the upper door Assy to the left, remove the right shaft from the device, and remove the upper door Assy.
Remove the spring at this point as well.



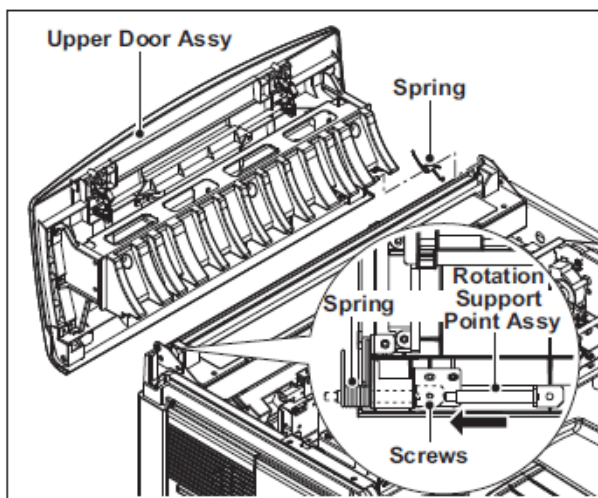
This concludes the upper door Assy removal process.

The attachment process

IMPORTANT

When changing the upper door assy, the locating plate's position needs to be adjusted. Refer to "The attachment process (when changing assemblies) (Page 5-24)".

1. Place the spring into the support pillar (Rotation Support Point Assy) on the right side of the upper door assy, then insert the support pillar into the device.
2. Attach the removed Rotation Support Point Assy to the spring, then fix it to the device.
 - TP hexagonal Phillips-head screw × 1 (M3 × 6)

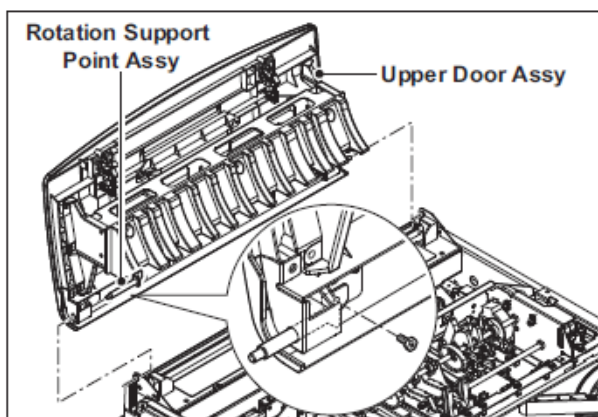


3. Refer to "5.2.3 Removing & attaching the upper cover assy (Page 5-5)" and attach the upper cover assy.

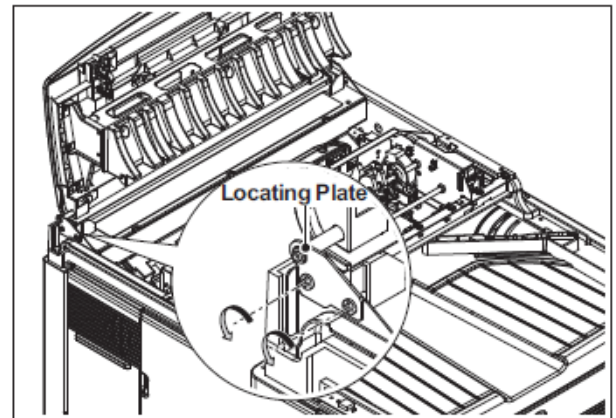
This concludes the upper cover assy attachment process.

The attachment process (when changing assemblies)

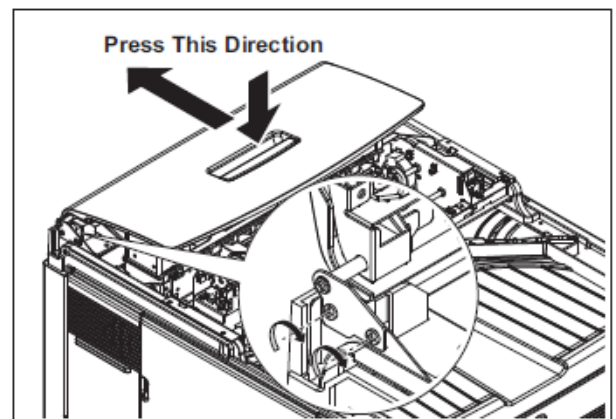
1. Attach the upper door assy to the device without a spring.
 - TP hexagonal Phillips-head screw × 1 (M3 × 6)



2. Loosen the screws on each side of the locating plate.
 - Screws on each side × 2



3. Close the upper door, push in the direction indicated below, then tighten the locating plate's screws.
 - Screws on each side × 2



4. Remove the upper door assy, insert the spring, then attach it to the device.
 - TP hexagonal Phillips-head screw × 1 (M3 × 6)
5. Refer to "5.2.3 Removing & attaching the upper cover assy (Page 5-5)" and attach the upper cover assy.

IMPORTANT

After attaching the upper cover assy, confirm that there is no noticeable difference in level between it and the upper door assy. If the upper door assy is raised, readjust its positioning board.

This concludes the upper door assy attachment process for changing assemblies.

5.3 Changing electrical parts

5.3.1 Changing the status LED board

Follow the steps below to change the status LED board.

⚠ CAUTION

- Before disassembling/assembling this device, be sure to switch OFF the power supply circuit breaker and disconnect the power cable from the device.
- In order to avoid potential base board damage while discharging it, refer to "5.1.1 Disassembly/Assembly Cautions (Page 5-2)" and wear a grounding strap during the process.

Process Summary

This process requires the following number of employees and estimated number of man-hours to perform.

Employees	Man-hours
1	

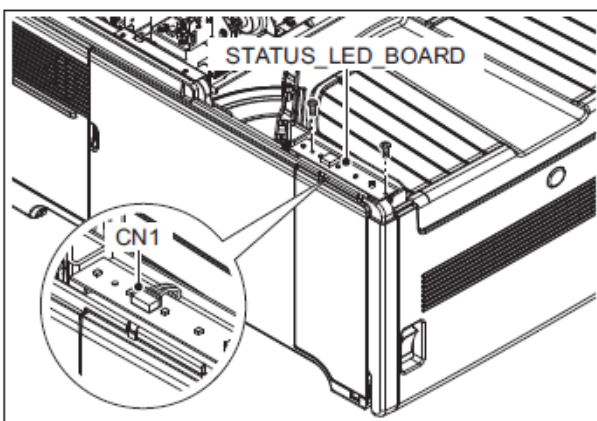
Things to prepare

Standard tools are required to perform the following steps.

The changing process

1. Refer to "5.2.3 Removing & attaching the upper cover assy (Page 5-5)", and remove the upper cover assy.
2. Change the status LED board.
 - Screws × 2
 - CN1
(Status LED board connector name.)

Pass the cables underneath the status LED board to connect them.



3. Refer to "5.2.3 Removing & attaching the upper cover assy (Page 5-5)" and attach the upper cover assy.

This concludes the status LED board changing process.

5.3.2 Changing the LED drive2 board

Follow the steps below to change the LED drive2 board.

⚠ CAUTION

- Before disassembling/assembling this device, be sure to switch OFF the power supply circuit breaker and disconnect the power cable from the device.
- In order to avoid potential base board damage while discharging it, refer to "5.1.1 Disassembly/Assembly Cautions (Page 5-2)" and wear a grounding strap during the process.

Process Summary

This process requires the following number of employees and estimated number of man-hours to perform.

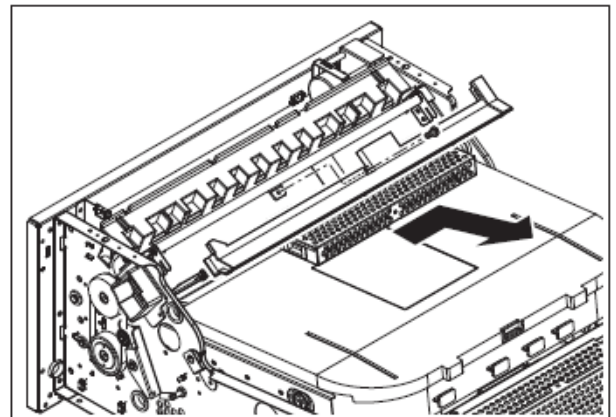
Employees	Man-hours
1	

Things to prepare

Standard tools are required to perform the following steps.

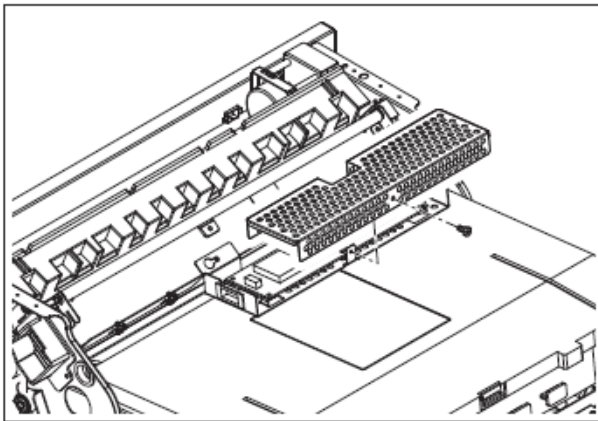
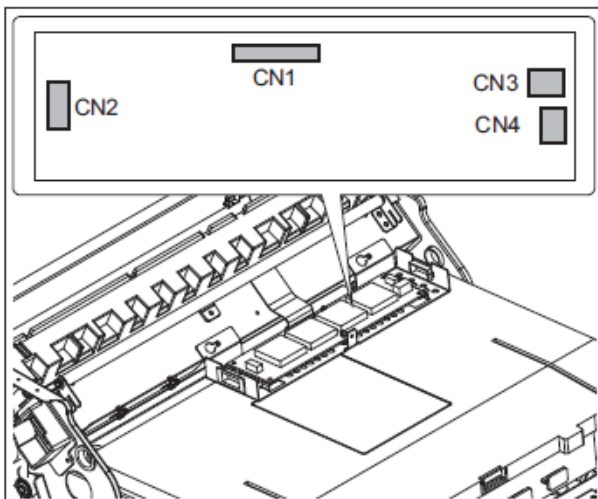
The changing process

1. Refer to "5.2.10 Removing & attaching the Sub-scan unit (Page 5-12)", and remove the sub-scan unit.
2. Remove the dustproof cover.
 - TP hexagonal Phillips-head screw × 1 (M3 × 6)



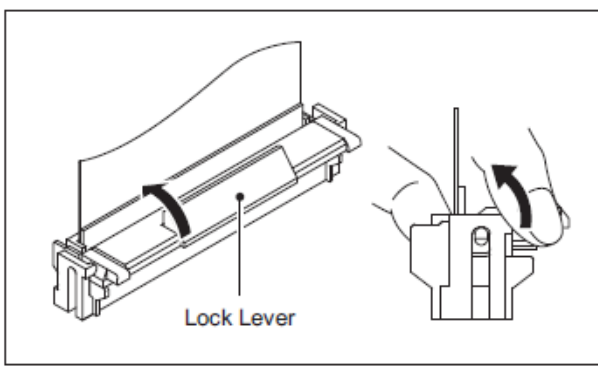
3. Remove the base board cover.

- TP hexagonal Phillips-head screw × 1 (M3 × 6)

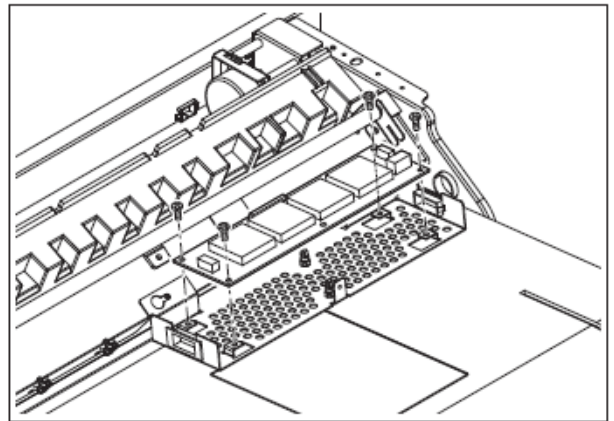
**4. Remove all cables connected to the LED drive2 board.****IMPORTANT**

How to remove the FFC cable (CN1)

Remove the FFC cable after unlocking the lock by pushing up the CN1 lock lever.

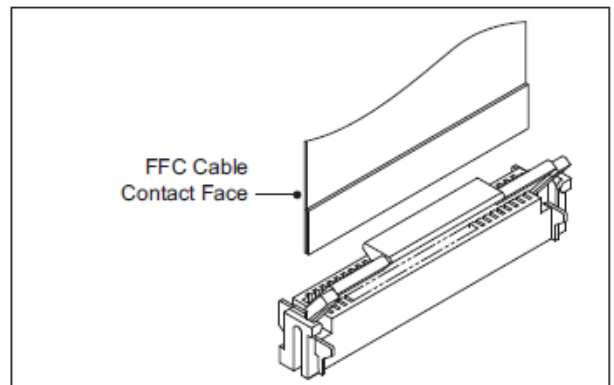
**5. Change the LED drive2 board.**

- Screws × 4

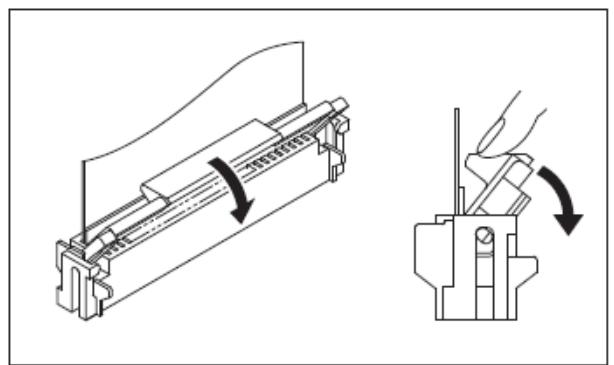
**6. Connect cables to all LED drive2 board connectors.****IMPORTANT**

How to connect the FFC cable (CN1)

Insert the FFC cable placing the contact face toward the opposite side of the lock lever and insert it vertically. When doing so, make sure to firmly insert the FFC cable to the connector vertically until it touches the body.



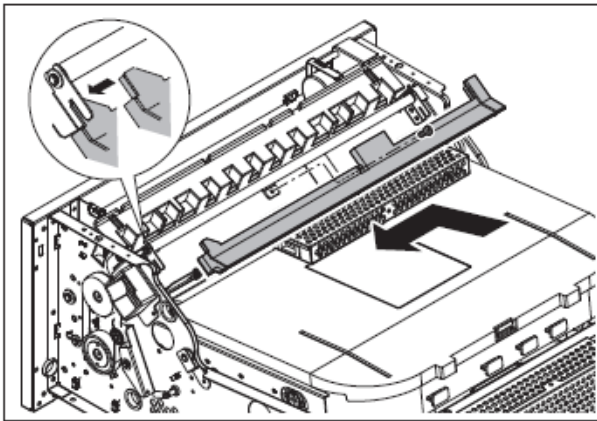
Push up the vertically standing lock lever toward the rotation direction to lock it. After connecting it, confirm that it is firmly locked by gently pulling up the FFC cable.

**7. Attach the base board cover.**

- TP hexagonal Phillips-head screw × 1 (M3 × 6)

8. Attach the dustproof cover.

- TP hexagonal Phillips-head screw × 1 (M3 × 6)

**9. Refer to "5.2.10 Removing & attaching the Sub-scan unit (Page 5-12)" and attach the sub-scan unit.**

This concludes the LED drive2 board changing process.

5.3.3 Changing the Eraser Unit Assy (Eraser LED)

Follow the steps below to change the Eraser Unit Assy.

⚠ CAUTION

- Before disassembling/assembling this device, be sure to switch OFF the power supply circuit breaker and disconnect the power cable from the device.
- In order to avoid potential base board damage while discharging it, refer to "5.1.1 Disassembly/Assembly Cautions (Page 5-2)" and wear a grounding strap during the process.

Process Summary

This process requires the following number of employees and estimated number of man-hours to perform.

Employees	Man-hours
1	

Things to prepare

Standard tools are required to perform the following steps.

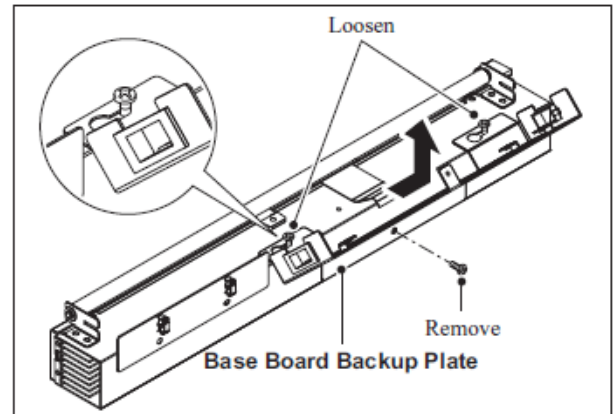
The changing process

1. Refer to "5.2.10 Removing & attaching the Sub-scan unit (Page 5-12)", and remove the sub-scan unit.
2. Refer to "5.3.2 Changing the LED drive2 board (Page 5-25)", and remove the LED drive2 board.
3. Refer to "5.2.11 Removing and attaching the eraser unit (Page 5-17)", and remove the eraser unit from the sub-scan unit.

4. Remove the substrate backing plate.

- Screw × 3

Loosen the 2 screws on the top, and remove 1 screw on the bottom.

**5. Attach the substrate backing plate to the new Eraser Unit Assy.**

- Screw × 3

6. Refer to "5.2.11 Removing and attaching the eraser unit (Page 5-17)", and attach the eraser unit to the sub-scan unit.

In addition, refer to the attachment process covered in "5.2.11 Removing and attaching the eraser unit", and when "connecting cables to the LED drive2 board", attach the previously removed LED drive2 board, then connect the cables to all LED drive2 board connectors.

7. Refer to "5.2.10 Removing & attaching the Sub-scan unit (Page 5-12)" and attach the sub-scan unit.

This concludes the eraser unit assy changing process.

Confirmation/settings needed after changing the Eraser Unit Assy

- Confirm that the Eraser LED lights up
Use the service tool's [Unit test] to access the [Unit Test] (Check) screen, operate the Eraser LED, and confirm visually that it lights up.

- If the LED flickers or does not light up,
Cable connection may be defective.

For a detailed explanation of the [Unit Test] (Check) function, refer to "4.2 [Unit Test] (Check) Screen Display (Page 4-2)".

- Clear the Erase Counter
Use the service tool's [Processing Counter] function to perform this task.

Refer to the ImagePilot Installation & Repair Manual for details.

HINT

When the Eraser LED has lit up more than 42,000 times, a warning message is displayed.

5.3.4 Changing the main board

All of the device's setting data is saved to the main board. So when changing the main board, it's important to back up any data saved to the current main board before progressing. After changing it, the backup data must then be restored to the new main board.

A reader program is also written to the main board. A program written to the main board may become old when the main board is changed, and require overwrite to a new program.

Use the service tool of the ImagePilot connected to this device to backup or restore data or overwrite a program.

The steps needed to change the main board are summarized as follows.

- For details on service tool operation, refer to the ImagePilot Installation & Service Manual.

⚠ CAUTION

- Before disassembling/assembling this device, be sure to switch OFF the power supply circuit breaker and disconnect the power cable from the device.
- In order to avoid potential base board damage while discharging it, refer to "5.1.1 Disassembly/Assembly Cautions (Page 5-2)" and wear a grounding strap during the process.

📌 IMPORTANT

The main board must be changed when a problem is thought to have occurred from within the main board. In most cases, when a problem occurs with the main board, connection is cut off between the device and ImagePilot, and backup data can no longer be created.

The backup option is included while adjusting settings, so be sure to create backup data each time settings are changed while the device is being serviced, etc.

Process Summary

This process requires the following number of employees and estimated number of man-hours to perform.

Employees	Man-hours
1	

Things to prepare

Standard tools are required to perform the following steps.

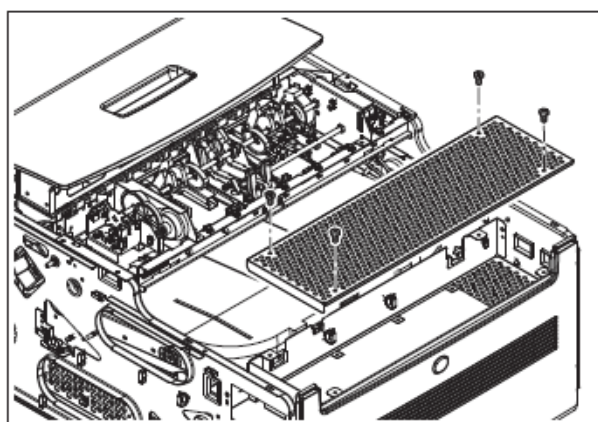
The changing process

- If communications are possible between this device and the ImagePilot, access the service tool's [Change Base Board] screen to back up all data saved to the main board into the ImagePilot.

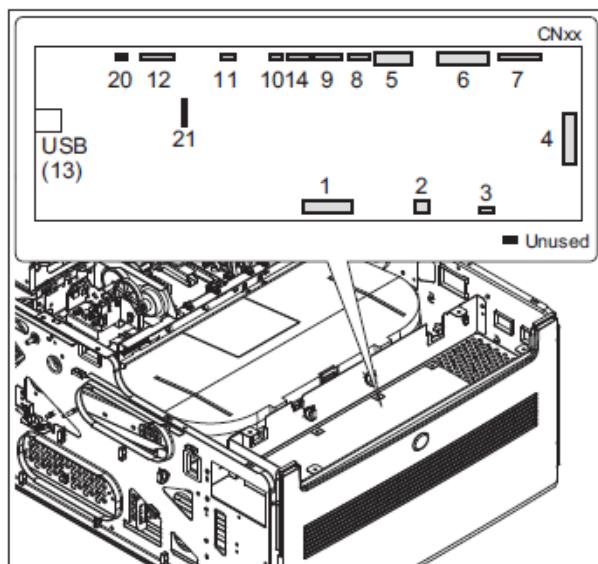
💡 HINT

When using the REGIUS MODEL 110 etc. to back up ImagePilot setting information (system and JM information), ImagePilot settings had to be returned to default. However, since USB connection is used for the REGIUS Σ/ΣII, this is no longer necessary.

- Exit the ImagePilot and device, then turn OFF their power sources.
- Refer to "5.2.3 Removing & attaching the upper cover assy (Page 5-5)", and remove the upper cover assy.
- Refer to "5.2.5 Removing & attaching the left side panel (Page 5-7)" and remove the left side panel.
- Refer to "5.2.6 Removing & attaching the right side panel (Page 5-8)" and remove the right side panel.
- Refer to "5.2.8 Removing & attaching the lower insertion unit (Page 5-9)", and remove the lower insertion unit.
- Remove the frame's anterior cover.
 - TP hexagonal Phillips-head screw × 4 (M3 × 6)

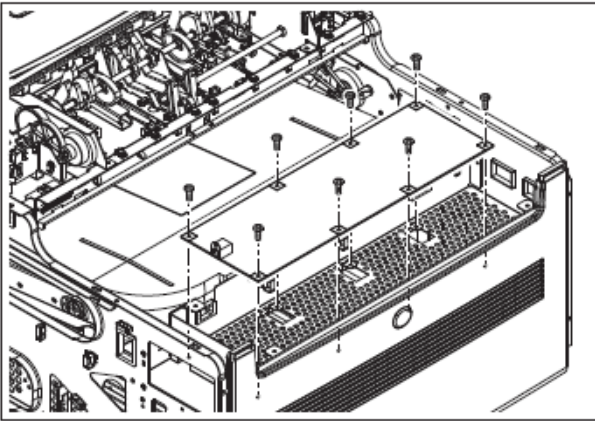


- Remove all cables connected to the main board.

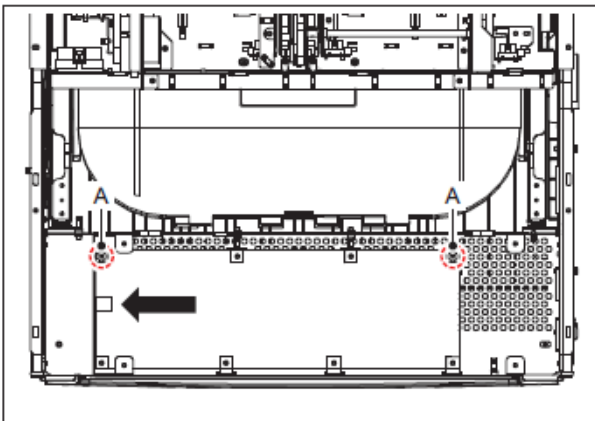


9. Change the main board.

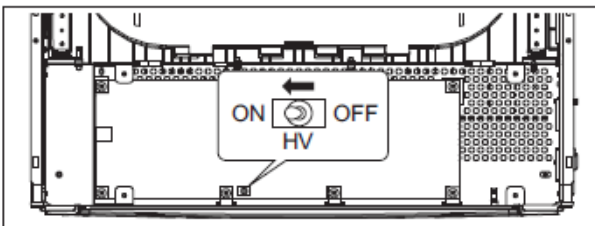
- Screw × 8



When attaching the main board, loosely fasten the 2 screws marked "A" in the illustration, then tighten them after sliding it to the left. After that, tighten it down with the remaining 6 screws.



10. Confirm that the HV (high voltage) switch of the MAIN_BOARD is set to ON.

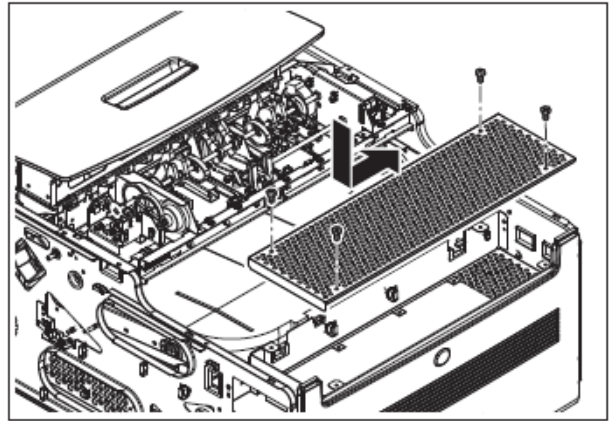


Turn it on ON if it is OFF.

11. Connect the cables removed in [step 8](#) to the main board.

12. Attach the frame's anterior cover.

- TP hexagonal Phillips-head screw × 4 (M3 × 6)



13. Refer to ["5.2.8 Removing & attaching the lower insertion unit \(Page 5-9\)"](#) and attach the lower insertion unit.

14. Refer to ["5.2.6 Removing & attaching the right side panel \(Page 5-8\)"](#) and attach the right side panel.

15. Refer to ["5.2.5 Removing & attaching the left side panel \(Page 5-7\)"](#) and attach the left side panel.

16. Refer to ["5.2.3 Removing & attaching the upper cover assy \(Page 5-5\)"](#) and attach the upper cover assy.

17. Turn ON the ImagePilot and device to start them up.

18. Overwrite the MAIN_BOARD program with the latest version on the [Install] screen and the [Program] screen of the service tool.

Refer to the ImagePilot Installation & Service Manual for details.

19. Restore the backup data, which is created in [step 1](#) at the [PCB] screen (or created when setting or servicing the device, etc.) of the service tool, to the new main board.

If no data is available to restore

If you were unable to back up the previous main board's data, and there is no backup data created when setting or servicing, the following recheck/readjustment process must be performed.

- Confirm the number of read pixels and read starting point
- Adjust the image size and image position
- Calibrate irregularity
- Perform sensitivity calibration

Refer to the ImagePilot Installation & Service Manual for details.

IMPORTANT

After making these settings, make sure to back up the main board data.

20. Restart the device.

This concludes the main board changing process.

5.3.5 Changing the SW power supply

Follow the steps below to change the SW power supply.

CAUTION

Before disassembling/assembling this device, be sure to switch OFF the power supply circuit breaker and disconnect the power cable from the device.

Process Summary

This process requires the following number of employees and estimated number of man-hours to perform.

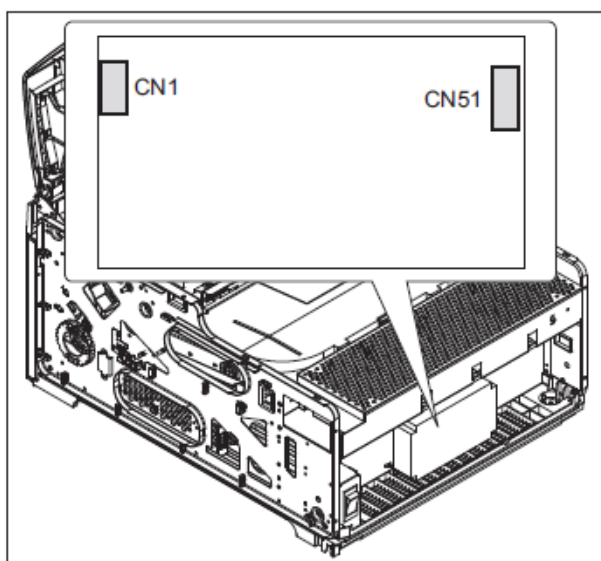
Employees	Man-hours
1	

Things to prepare

Standard tools are required to perform the following steps.

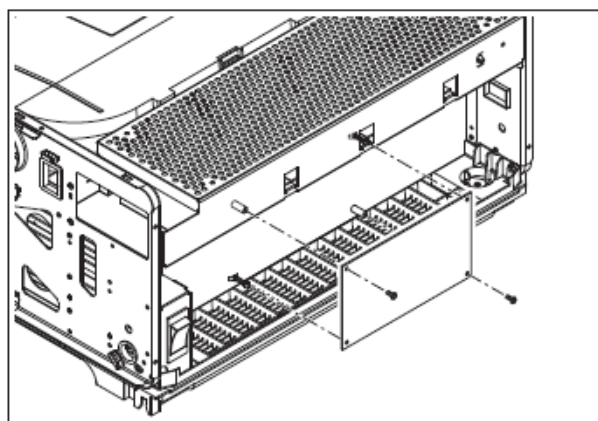
The changing process

1. Refer to "5.2.3 Removing & attaching the upper cover assy (Page 5-5)", and remove the upper cover assy.
2. Refer to "5.2.5 Removing & attaching the left side panel (Page 5-7)" and remove the left side panel.
3. Refer to "5.2.6 Removing & attaching the right side panel (Page 5-8)" and remove the right side panel.
4. Refer to "5.2.8 Removing & attaching the lower insertion unit (Page 5-9)", and remove the lower insertion unit.
5. Refer to "5.2.9 Removing/attaching the front panel assy (Page 5-11)" and remove the front panel assy.
6. Remove all cables connected to the SW power supply.



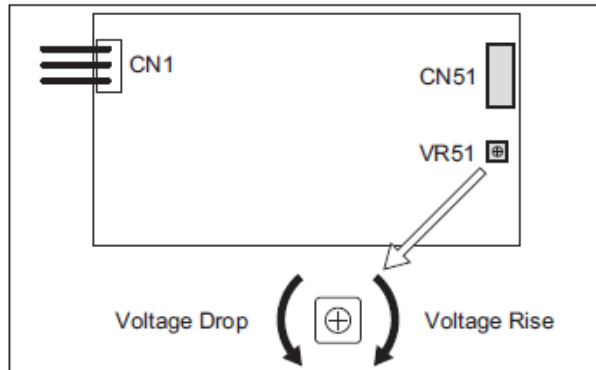
7. Change the SW power supply.

- Screws × 2



8. Connect the power supply circuit breaker cable to the SW power supply.
 - CN1
9. Temporarily connect the power cable to the device and plug it into an outlet.
10. Turn the device ON, confirm the SW power supply's output voltage, and adjust if out of range.

Signal name	Point of measurement		Voltage range
+24 V	CN51-6 (+ side)	CN51-1 (- side)	23.50 - 24.50V



11. Unplug the power cable from its outlet and disconnect it from the device.
12. Connect the cables to the SW power supply.
 - CN51
13. Refer to "5.2.9 Removing/attaching the front panel assy (Page 5-11)" and attach the front panel assy.
14. Refer to "5.2.8 Removing & attaching the lower insertion unit (Page 5-9)" and attach the lower insertion unit.
15. Refer to "5.2.6 Removing & attaching the right side panel (Page 5-8)" and attach the right side panel.
16. Refer to "5.2.5 Removing & attaching the left side panel (Page 5-7)" and attach the left side panel.

17. Refer to "5.2.3 Removing & attaching the upper cover assy (Page 5-5)" and attach the upper cover assy.

This concludes the SW power supply changing process.

5.3.6 Changing the power supply circuit breaker

Follow the steps below to change the power supply circuit breaker.

⚠ CAUTION

Before disassembling/assembling this device, be sure to switch OFF the power supply circuit breaker and disconnect the power cable from the device.

Process Summary

This process requires the following number of employees and estimated number of man-hours to perform.

Employees	Man-hours
1	

Things to prepare

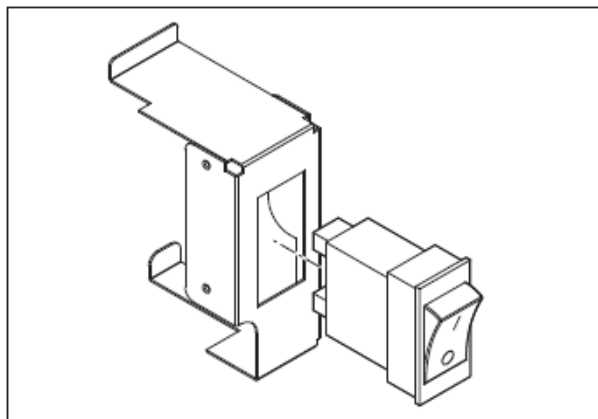
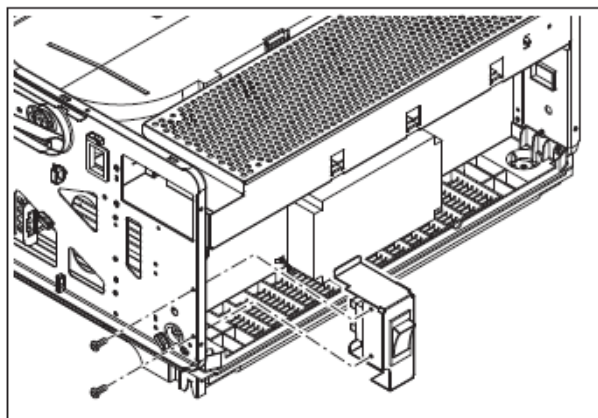
Standard tools are required to perform the following steps.

The changing process

- Refer to "5.2.3 Removing & attaching the upper cover assy (Page 5-5)", and remove the upper cover assy.
- Refer to "5.2.5 Removing & attaching the left side panel (Page 5-7)" and remove the left side panel.
- Refer to "5.2.6 Removing & attaching the right side panel (Page 5-8)" and remove the right side panel.
- Refer to "5.2.8 Removing & attaching the lower insertion unit (Page 5-9)", and remove the lower insertion unit.
- Refer to "5.2.9 Removing/attaching the front panel assy (Page 5-11)" and remove the front panel assy.
- Remove the brackets attached to the power supply circuit breaker.
 - Screws x 2
- Change the power supply circuit breaker.

Check the switch's ON/OFF symbols to make sure it is pointing in the direction shown in the illustration when attaching it.
- Attach the brackets.
 - Screws x 2
- Refer to "5.2.9 Removing/attaching the front panel assy (Page 5-11)" and attach the front panel assy.
- Refer to "5.2.8 Removing & attaching the lower insertion unit (Page 5-9)" and attach the lower insertion unit.
- Refer to "5.2.6 Removing & attaching the right side panel (Page 5-8)" and attach the right side panel.
- Refer to "5.2.5 Removing & attaching the left side panel (Page 5-7)" and attach the left side panel.
- Refer to "5.2.3 Removing & attaching the upper cover assy (Page 5-5)" and attach the upper cover assy.

This concludes the power supply circuit breaker changing process.



5.3.7 Changing the Inlet-NF wiring Assy

Follow the steps below to change the Inlet-NF wiring Assy.

⚠ CAUTION

Before disassembling/assembling this device, be sure to switch OFF the power supply circuit breaker and disconnect the power cable from the device.

Process Summary

This process requires the following number of employees and estimated number of man-hours to perform.

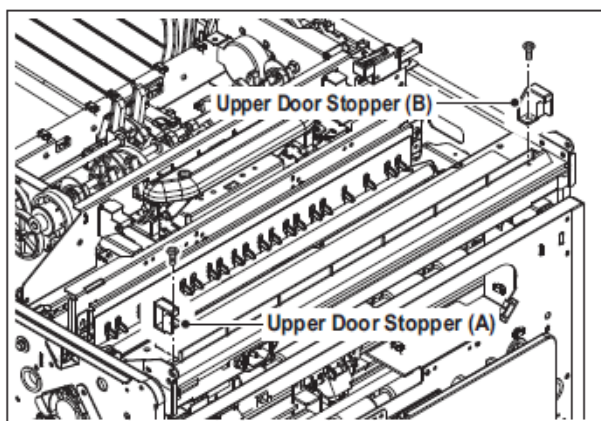
Employees	Man-hours
1	

Things to prepare

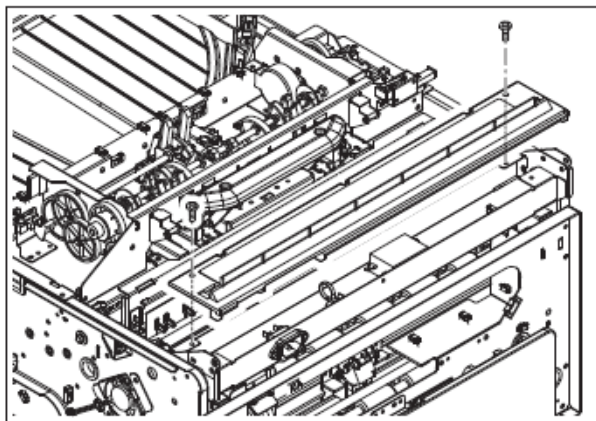
Standard tools are required to perform the following steps.

The changing process

1. Refer to "5.2.3 Removing & attaching the upper cover assy (Page 5-5)", and remove the upper cover assy.
2. Refer to "5.2.14 Removing & attaching the upper door assy (Page 5-23)", and remove the upper door assy.
3. Refer to "5.2.5 Removing & attaching the left side panel (Page 5-7)" and remove the left side panel.
4. Refer to "5.2.6 Removing & attaching the right side panel (Page 5-8)" and remove the right side panel.
5. Refer to "5.2.7 Removing & attaching the rear panel patch assy. (Page 5-9)" and remove the rear panel patch assy.
6. Remove upper door stoppers A and B.
 - Screw × 1 each

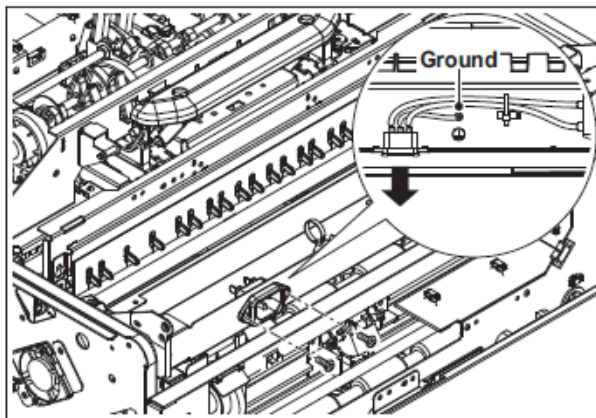


7. Remove the box cover patch assembly.
 - Screws × 2



8. Change the Inlet-NF wiring Assy.
 - Screws × 2
 - Screw × 1 (ground)

Attach so the inlet's central terminal is facing downward.



9. Attach the box cover patch assembly.
 - Screws × 2
10. Attach upper door stoppers A and B.
 - Screw × 1 each
11. Refer to "5.2.7 Removing & attaching the rear panel patch assy. (Page 5-9)" and attach the rear panel patch assy.
12. Refer to "5.2.6 Removing & attaching the right side panel (Page 5-8)" and attach the right side panel.
13. Refer to "5.2.5 Removing & attaching the left side panel (Page 5-7)" and attach the left side panel.
14. Refer to "5.2.14 Removing & attaching the upper door assy (Page 5-23)" and attach the upper door assy.
15. Refer to "5.2.3 Removing & attaching the upper cover assy (Page 5-5)" and attach the upper cover assy.

This concludes the Inlet-NF wiring Assy changing process.

5.3.8 Changing size detection sensors 1 and 2

Follow the steps below to change size detection sensor 1 and size detection sensor 2.

⚠ CAUTION

- Before disassembling/assembling this device, be sure to switch OFF the power supply circuit breaker and disconnect the power cable from the device.
- In order to avoid potential base board damage while discharging it, refer to "5.1.1 Disassembly/Assembly Cautions (Page 5-2)" and wear a grounding strap during the process.

Process Summary

This process requires the following number of employees and estimated number of man-hours to perform.

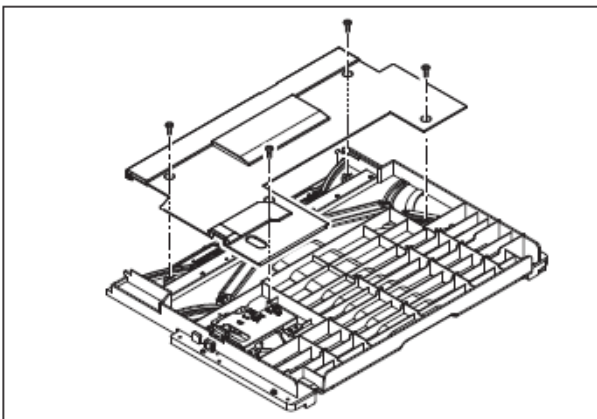
Employees	Man-hours
1	

Things to prepare

Standard tools are required to perform the following steps.

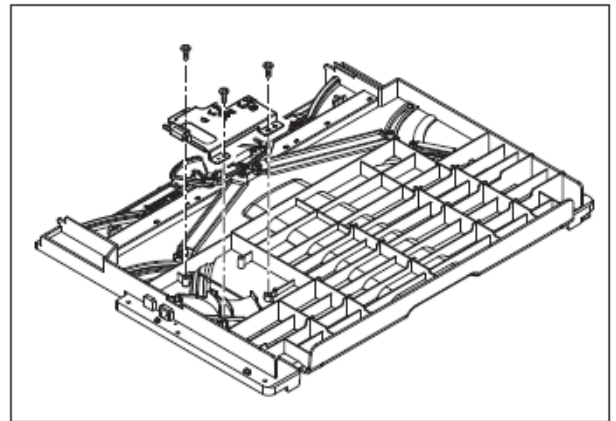
The changing process

1. Refer to "5.2.3 Removing & attaching the upper cover assy (Page 5-5)", and remove the upper cover assy.
2. Refer to "5.2.5 Removing & attaching the left side panel (Page 5-7)" and remove the left side panel.
3. Refer to "5.2.6 Removing & attaching the right side panel (Page 5-8)" and remove the right side panel.
4. Refer to "5.2.8 Removing & attaching the lower insertion unit (Page 5-9)", and remove the lower insertion unit.
5. Turn over the lower insertion unit and remove its cover.
 - P Tight × 4 (M3 × 8)

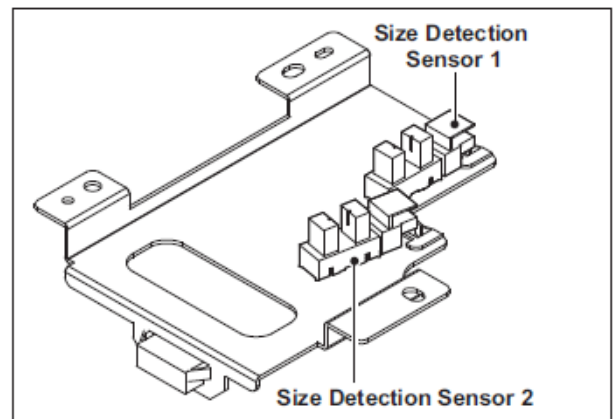


6. Remove the sensor assembly.

- Screws × 3



7. Change the sensors.



8. Attach the sensor assembly.

- Screws × 3

9. Attach the lower insertion unit cover.

- P Tight × 4 (M3 × 8)

10. Refer to "5.2.8 Removing & attaching the lower insertion unit (Page 5-9)" and attach the lower insertion unit.

11. Refer to "5.2.6 Removing & attaching the right side panel (Page 5-8)" and attach the right side panel.

12. Refer to "5.2.5 Removing & attaching the left side panel (Page 5-7)" and attach the left side panel.

13. Refer to "5.2.3 Removing & attaching the upper cover assy (Page 5-5)" and attach the upper cover assy.

This concludes the changing process for size detection sensors 1 and 2.

5.3.9 Changing the cassette sensor

Follow the steps below to change the cassette sensor.

⚠ CAUTION

- Before disassembling/assembling this device, be sure to switch OFF the power supply circuit breaker and disconnect the power cable from the device.
- In order to avoid potential base board damage while discharging it, refer to "5.1.1 Disassembly/Assembly Cautions (Page 5-2)" and wear a grounding strap during the process.

Process Summary

This process requires the following number of employees and estimated number of man-hours to perform.

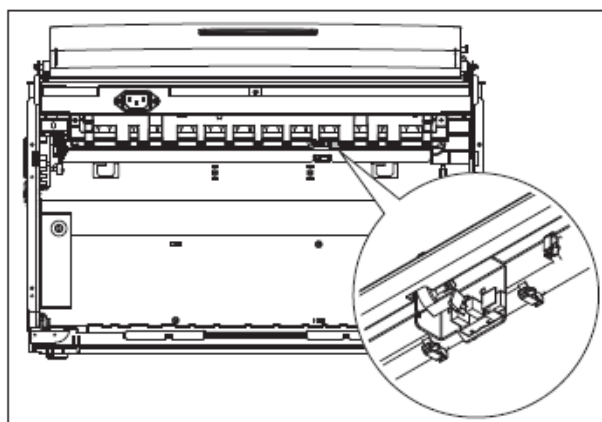
Employees	Man-hours
1	

Things to prepare

Standard tools are required to perform the following steps.

The changing process

1. Refer to "5.2.10 Removing & attaching the Sub-scan unit (Page 5-12)", and remove the sub-scan unit.
2. Change the cassette sensor.



3. Refer to "5.2.10 Removing & attaching the Sub-scan unit (Page 5-12)" and attach the sub-scan unit.

This concludes the cassette sensor changing process.

5.3.10 Changing the plate sensor (light receiving)

Follow the steps below to change the plate sensor (light receiving).

⚠ CAUTION

- Before disassembling/assembling this device, be sure to switch OFF the power supply circuit breaker and disconnect the power cable from the device.
- In order to avoid potential base board damage while discharging it, refer to "5.1.1 Disassembly/Assembly Cautions (Page 5-2)" and wear a grounding strap during the process.

Process Summary

This process requires the following number of employees and estimated number of man-hours to perform.

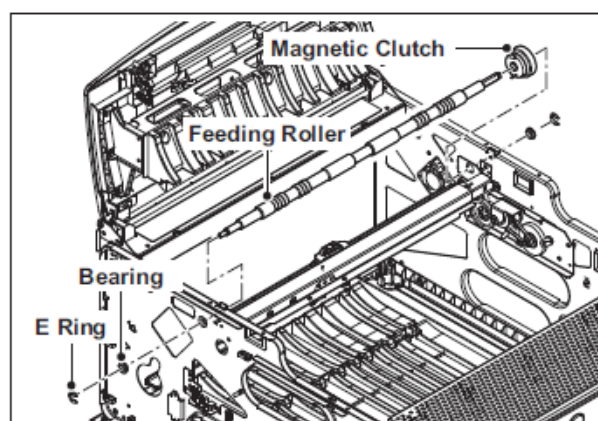
Employees	Man-hours
1	

Things to prepare

Standard tools are required to perform the following steps.

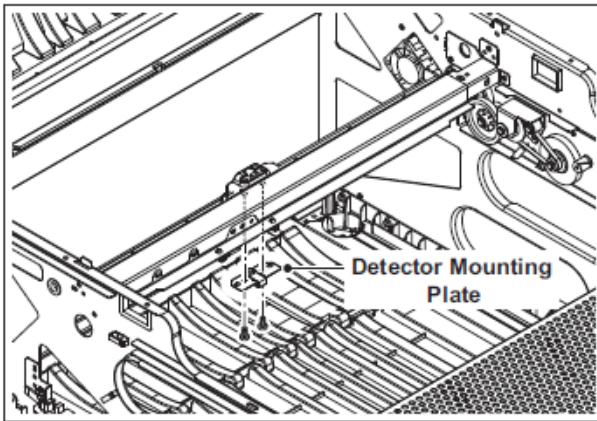
The changing process

1. Refer to "5.2.10 Removing & attaching the Sub-scan unit (Page 5-12)" and "5.2.4 Removing & attaching the upper insertion unit (Page 5-6)" to remove the sub-scan unit and upper insertion unit.
At this point, only the front panel assy and upper door assy are attached to the frame.
2. Refer to "5.4.5 Changing the magnetic clutch (Page 5-45)", and remove the feeding roller.

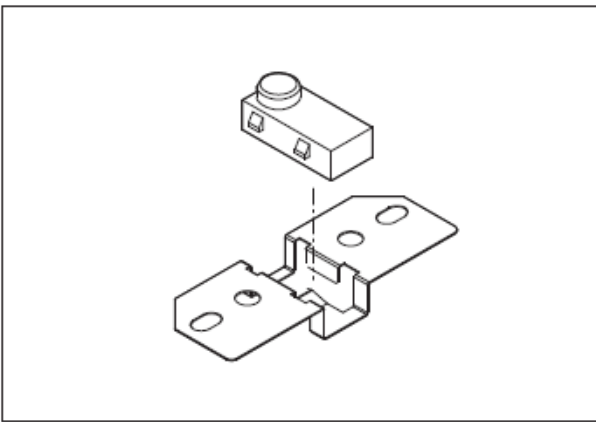


3. Remove the detector mounting plate.

- Screws × 2



4. Change the plate sensor (light receiving).



5. Attach the detector mounting plate.

- Screws × 2

6. Refer to "5.4.5 Changing the magnetic clutch (Page 5-45)", and attach the feeding Roller.

7. Refer to "5.2.4 Removing & attaching the upper insertion unit (Page 5-6)" and "5.2.10 Removing & attaching the Sub-scan unit (Page 5-12)" to attach the upper insertion unit and sub-scan unit.

This concludes the plate sensor (light receiving) changing process.

5.3.11 Changing the plate sensor (light emitting)

Follow the steps below to change the plate sensor (light emitting).

⚠ CAUTION

Before disassembling/assembling this device, be sure to switch OFF the power supply circuit breaker and disconnect the power cable from the device.

Process Summary

This process requires the following number of employees and estimated number of man-hours to perform.

Employees	Man-hours
1	

Things to prepare

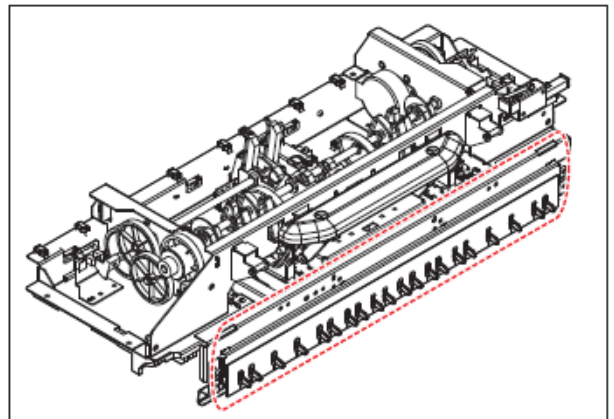
Standard tools are required to perform the following steps.

The changing process

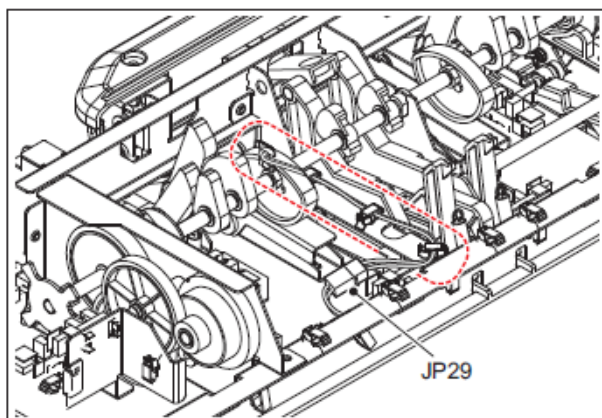
1. Refer to "5.2.3 Removing & attaching the upper cover assy (Page 5-5)", and remove the upper cover assy.
2. Refer to "5.2.4 Removing & attaching the upper insertion unit (Page 5-6)" and remove the upper insertion unit.

📌 IMPORTANT

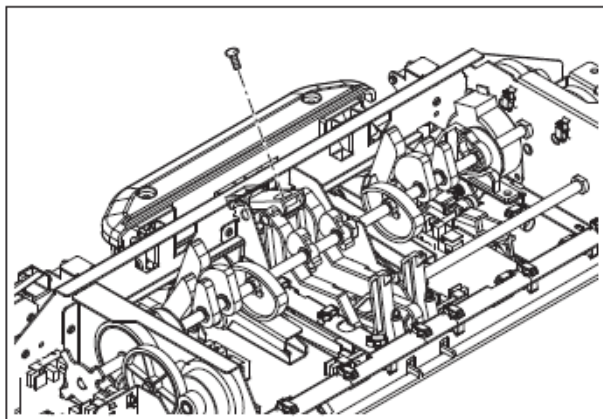
Remove the plate sensor (light emitting) from the upper insertion unit. Make sure not to scratch the antistatic sheet Assy (circled below) during this process.



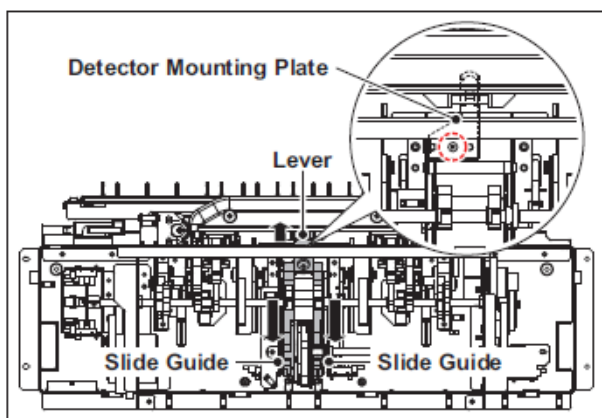
3. Remove the connector (JP29), then remove the cable connected to the plate sensor (light emitting) from its wire saddles (× 4).



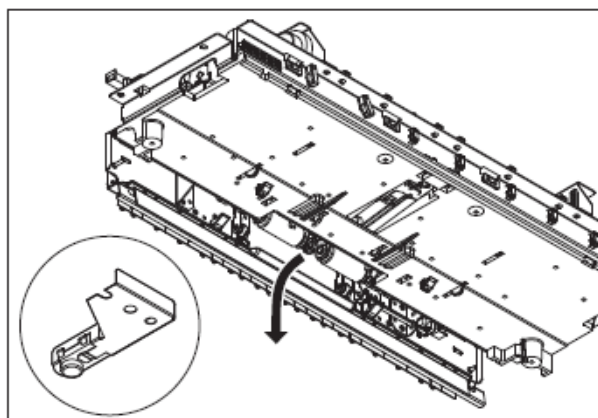
4. Remove screw from the lever.
• Screws × 1



5. Remove screw from the detector mounting plate.
• Screws × 1
Rotate the cam part, then move the slide guide as indicated by the arrow below. Once the lever is loose, the screws can be accessed.



6. Take the detector mounting plate out from under the upper insertion unit, then change the plate sensor (light emitting).



7. Place the detector mounting plate and run its cable through the wire saddles.



Once the detector mounting plate has been secured, running its cable through the wire saddles becomes difficult. Run the cable through the 2 wire saddles on the sensor side here.

8. Attach the detector mounting plate.
• Screws × 1
9. Connect the cable and run it through the wire saddles.
• JP29
10. Attach the lever screw.
• Screws × 1
11. Refer to "5.2.4 Removing & attaching the upper insertion unit (Page 5-6)" and attach the upper insertion unit.
12. Refer to "5.2.3 Removing & attaching the upper cover assy (Page 5-5)" and attach the upper cover assy.

This concludes the plate sensor (light emitting) changing process.

5.3.12 Changing the V-SYNC sensor and V-SYNC assembly

Follow the steps below to change the V-SYNC sensor and V-SYNC assembly.

⚠ CAUTION

Before disassembling/assembling this device, be sure to switch OFF the power supply circuit breaker and disconnect the power cable from the device.

Process Summary

This process requires the following number of employees and estimated number of man-hours to perform.

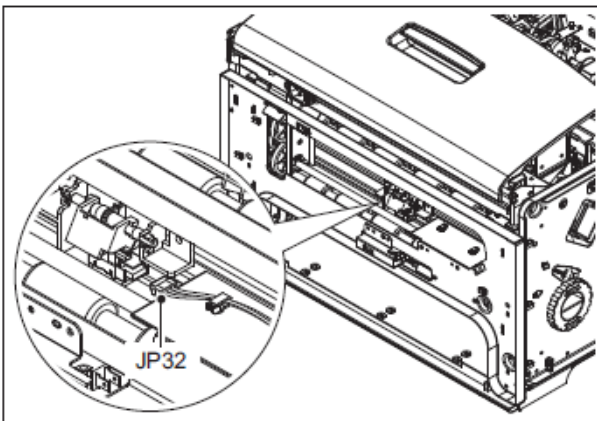
Employees	Man-hours
1	

Things to prepare

Standard tools are required to perform the following steps.

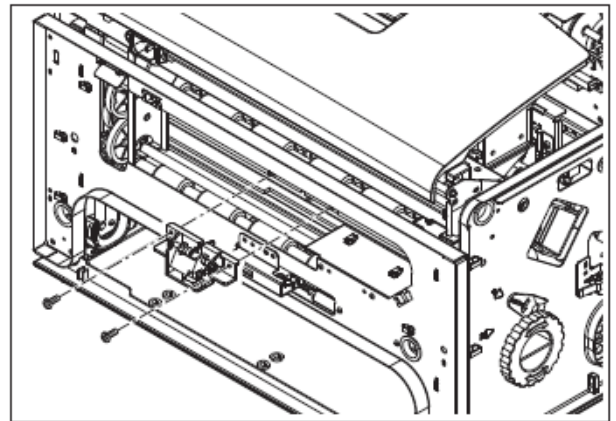
The changing process

1. Refer to "5.2.3 Removing & attaching the upper cover assy (Page 5-5)", and remove the upper cover assy.
2. Refer to "5.2.5 Removing & attaching the left side panel (Page 5-7)" and remove the left side panel.
3. Refer to "5.2.6 Removing & attaching the right side panel (Page 5-8)" and remove the right side panel.
4. Refer to "5.2.7 Removing & attaching the rear panel patch assy. (Page 5-9)" and remove the rear panel patch assy.
5. Remove the cable from the V-SYNC sensor.
 - JP32

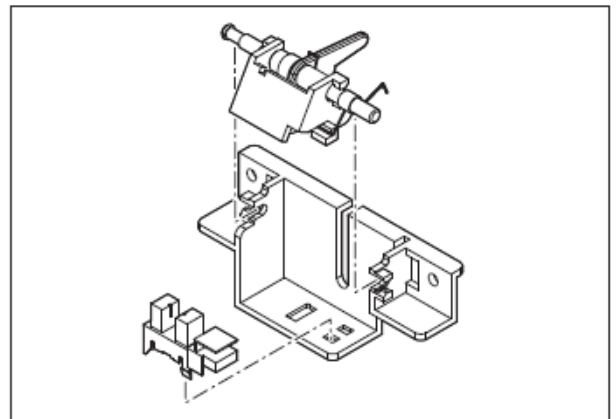


6. Remove the V-SYNC assembly.

- Screws × 2



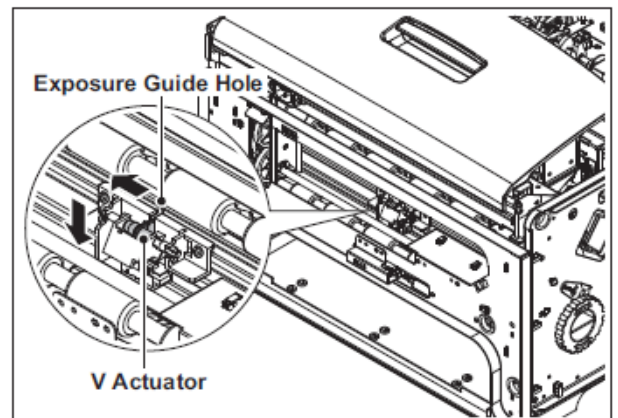
7. If you are changing the V-SYNC sensor, replace the old one.



8. Attach the V-SYNC assembly.

- Screws × 2

Insert the V actuator into the exposure guide holes, then attach so it is abutted as shown in the illustration.



9. Connect the cable to the V-SYNC sensor.

- JP32

10. Refer to "5.2.7 Removing & attaching the rear panel patch assy. (Page 5-9)"

11. Refer to "5.2.6 Removing & attaching the right side panel (Page 5-8)" and attach the right side panel.

12. Refer to "5.2.5 Removing & attaching the left side panel (Page 5-7)" and attach the left side panel.
13. Refer to "5.2.3 Removing & attaching the upper cover Assy (Page 5-5)" and attach the upper cover Assy.

This concludes the V-SYNC sensor and V-SYNC assembly changing process.

5.3.13 Changing the entrance NIP sensor

Follow the steps below to change the entrance NIP sensor.

CAUTION

- Before disassembling/assembling this device, be sure to switch OFF the power supply circuit breaker and disconnect the power cable from the device.
- In order to avoid potential base board damage while discharging it, refer to "5.1.1 Disassembly/Assembly Cautions (Page 5-2)" and wear a grounding strap during the process.

Process Summary

This process requires the following number of employees and estimated number of man-hours to perform.

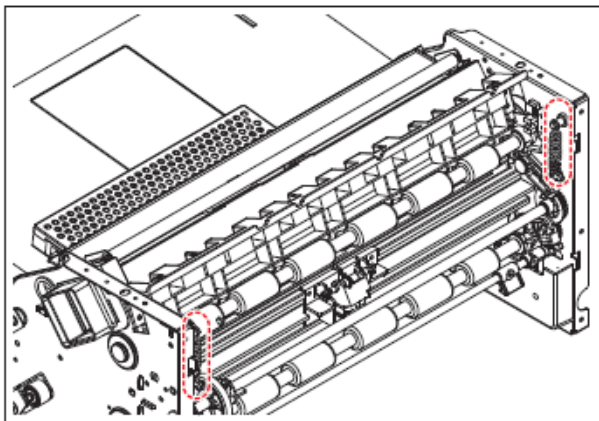
Employees	Man-hours
1	

Things to prepare

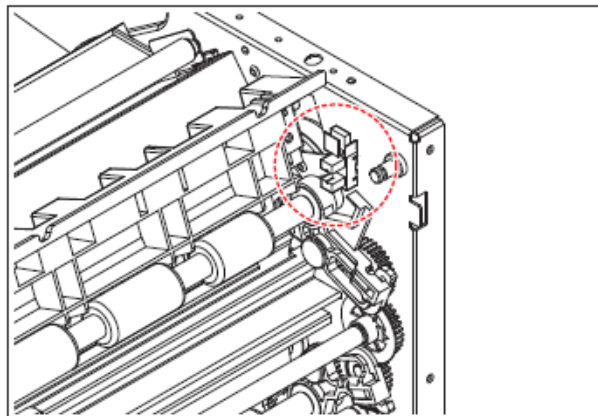
Standard tools are required to perform the following steps.

The changing process

1. Refer to "5.2.10 Removing & attaching the Sub-scan unit (Page 5-12)", and remove the sub-scan unit.
2. Refer to "5.4.2 Changing the sub-scan NIP motor and sub-scan drive Assy (Page 5-41)", and remove the sub-scan drive Assy.
3. Remove the upper open/close springs (× 2).



4. Change the entrance NIP sensor.



5. Refer to "5.4.2 Changing the sub-scan NIP motor and sub-scan drive Assy" starting from Process 8 on Page 5-42, and attach the other parts that were removed from the sub-scan drive Assy during the sub-scan unit removal process.
6. Refer to "5.2.10 Removing & attaching the Sub-scan unit (Page 5-12)" and attach the sub-scan unit.

This concludes the entrance NIP sensor changing process.

5.3.14 Changing the exit NIP sensor

Follow the steps below to change the exit NIP sensor.

⚠ CAUTION

- Before disassembling/assembling this device, be sure to switch OFF the power supply circuit breaker and disconnect the power cable from the device.
- In order to avoid potential base board damage while discharging it, refer to "5.1.1 Disassembly/Assembly Cautions (Page 5-2)" and wear a grounding strap during the process.

Process Summary

This process requires the following number of employees and estimated number of man-hours to perform.

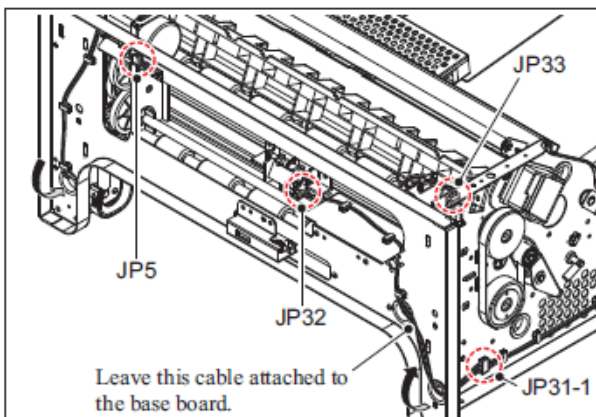
Employees	Man-hours
1	

Things to prepare

Standard tools are required to perform the following steps.

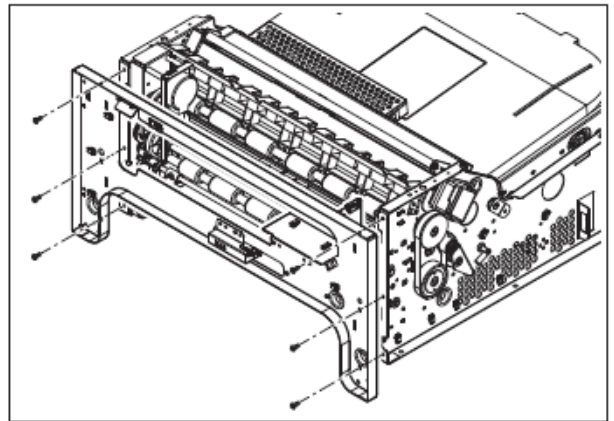
The changing process

1. Refer to "5.2.10 Removing & attaching the Sub-scan unit (Page 5-12)", and remove the sub-scan unit.
2. Remove the connectors (× 4) circled in the illustration, then remove their cables from the base board's cable hole.
 - JP5
 - JP32 (V-SYNC sensor)
 - JP33 (entrance NIP sensor)
 - JP31-1



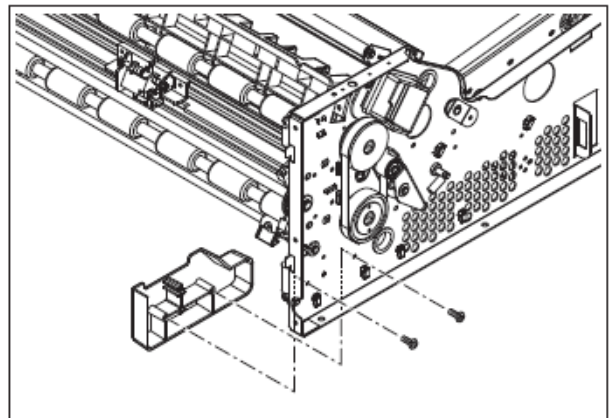
3. Remove the base board assembly.

- Screw × 6

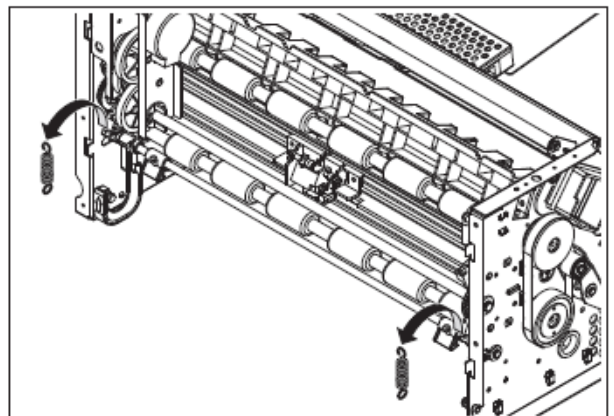


4. Remove the protective cover.

- Screws × 2

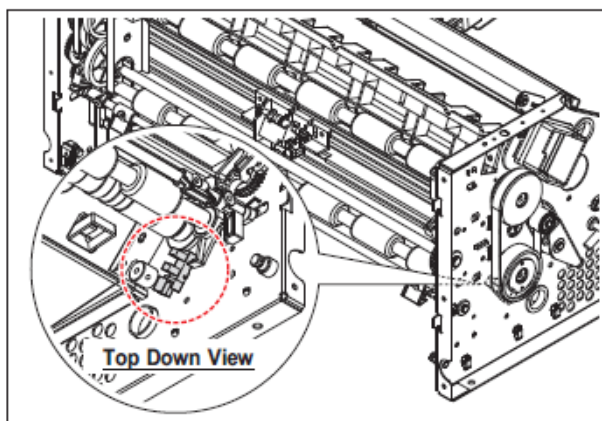


5. Remove the lower open/close springs (× 2).



6. Change the exit NIP sensor.

- JP34

**7. Attach the lower open/close springs (× 2).****8. Attach the protective cover.**

- Screws × 2

9. Attach the base board assembly.

- Screw × 6

10. Refer to [step 2](#) and connect the cables.

- JP5
- JP32 (V-SYNC sensor)
- JP33 (entrance NIP sensor)
- JP31-1

11. Refer to "[5.2.10 Removing & attaching the Sub-scan unit \(Page 5-12\)](#)" and attach the sub-scan unit.

This concludes the exit NIP sensor changing process.

5.4 Changing mechanical parts**5.4.1 Changing the loading/feeding motor**

Follow the steps below to change the loading/feeding motor.

⚠ CAUTION

- Before disassembling/assembling this device, be sure to switch OFF the power supply circuit breaker and disconnect the power cable from the device.
- In order to avoid potential base board damage while discharging it, refer to "[5.1.1 Disassembly/Assembly Cautions \(Page 5-2\)](#)" and wear a grounding strap during the process.

Process Summary

This process requires the following number of employees and estimated number of man-hours to perform.

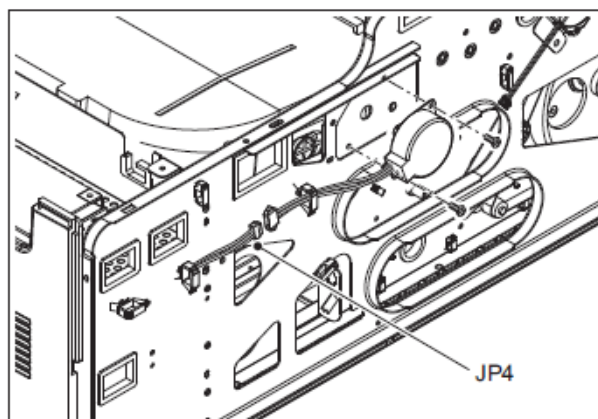
Employees	Man-hours
1	

Things to prepare

Standard tools are required to perform the following steps.

The changing process

1. Refer to "[5.2.3 Removing & attaching the upper cover assy \(Page 5-5\)](#)", and remove the upper cover assy.
2. Refer to "[5.2.5 Removing & attaching the left side panel \(Page 5-7\)](#)" and remove the left side panel.
3. Refer to "[5.2.6 Removing & attaching the right side panel \(Page 5-8\)](#)" and remove the right side panel.
4. Refer to "[5.2.8 Removing & attaching the lower insertion unit \(Page 5-9\)](#)", and remove the lower insertion unit.
5. Change the loading/feeding motor (stepping motor).
 - Screws × 2
 - JP4



6. Refer to "5.2.8 Removing & attaching the lower insertion unit (Page 5-9)" and attach the lower insertion unit.
7. Refer to "5.2.6 Removing & attaching the right side panel (Page 5-8)" and attach the right side panel.
8. Refer to "5.2.5 Removing & attaching the left side panel (Page 5-7)" and attach the left side panel.
9. Refer to "5.2.3 Removing & attaching the upper cover Assy (Page 5-5)" and attach the upper cover Assy.

This concludes the loading/feeding motor changing process.

5.4.2 Changing the sub-scan NIP motor and sub-scan drive Assy

Follow the steps below to change the sub-scan NIP motor.

The sub-scan drive Assy is removed when changing the sub-scan NIP motor.

⚠ CAUTION

- Before disassembling/assembling this device, be sure to switch OFF the power supply circuit breaker and disconnect the power cable from the device.
- In order to avoid potential base board damage while discharging it, refer to "5.1.1 Disassembly/Assembly Cautions (Page 5-2)" and wear a grounding strap during the process.

Process Summary

This process requires the following number of employees and estimated number of man-hours to perform.

Employees	Man-hours
1	

Things to prepare

Standard tools are required to perform the following steps.

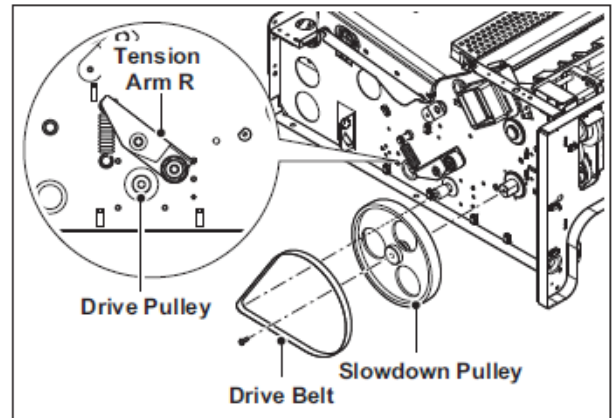
The changing process

1. Refer to "5.2.10 Removing & attaching the Sub-scan unit (Page 5-12)", and remove the sub-scan unit.

2. Take off the tension arm R spring, then remove the drive belt and slowdown pulley.

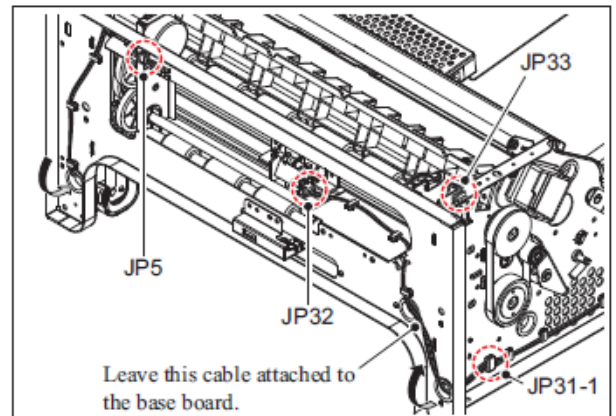
- Screw × 1

After removal, move the roller part of tension arm R downward (as shown in the illustration bubble) so that it does not make contact with the drive pulley.



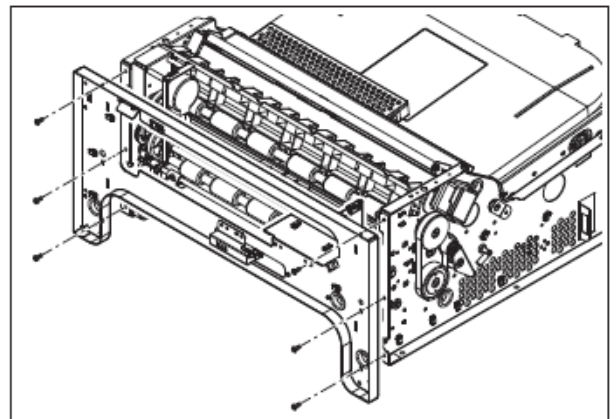
3. Remove the connectors (x4) circled in the illustration, then remove their cables from the base board's cable hole.

- JP5
- JP32 (V-SYNC sensor)
- JP33 (entrance NIP sensor)
- JP31-1



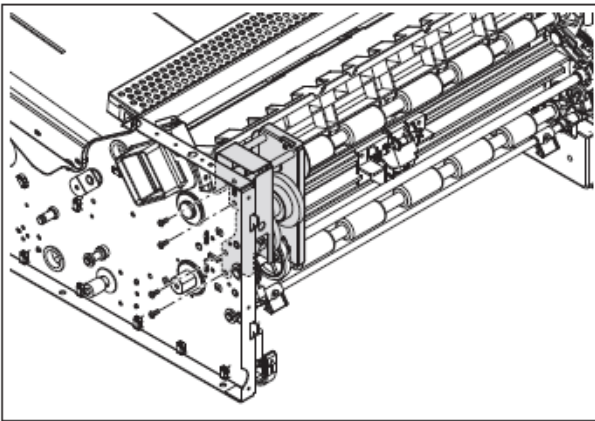
4. Remove the base board assembly.

- Screw × 6

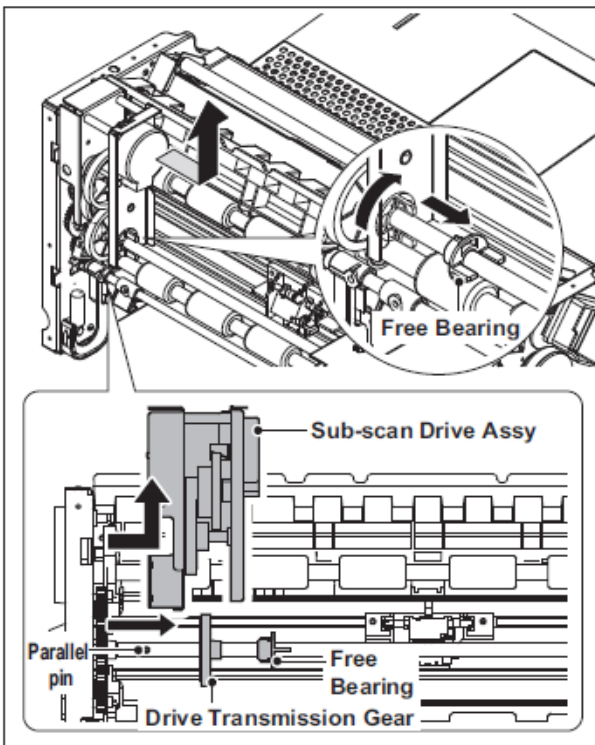


5. Remove the sub-scan drive Assy screws.

- Screw x 4

**6. Remove the sub-scan drive Assy.**

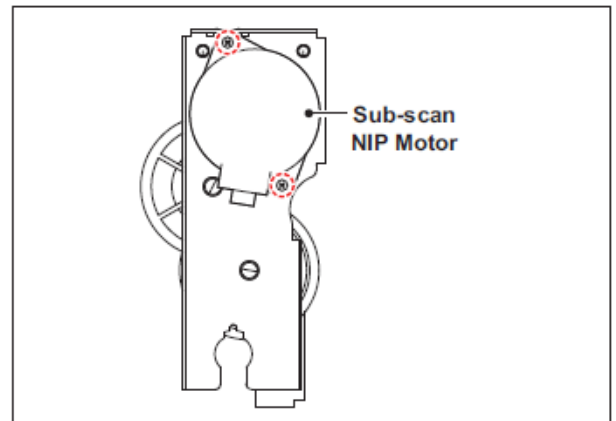
- 1) Rotate the free shaft holder 90 degrees, release the lock, and remove it from the sub-scan drive Assy.
- 2) Slide the sub-scan drive Assy together with the drive transmission gear, then pull them up and out.
- 3) Remove the drive transmission gear's parallel pin for baffling.



If you are not going to change the sub-scan NIP motor, skip to [step 8](#).

7. Change the sub-scan NIP motor.

- Screws x 2

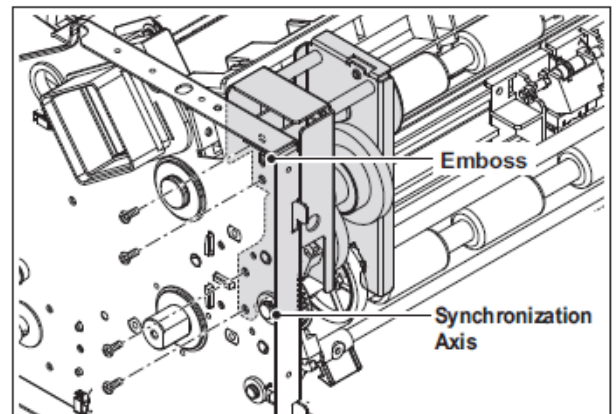
**8. Refer to [step 6](#), then assemble and attach the sub-scan drive Assy.**

Align it with the sub-scan unit's side plate and emboss.

9. Secure the sub-scan drive Assy by attaching the screws.

- Screw x 4

Start with the screws closest to the synchronization axis.

**10. Attach the base board assembly.**

- Screw x 6

11. Refer to [step 3](#) and connect the cables.

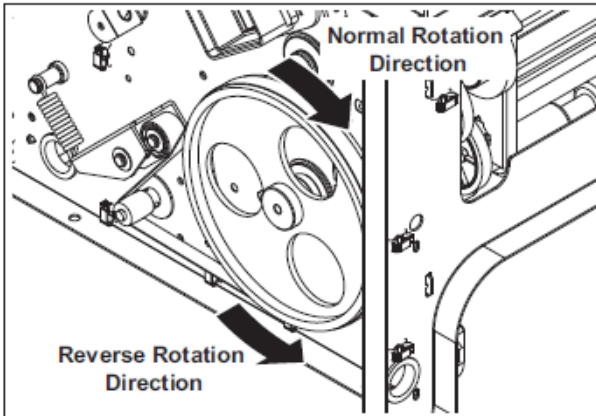
- JP5
- JP32 (V-SYNC sensor)
- JP33 (entrance NIP sensor)
- JP31-1

12. Attach the slowdown pulley and drive belt, then connect the tension arm R spring.

- Screw x 1

13. Adapt the drive belt to the pulley.

Rotate the slowdown pulley in reverse and normal directions at least 5 times each.

**14. Refer to "5.2.10 Removing & attaching the Sub-scan unit (Page 5-12)" and attach the sub-scan unit.**

This concludes the sub-scan NIP motor and sub-scan drive Assy changing process.

5.4.3 Changing the sub-scan feeding motor

Follow the steps below to change the sub-scan feeding motor.

⚠ CAUTION

- Before disassembling/assembling this device, be sure to switch OFF the power supply circuit breaker and disconnect the power cable from the device.
- In order to avoid potential base board damage while discharging it, refer to "5.1.1 Disassembly/Assembly Cautions (Page 5-2)" and wear a grounding strap during the process.

Process Summary

This process requires the following number of employees and estimated number of man-hours to perform.

Employees	Man-hours
1	

Things to prepare

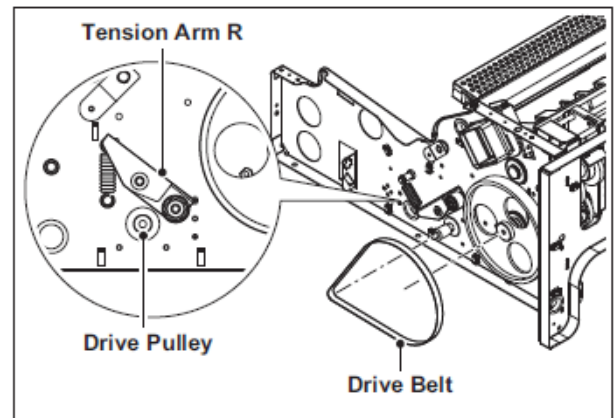
Standard tools are required to perform the following steps.

The changing process

1. Refer to "5.2.10 Removing & attaching the Sub-scan unit (Page 5-12)", and remove the sub-scan unit.
2. Refer to "5.2.12 Removing & attaching the exposure unit (Page 5-20)", and remove the exposure unit.
3. Refer to "5.2.13 Removing & attaching light collection unit (Page 5-21)", and remove the light collection unit.

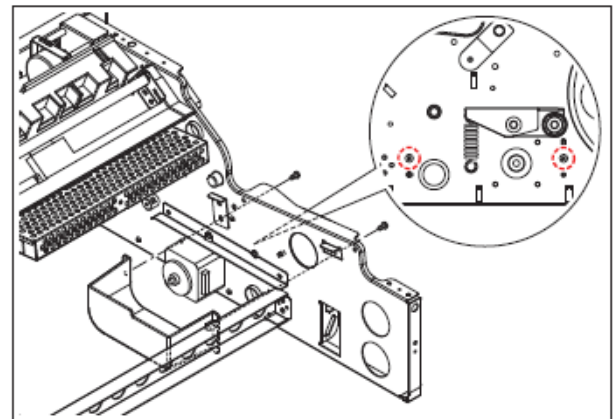
4. Take off the tension arm R spring, then remove the drive belt.

After removal, move the roller part of tension arm R downward (as shown in the illustration bubble) so that it does not make contact with the drive pulley.

**5. Remove the protective cover.**

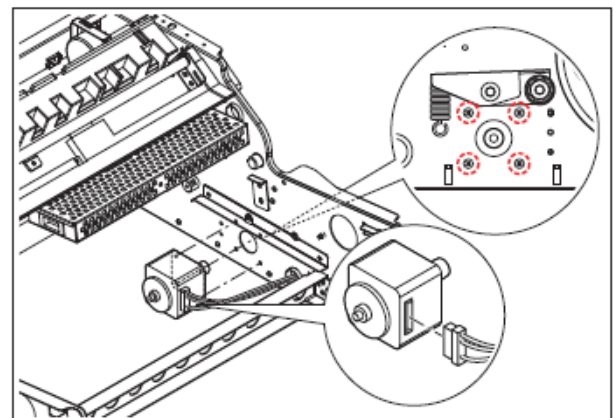
- Screws × 2

When removing the rear side screws, hold tension arm R in your hand and move it for accessibility.

**6. Change the sub-scan feeding motor (stepping motor Assy).**

- Screws × 4
- JP6

When removing the screws, hold tension arm R in your hand and move it for accessibility.



IMPORTANT

Do NOT remove the drive pulley attached to the stepping motor.

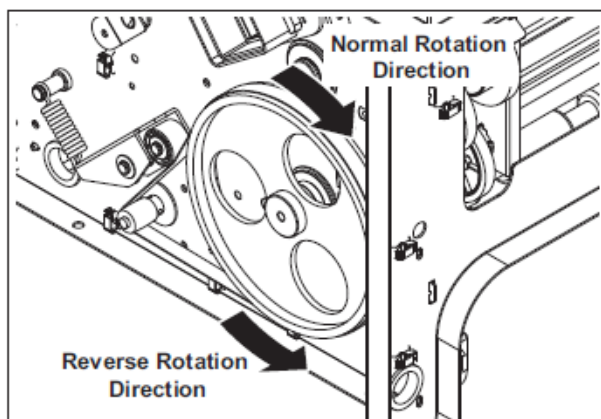
7. Attach the protective cover.

- Screws × 2

8. Attach the drive belt and fasten the tension arm R spring.

9. Adapt the drive belt to the pulley.

Rotate the slowdown pulley in reverse and normal directions at least 5 times each.



10. Refer to "5.2.13 Removing & attaching light collection unit (Page 5-21)" and attach the light collection unit.

11. Refer to "5.2.12 Removing & attaching the exposure unit (Page 5-20)" and attach the exposure unit.

12. Refer to "5.2.10 Removing & attaching the Sub-scan unit (Page 5-12)" and attach the sub-scan unit.

This concludes the sub-scan feeding motor changing process.

5.4.4 Changing the Eraser LED cooling fan

Follow the steps below to change the Eraser LED cooling fan.

CAUTION

Before disassembling/assembling this device, be sure to switch OFF the power supply circuit breaker and disconnect the power cable from the device.

Process Summary

This process requires the following number of employees and estimated number of man-hours to perform.

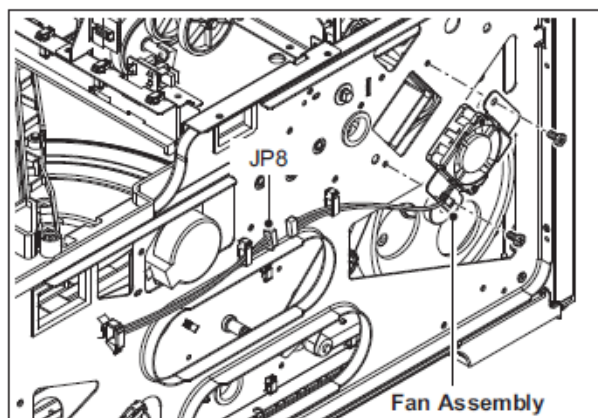
Employees	Man-hours
1	

Things to prepare

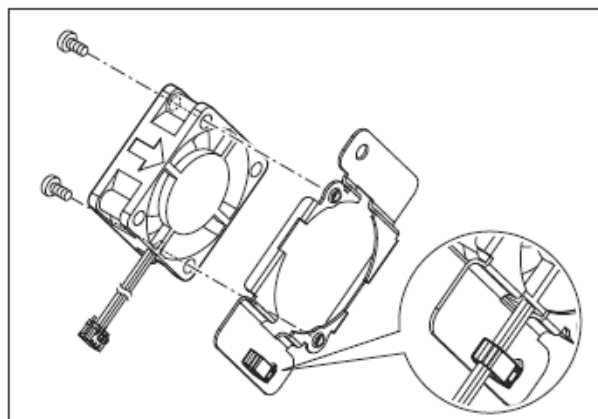
Standard tools are required to perform the following steps.

The changing process

1. Refer to "5.2.3 Removing & attaching the upper cover assy (Page 5-5)", and remove the upper cover assy.
2. Refer to "5.2.6 Removing & attaching the right side panel (Page 5-8)" and remove the right side panel.
3. Remove the Eraser LED cooling fan cable.
 - JP8
4. Remove the fan assembly.
 - TP hexagonal Phillips-head screw × 4 (M3 × 6)



5. Change the Eraser LED cooling fan (fan motor).
 - Screws × 2



6. Attach the fan assembly.
 - TP hexagonal Phillips-head screw × 4 (M3 × 6)
7. Connect the Eraser LED cooling fan cable.
 - JP8
8. Refer to "5.2.6 Removing & attaching the right side panel (Page 5-8)" and attach the right side panel.
9. Refer to "5.2.3 Removing & attaching the upper cover assy (Page 5-5)" and attach the upper cover assy.

This concludes the Eraser LED cooling fan changing process.

5.4.5 Changing the magnetic clutch

Follow the steps below to change the magnetic clutch.

⚠ CAUTION

Before disassembling/assembling this device, be sure to switch OFF the power supply circuit breaker and disconnect the power cable from the device.

Process Summary

This process requires the following number of employees and estimated number of man-hours to perform.

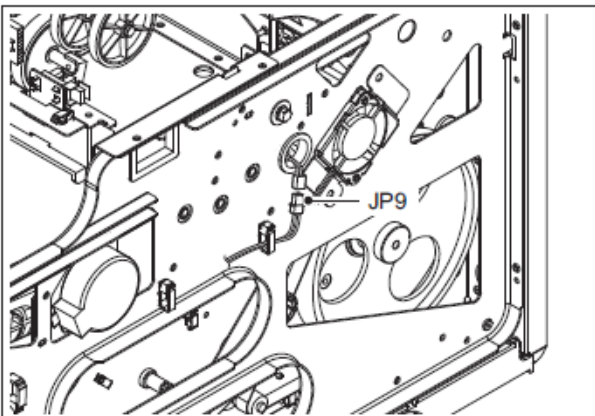
Employees	Man-hours
1	

Things to prepare

Standard tools are required to perform the following steps.

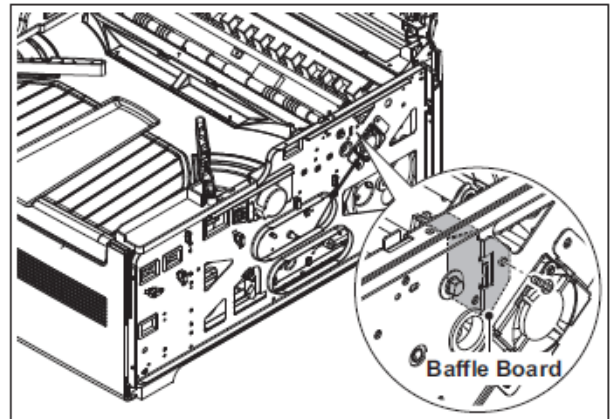
The changing process

1. Refer to "5.2.3 Removing & attaching the upper cover assy (Page 5-5)", and remove the upper cover assy.
2. Refer to "5.2.5 Removing & attaching the left side panel (Page 5-7)" and remove the left side panel.
3. Refer to "5.2.6 Removing & attaching the right side panel (Page 5-8)" and remove the right side panel.
4. Refer to "5.2.4 Removing & attaching the upper insertion unit (Page 5-6)" and remove the upper insertion unit.
5. Remove the magnetic clutch cable.
 - JP9



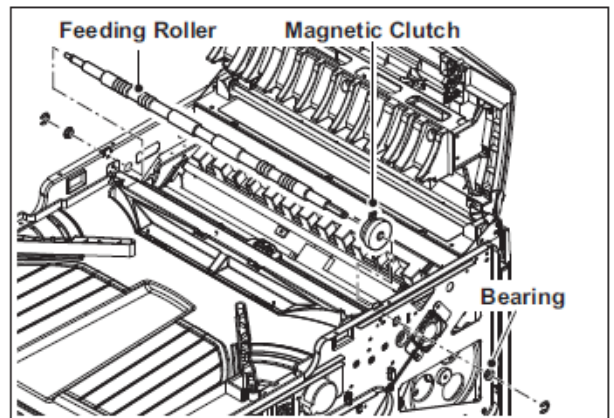
6. Remove the baffle board.

- Screws × 1



7. Remove the shaft holders (× 2), feeding Roller, and magnetic clutch.

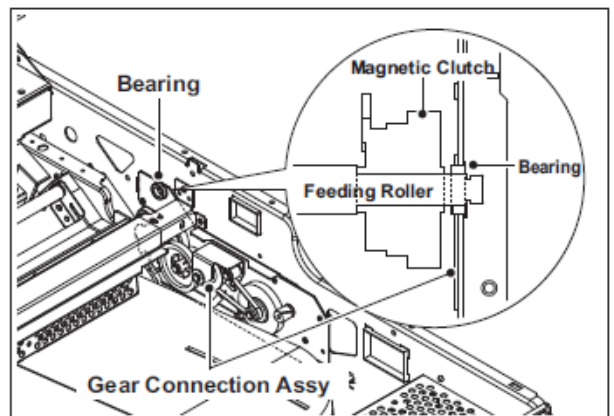
- E Rings × 2



8. Attach the shaft holders (× 2), feeding Roller, and new magnetic clutch.

- E Rings × 2

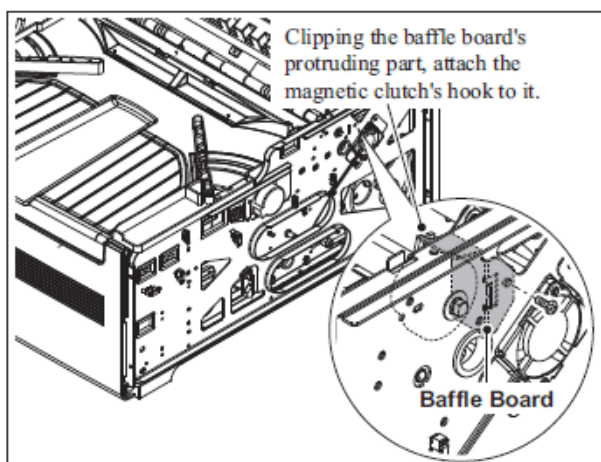
Make sure to insert the right-side shaft holder into the gear connection assy hole.



(The lower insertion unit is removed in this illustration.)

9. Attach the baffle board.

- Screws × 1



10. Connect the magnetic clutch cable.

- JP9

11. Refer to "5.2.4 Removing & attaching the upper insertion unit (Page 5-6)" and attach the upper insertion unit.

12. Refer to "5.2.6 Removing & attaching the right side panel (Page 5-8)" and attach the right side panel.

13. Refer to "5.2.5 Removing & attaching the left side panel (Page 5-7)" and attach the left side panel.

14. Refer to "5.2.3 Removing & attaching the upper cover Assy (Page 5-5)" and attach the upper cover Assy.

This concludes the magnetic clutch changing process.

5.4.6 Changing the upper door guide

Follow the steps below to change the upper door guide.

⚠ CAUTION

Before disassembling/assembling this device, be sure to switch OFF the power supply circuit breaker and disconnect the power cable from the device.

Process Summary

This process requires the following number of employees and estimated number of man-hours to perform.

Employees	Man-hours
1	

Things to prepare

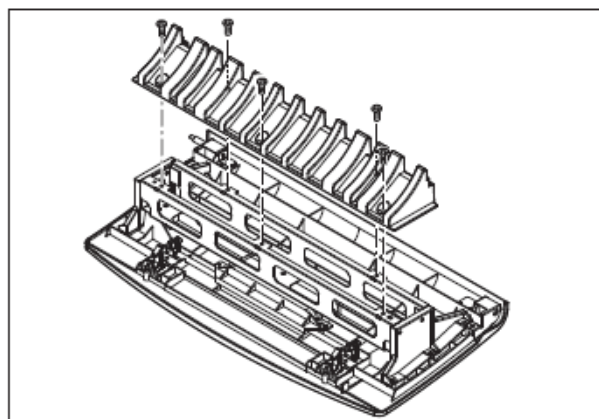
Standard tools are required to perform the following steps.

The changing process

HINT

After changing the upper door guide, the locating plate which secures the upper door support pillar needs a position adjustment. The upper door Assy must be removed to perform this task.

1. Refer to "5.2.3 Removing & attaching the upper cover Assy (Page 5-5)", and remove the upper cover Assy.
2. Refer to "5.2.14 Removing & attaching the upper door Assy (Page 5-23)", and remove the upper door Assy.
3. Change the guide.
 - Screws × 5



4. Refer to "5.2.14 Removing & attaching the upper door Assy (Page 5-23)" and attach the upper door Assy.
5. Refer to "5.2.3 Removing & attaching the upper cover Assy (Page 5-5)" and attach the upper cover Assy.

This concludes the upper door guide changing process.

5.4.7 Changing the link assemblies

Follow the steps below to change Link Assy.1 and Link Assy.2.

⚠ CAUTION

- Before disassembling/assembling this device, be sure to switch OFF the power supply circuit breaker and disconnect the power cable from the device.
- In order to avoid potential base board damage while discharging it, refer to "5.1.1 Disassembly/Assembly Cautions (Page 5-2)" and wear a grounding strap during the process.

Process Summary

This process requires the following number of employees and estimated number of man-hours to perform.

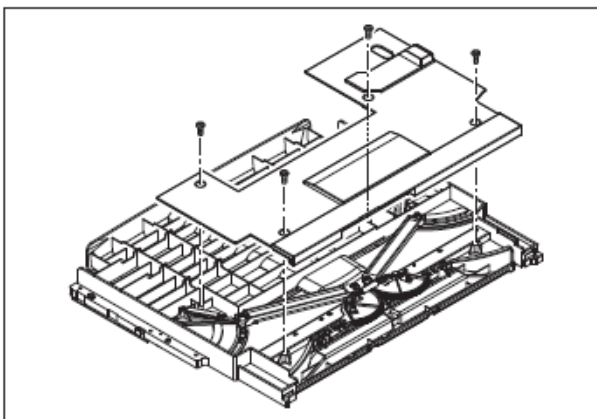
Employees	Man-hours
1	

Things to prepare

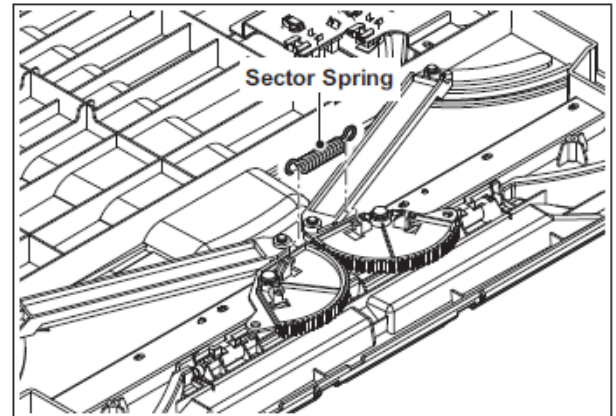
Standard tools are required to perform the following steps.

The changing process

1. Refer to "5.2.3 Removing & attaching the upper cover assy (Page 5-5)", and remove the upper cover assy.
2. Refer to "5.2.5 Removing & attaching the left side panel (Page 5-7)" and remove the left side panel.
3. Refer to "5.2.6 Removing & attaching the right side panel (Page 5-8)" and remove the right side panel.
4. Refer to "5.2.8 Removing & attaching the lower insertion unit (Page 5-9)", and remove the lower insertion unit.
5. Turn over the lower insertion unit and remove its cover.
 - P Tight × 4 (M3 × 8)



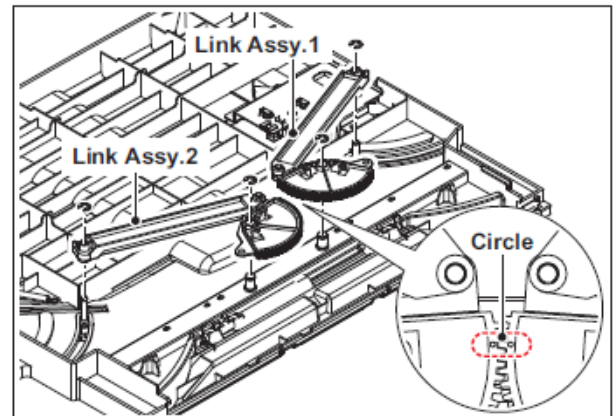
6. Remove the sector spring.



7. Change the link assemblies.

- E Rings × 2 per assembly

Attach them in alignment with the sector gear circles.



8. Attach the sector spring.

9. Attach the lower insertion unit cover.

- P Tight × 4 (M3 × 8)

10. Refer to "5.2.8 Removing & attaching the lower insertion unit (Page 5-9)" and attach the lower insertion unit.

11. Refer to "5.2.6 Removing & attaching the right side panel (Page 5-8)" and attach the right side panel.

12. Refer to "5.2.5 Removing & attaching the left side panel (Page 5-7)" and attach the left side panel.

13. Refer to "5.2.3 Removing & attaching the upper cover assy (Page 5-5)" and attach the upper cover assy.

This concludes the changing process for link assemblies 1 and 2.

5.4.8 Changing the entrance guide

Follow the steps below to change the entrance guide.

⚠ CAUTION

- Before disassembling/assembling this device, be sure to switch OFF the power supply circuit breaker and disconnect the power cable from the device.
- In order to avoid potential base board damage while discharging it, refer to "5.1.1 Disassembly/Assembly Cautions (Page 5-2)" and wear a grounding strap during the process.

Process Summary

This process requires the following number of employees and estimated number of man-hours to perform.

Employees	Man-hours
1	

Things to prepare

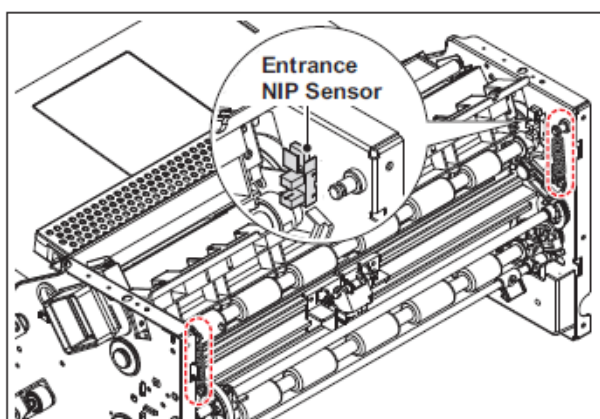
Standard tools are required to perform the following steps.

The changing process

🔑 IMPORTANT

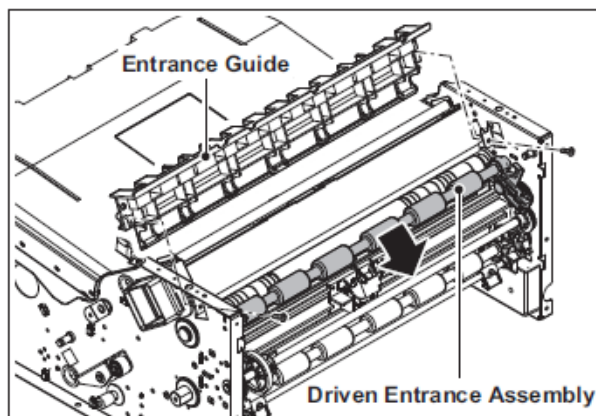
When removing and attaching an entrance guide, make sure the eraser unit is attached to the sub-scan unit.

1. Refer to "5.2.10 Removing & attaching the Sub-scan unit (Page 5-12)", and remove the sub-scan unit.
2. Refer to "5.4.2 Changing the sub-scan NIP motor and sub-scan drive Assy (Page 5-41)", and remove the sub-scan drive Assy.
3. Remove the upper open/close springs (× 2) and entrance NIP sensor.

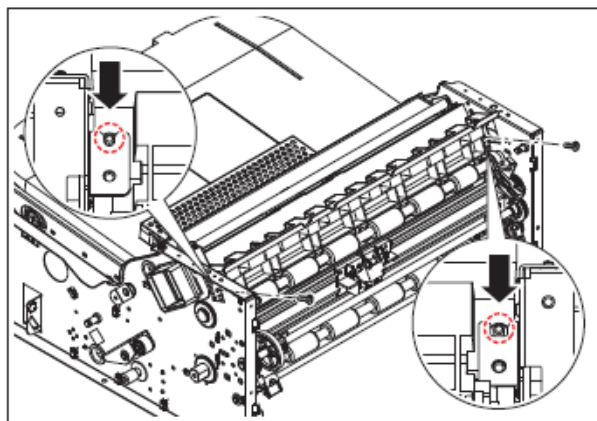


4. Pull down the driven entrance assembly, release the feeding Roller clamp, and change the entrance guide.

• Screws × 2



When attaching it, abut its locating boss to the sub-scan unit's notches, then secure with screws while bringing it in.



5. Attach the upper open/close springs (× 2) and entrance NIP sensor.
6. Refer to "5.4.2 Changing the sub-scan NIP motor and sub-scan drive Assy" starting from Process 8 on Page 5-42, and attach the other parts that were removed from the sub-scan drive Assy during the sub-scan unit removal process.
7. Refer to "5.2.10 Removing & attaching the Sub-scan unit (Page 5-12)" and attach the sub-scan unit.

This concludes the entrance guide changing process.

5.4.9 Changing the drive rollers

Follow the steps below to change drive roller U and drive roller D.

⚠ CAUTION

- Before disassembling/assembling this device, be sure to switch OFF the power supply circuit breaker and disconnect the power cable from the device.
- In order to avoid potential base board damage while discharging it, refer to "5.1.1 Disassembly/Assembly Cautions (Page 5-2)" and wear a grounding strap during the process.

Process Summary

This process requires the following number of employees and estimated number of man-hours to perform.

Employees	Man-hours
1	

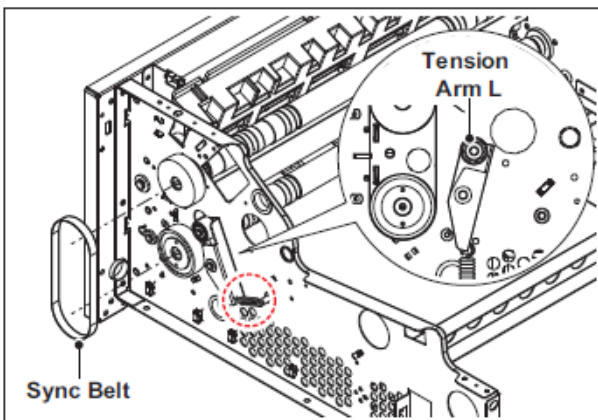
Things to prepare

Standard tools are required to perform the following steps.

The changing process

1. Refer to "5.2.10 Removing & attaching the Sub-scan unit (Page 5-12)", and remove the sub-scan unit.
2. Refer to "5.2.11 Removing and attaching the eraser unit (Page 5-17)", and remove the eraser unit.
3. Refer to "5.2.12 Removing & attaching the exposure unit (Page 5-20)", and remove the exposure unit.
4. Refer to "5.2.13 Removing & attaching light collection unit (Page 5-21)", and remove the light collection unit.
5. Take off the tension arm L spring, then remove the sync belt.

After removing, pull tension arm L's roller side down to the right as shown in the illustration bubble.

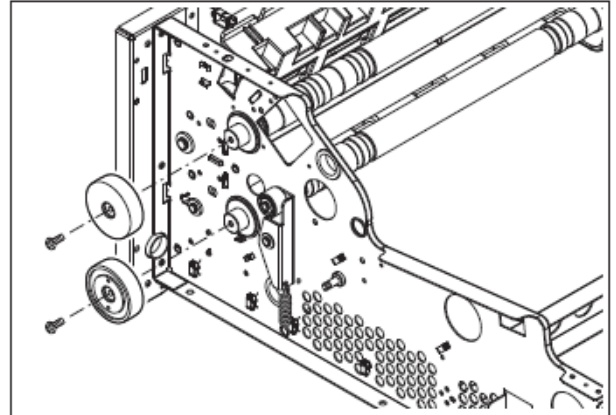


6. Remove the flywheel of the drive roller to be changed.

- Screw × 1 each

🔧 HINT

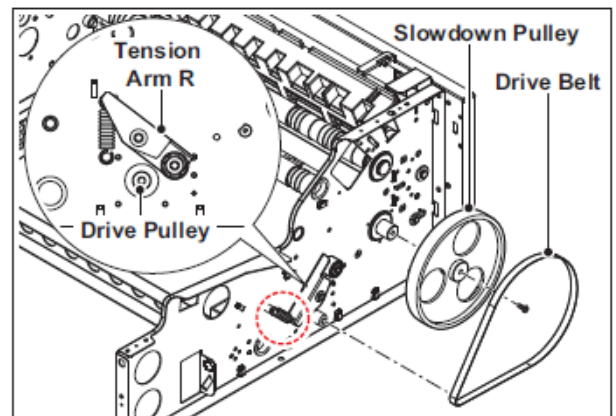
When removing the screw for flywheel D (bottom), holding the reverse side slowdown pulley secure will make it easier.



7. Take off the tension arm R spring, then remove the drive belt and slowdown pulley.

- Screw × 1

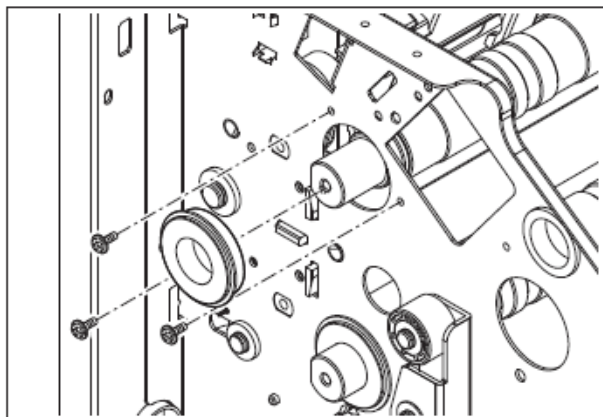
After removal, move the roller part of tension arm R downward (as shown in the illustration bubble) so that it does not make contact with the drive pulley.



8. Change drive roller U. (If only changing drive roller D, these steps are unnecessary.)

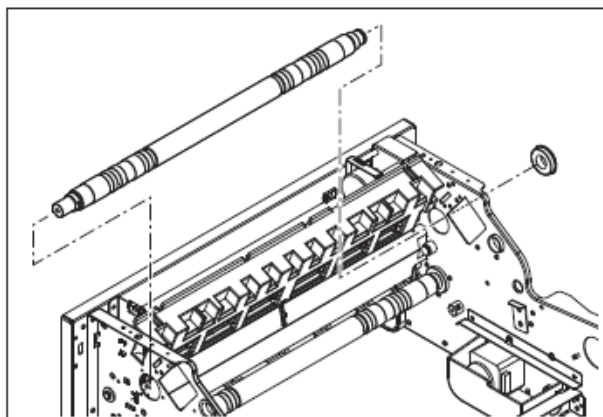
1) Remove drive roller U's left shaft holder.

- Screws × 3



2) Change drive roller U.

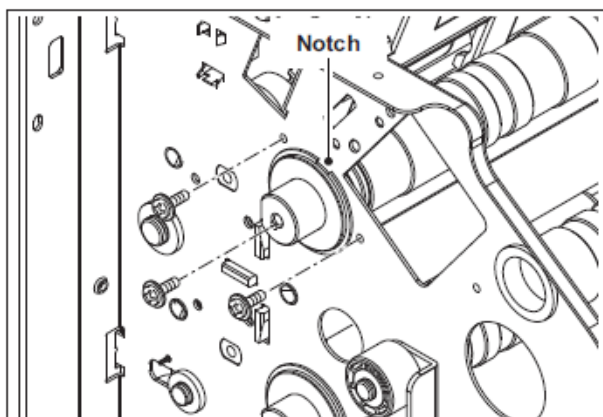
The right shaft holder is not secure, so it will come off when sliding the roller to the right.



3) Attach drive roller U's left shaft holder.

- Screws × 3

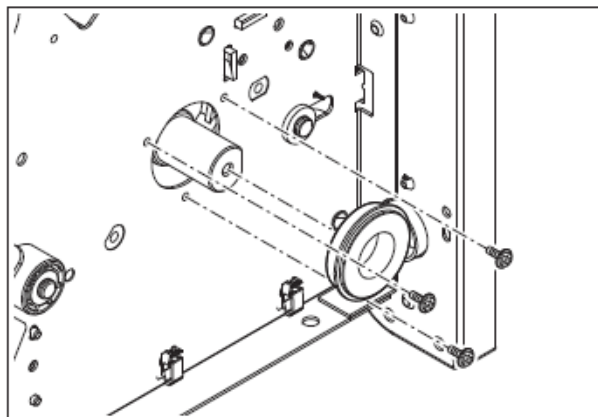
Make sure no screws fall into the shaft holder notches.



9. Change drive roller D. (If only changing drive roller U, these steps are unnecessary.)

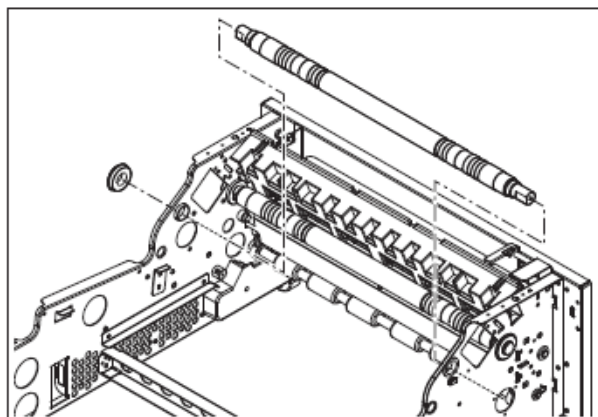
1) Remove drive roller D's right shaft holder.

- Screws × 3



2) Change drive roller D.

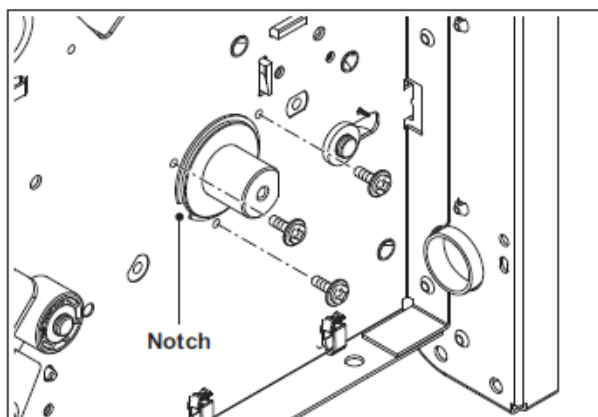
The left shaft holder is not secure, so it will come off when sliding the roller to the left.



3) Attach drive roller D's right shaft holder.

- Screws × 3

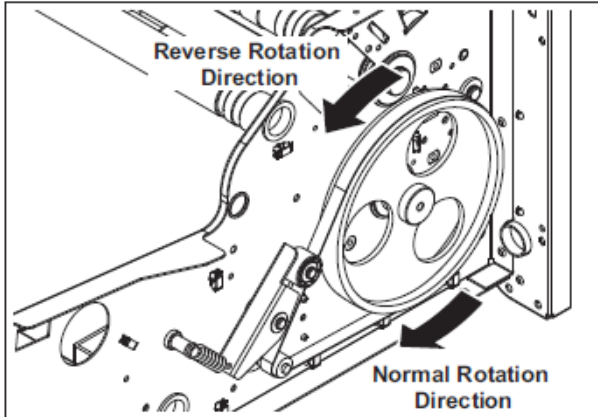
Make sure no screws fall into the shaft holder notches.



10. Refer to [step 7](#) to attach the slowdown pulley and drive belt, then connect the tension arm R spring.
- Screw x 1

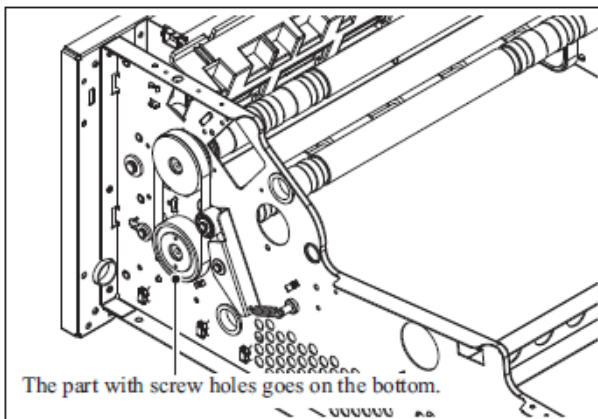
11. Adapt the drive belt to the pulley.

Rotate the slowdown pulley in reverse and normal directions at least 5 times each.



12. Refer to [step 6](#) and [step 5](#), attach the flywheel and sync belt, then connect the tension arm L spring.

• Flywheels: 1 screw each



13. Perform the same operation done in [step 11](#) (rotate the slowdown pulley in reverse and normal directions at least 5 times each), then adapt the sync Belt.
14. Refer to "[5.2.13 Removing & attaching light collection unit \(Page 5-21\)](#)" and attach the light collection unit.
15. Refer to "[5.2.12 Removing & attaching the exposure unit \(Page 5-20\)](#)" and attach the exposure unit.
16. Refer to "[5.2.11 Removing and attaching the eraser unit \(Page 5-17\)](#)", and attach the eraser unit.
17. Refer to "[5.2.10 Removing & attaching the Sub-scan unit \(Page 5-12\)](#)" and attach the sub-scan unit.

This concludes the changing process for drive roller U and drive roller D.

5.5 Changing cassette parts

5.5.1 Disassembling a cassette

The following steps describe how to disassemble the cassette's front plate, back plate, and tray.

Process Summary

This process requires the following number of employees and estimated number of man-hours to perform.

Employees	Man-hours
1	

Things to prepare

In addition to standard tools, the following tool(s) will be needed to perform this task.

No.	Tool	No.	Tool
1	Opening tools for Cassette	2	Plate storage bag
3	Portable vacuum cleaner		

HINT

Screw removal and attachment requires a Phillips-head screwdriver (#1 bit).

The disassembly process

1. Eject the plate from the cassette.

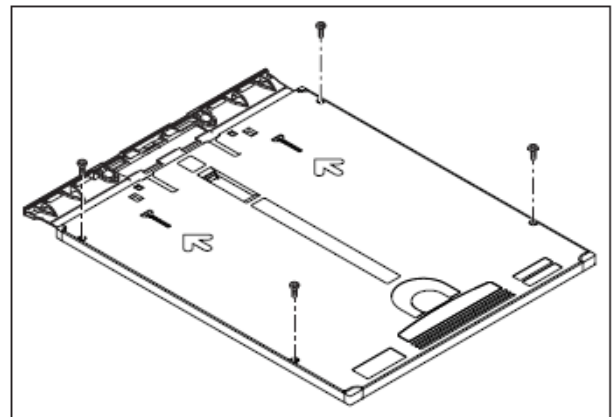
Leave the tray out of the cassette.

Put the plate in a storage bag.

For details on how to eject a plate, refer to the operation manual.

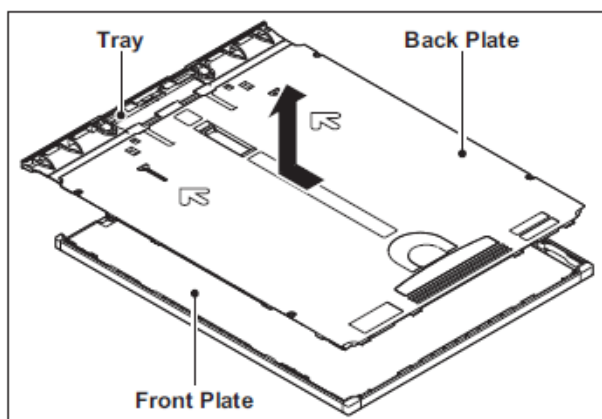
2. Position the back plate face up and remove the screws.

• Screws x 4



3. Detach the front and back plates from the cassette.

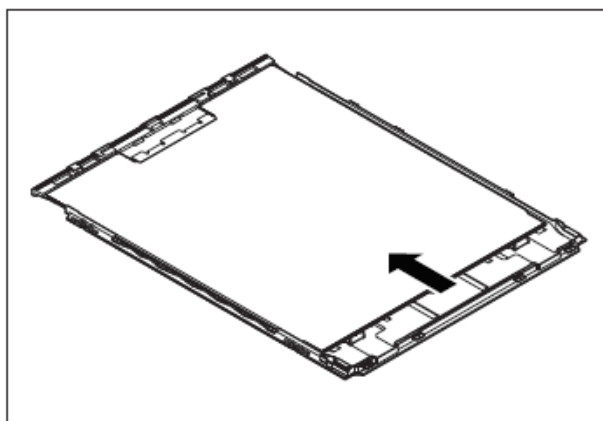
The tray is attached to the back plate.



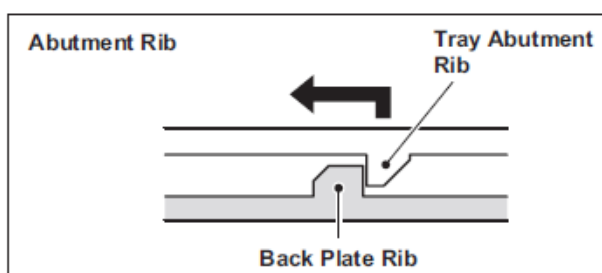
4. Position the back plate so that the tray faces upward.

5. Push the tray out as far as it will go.

It can only be pushed out a short distance at this point. Once caught on the abutment rib in the next step, it will no longer push out any farther.

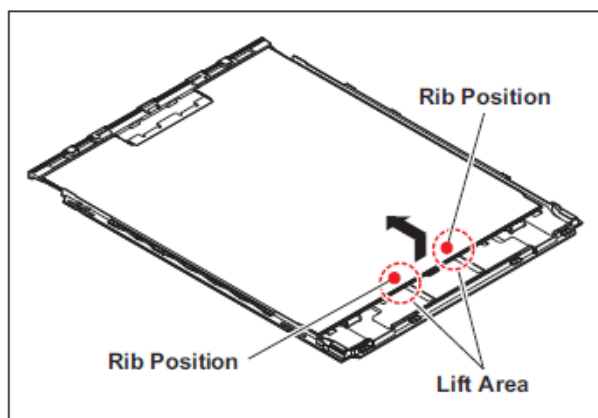


6. Lift up around the abutment rib so that the tray is no longer stuck on it, then continue pushing the tray out to its limit.

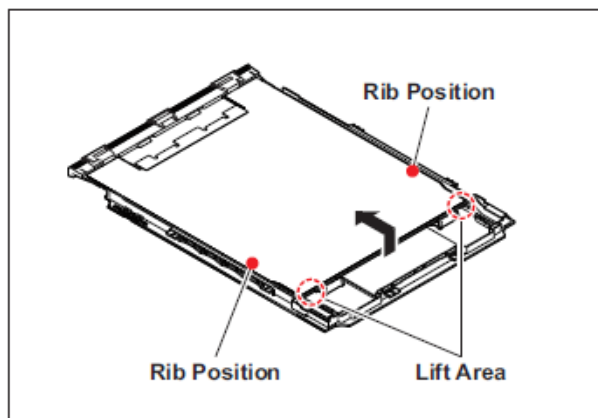


The abutment rib's position differs according to cassette size.

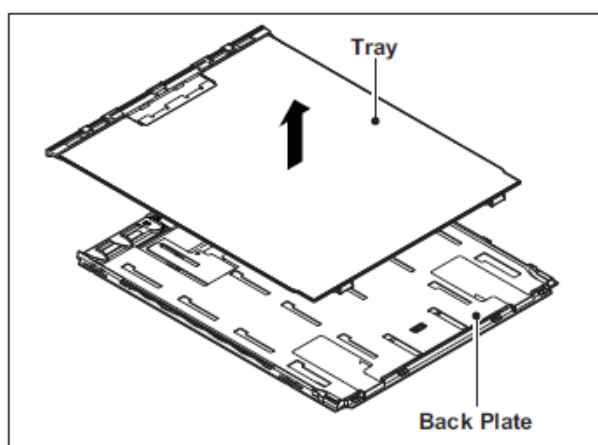
If the cassette size is 14 × 17, 14 × 14, 11 × 14, 10 × 12, 24 × 30 cm, or 15 × 30 cm (The illustration shows 14 × 17)



If the cassette size is 8 × 10 or 18 × 24 cm (illustration shows 8 × 10)



7. Remove the tray from the back plate.



This concludes the cassette disassembly process.

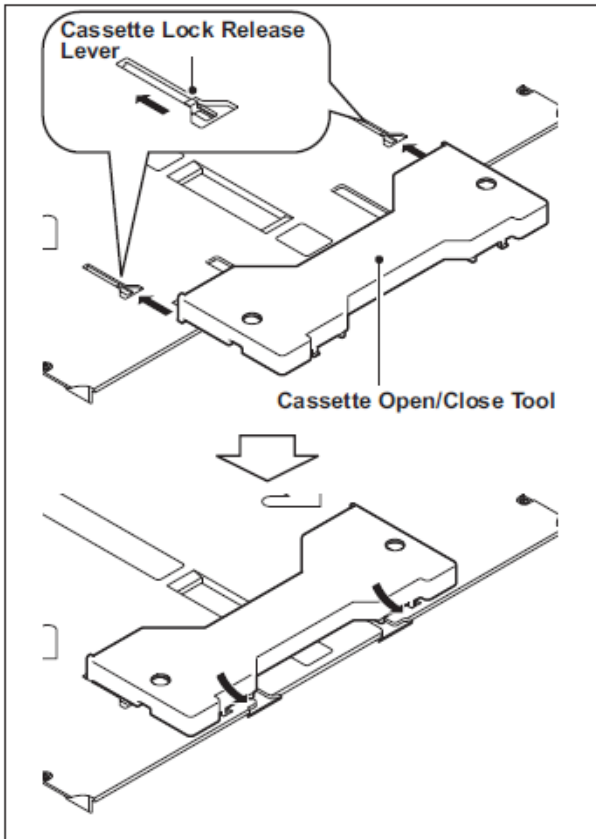
Things to note during assembly

⚠ CAUTION

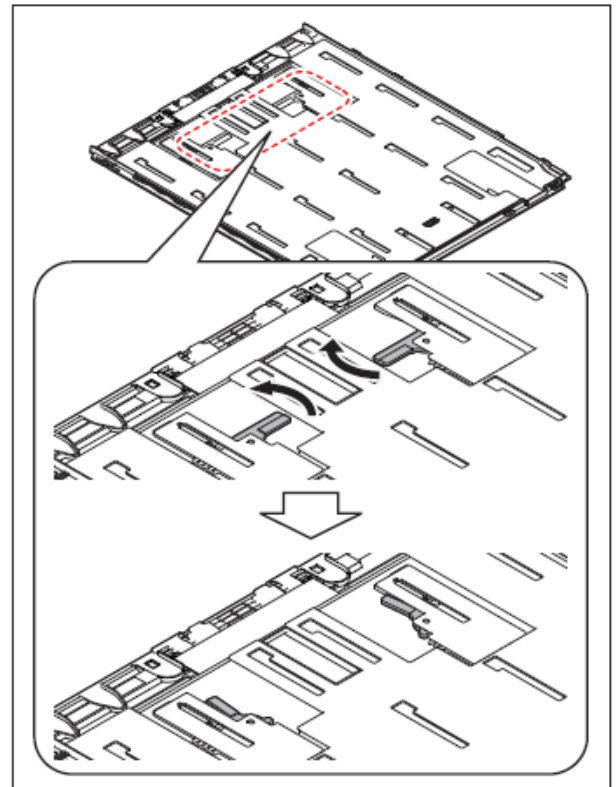
- When assembling the cassette, make sure there is no dust inside of it. If dust is found, clean it out with a mini vacuum cleaner.

- If the cassette in use has a cassette lock release lever, the tray lock (bolt mechanism) must be released before attaching the tray to the back plate. Use the opening tools for cassette to release the lock, then attach the tray to the back plate.

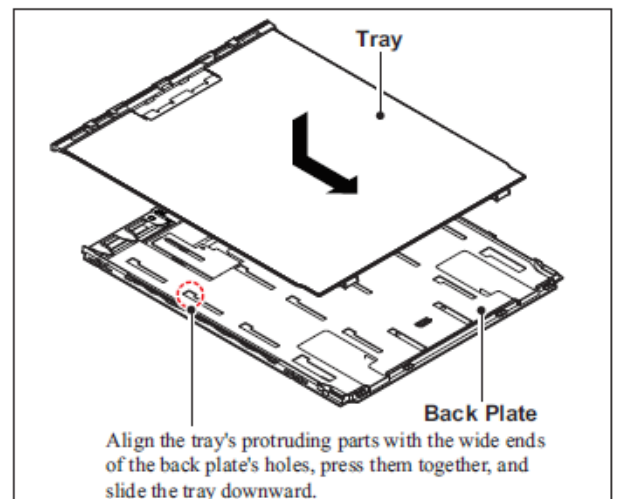
1) Attach the opening tools for cassette to the back plate.



2) Once the opening tools for cassette is attached, turn the back plate over and rotate the tray lock cams (x2) as shown in the diagram.



- When assembling and attaching the tray and back plate, line up all of the tray's protruding parts with the back plate's corresponding holes, and make sure they are pressed in correctly.



Things to do after assembly

- Erasing an image from the ejected plate

After re-attaching the tray to the back plate, check the following.

- Expose and check an X-ray image
When the image is shielded from light, the tray's protruding parts may not be properly aligned with the back plate's corresponding holes.

Chapter 6

Adjustments

An explanation of part adjustments necessary for repair and maintenance.

6.1 Oblique adjustments..... 6-2

6.1 Oblique adjustments

If the read image is oblique, adjust the tray abutment assembly flap.

Things to prepare

In addition to standard tools, the following tool(s) will be needed to perform this task.

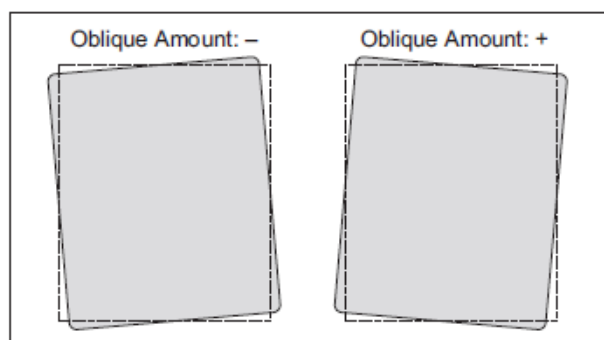
No.	Tool
1	A 14" × 17" size cassette and plate

Checking the oblique amount

1. Read a 14" × 17" sized image in normal read mode (175 μm), and measure the oblique amount.

If the oblique amount is more than ± 4 pixels, adjustment is necessary.

When an image is read using the service tool



IMPORTANT

If an image is read through normal ImagePilot operation, the image is vertically flipped.

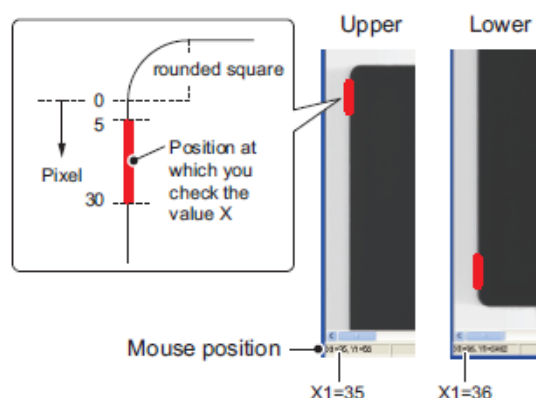
(Example) Reading an image with the service tool and confirming the amount of oblique motion.

1. Prepare a solid image.
 - Perform an appropriate amount of exposure (80kV, 2m, 10mAs) to the front side of the cassette.
2. Read the image by pressing the [Read Cassette] button on the [Device Settings] screen.
 - [Resolution]: 175μm

For details, refer to the ImagePilot Installation & Service Manual.
3. Display the image using the image tool.
 - [Display] [View Size] [No Pixel Skip]

4. Check the value of the green X on the upper screen and lower plate to measure the amount of oblique motion.

- The mouse position is shown on the status bar at the bottom of the screen. Position the cursor at the position indicated with the bold line in the figure to obtain the value X.



The following values are used in the figure above.

- X at the upper screen: 35

- X at the lower screen: 36

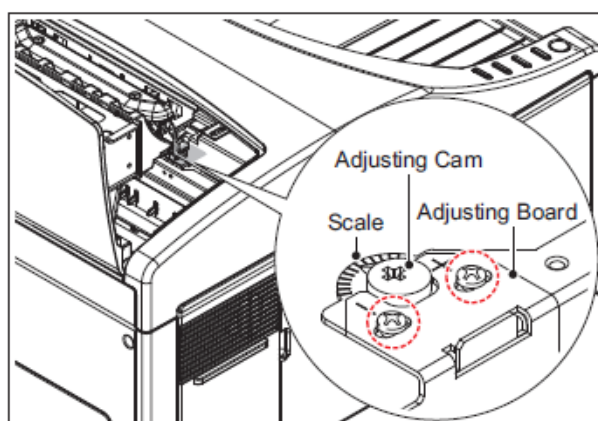
The oblique motion amount of the image, therefore, is -1 (35-36).

Because the value is within the ± 4 no adjustment is necessary.

Adjustment methods

Operate the upper insertion unit's adjusting cam. The adjusting cam can be accessed by opening the upper door.

1. Note the adjusting cam's current position (where it points on the scale).

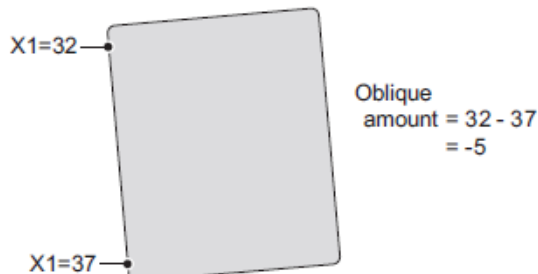


2. To check the post-adjustment position of the cam, check the motion amount of the adjustment cam to the oblique motion amount in the following table.

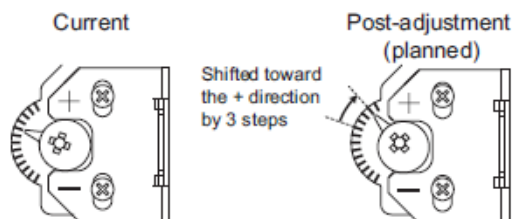
Oblique amount (pixels)	Movement from original position
-11 ~ -12	6 degrees in the "+" direction
-10 ~ -9	5 degrees in the "+" direction
-8 ~ -7	4 degrees in the "+" direction
-6 ~ -5	3 degrees in the "+" direction
+5 ~ +6	3 degrees in the "-" direction

Oblique amount (pixels)	Movement from original position
+7 ~ +8	4 degrees in the "-" direction
+9 ~ +10	5 degrees in the "-" direction
+11 ~ +12	6 degrees in the "-" direction

(Example)



1. Check the table to get the "distance moved from the original position".
Because in the case of the figure above, the motion amount is "-5", it is "shifted toward the + direction by 3 steps" from the current position.
2. Check the current position of the cam (which step on the scale it is pointing at) and the post-adjustment position.

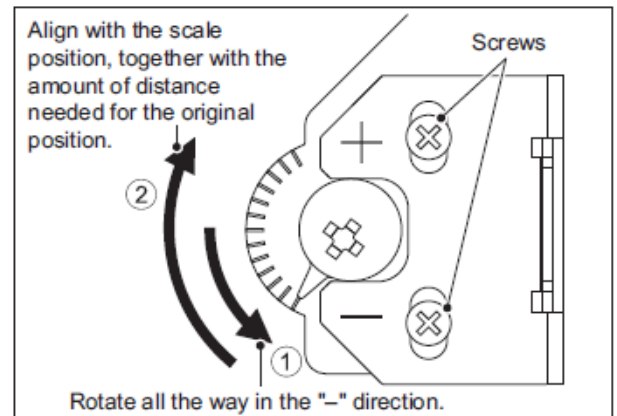


3. Loosen the adjusting board's screws (× 2).
4. Rotate the adjusting cam to its maximum "-" direction.

5. Check the adjusting cam's amount of movement for the oblique amount in the chart below, and slowly rotate it in the "+" direction.

IMPORTANT

- Rotate the adjusting cam equal to or less than 1 degree per second in the "+" direction. Rotating it too fast may damage the adjusting cam.
- If you have rotated the adjusting cam too much, start again from [process 4](#).

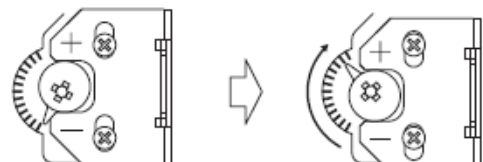


(Example)

1. Rotate the adjusting cam to its maximum "-" direction.



2. Slowly turn the adjustment cam to the planned post-adjustment position.



6. Tighten the adjusting board's screws (× 2) to secure it.

This concludes the adjustment process.

Make adjustments for all cassettes used in the facility to minimize the oblique amount as much as possible (to eliminate white streaks).

Rechecking/adjusting the oblique amount after initial adjustments

- Check irregularity correction data
If an error is found, perform irregularity correction.
For error details, refer to the ImagePilot Installation & Service Manual.

Chapter 7

Maintenance & Inspection

An explanation of how to maintain and inspect the device.

7.1	Maintenance/Inspection Schedule	7-2
7.2	Plate feeding path cleaning.....	7-2

7.1 Maintenance/Inspection Schedule

Maintenance/inspection should be performed according to the following schedule.

User duties

Maintenance/inspection content		Period
Plate	Plate cleaning	Once per week
Cassette	Cassette cleaning	Once per week
Reader	Cassette insertion slot cleaning	Once per week
	External cleaning of the device	Once per month
	Intake vent, exhaust outlet cleaning	Once per month
	Light Collection Unit cleaning (keeping them dust free)	If white lines error occurs
	Filter cleaning	When an error occurs

Refer to the operation manual for details.

Duties for maintenance service head

Maintenance/inspection content	Period
Roller cleaning (all rollers) Refer to "7.2 Plate feeding path cleaning (Page 7-2)".	When an error occurs
Antistatic Sheet cleaning Refer to "7.2 Plate feeding path cleaning (Page 7-2)".	When an error occurs

7.2 Plate feeding path cleaning

If dust gets into the plate feeding path, it should be cleaned as follows.

Cleaning area	Cleaning method
Rollers, guides, etc.	Use a vacuum cleaner etc. to suck out any dust.
Antistatic Sheet	Apply adhesive tape etc. to the antistatic sheet to remove any dust. ⚠ CAUTION Do not use the following items to clean the antistatic sheet, as they might scratch it. - An extremely sticky or adhesive substance - Vacuum cleaners etc. with powerful suction

Chapter 8

Appendix

A list of technical information for reference during device repair or maintenance.

8.1	Service Tool Screen (Unit test)	8-2
8.1.1	[Unit test] Screen	8-2
8.1.2	[Unit Test] (Check) Screen	8-3
8.1.3	[Error Hist] Screen	8-7
8.1.4	[Warning setting] Screen	8-8
8.1.5	[Report of reader] Screen	8-8
8.2	Device Wiring Diagram	8-9
8.2.1	REGIUS Σ Device Wiring Diagram	8-9
8.2.2	REGIUS Σ II Device Wiring Diagram	8-10

8.1 Service Tool Screen (Unit test)

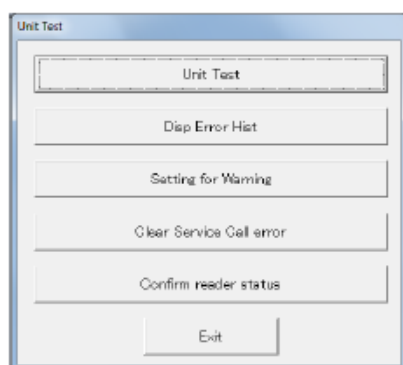
An explanation of the [Unit test] screen, part of a service tool which provides various maintenance functions for this unit.

8.1.1 [Unit test] Screen

[Unit test] Screen Display Process

1. Start up the Reader Settings Service Tool on an ImagePilot connected to this device.
The [Service Tool] screen will be displayed for reader settings.
2. Select [Reader Settings], then click the [Settings] button for [Unit Test].

[Unit test] Screen



Button	Content
[Unit Test] button	Allows you to perform an independent operation check for each device in the device. The [Unit Test] (Check) screen (Page 8-3) is displayed.
[Disp Error Hist] button	Allows you to check the error history. The [Error Hist] screen (Page 8-7) is displayed.
[Setting for Warning] button	Allows you to toggle the "No Test Registration" and image re-erase request dialogs ON/OFF. The [Warning setting] screen (Page 8-8) is displayed.
[Clear Service Call error] button	Used for a Service Call Error status. If a V-SYNC error occurs during the image read process, or a basic operation failure occurs in the base board (RAM/ROM cannot be accessed), the reader will be in Service Call Error status. Opening/closing the door or turning the power OFF/ON will not correct the error in this case. Repair/replace the part that caused the error, then click the [Clear Service Call error] button to clear the error. Refer to "3.4 Error Message Content & Countermeasures (Page 3-4)" for error codes applicable to Service Call Errors.
[Confirm reader status] button	Allows you to check reader adjustment results. The [Report of reader] screen (Page 8-8) is displayed.
[Exit] button	Allows you to exit the [Unit test] screen.

8.1.2 [Unit Test] (Check) Screen

IMPORTANT

- All [Unit Test] (Check) screen functions can even be executed during an error.
- If an error occurs while using this function (a blinking Error Lamp), use of "Unit Test" cannot resume until the error is corrected. If you want to continue using "Unit Test", open and close any door on the device to resolve the error status.

[Unit Test] (Check) screen display process

Click the [Unit Test] button at the [Unit test] screen (Page 8-2.)

[Unit Test] (Check) Screen

Unit Test

LED | Motor | Erase Unit

Ready LED
ON Blink OFF

Busy LED
ON Blink OFF

Error LED
ON Blink OFF

Erase LED
ON Blink OFF

ALL Setting
ON Blink OFF

Sensor status

Size detection S/1	ON	Unlock S	ON	AD converter error	OFF
Size detection S/2	ON	Insertion nip open close S	ON	Erase LED FAN error	ON
Size detection S/3	ON	Upper nip S	OFF	Erase LED breaking detection	OFF
Size detection S/4	ON	Lower nip S	OFF	SubScan connection detection	OFF
Cassette Lock S/R	OFF	V-sync S	OFF	PMT error	OFF
Cassette Lock S/L	OFF	Upper door S	OFF	Chassis Identifier	II
Cassette detection S	ON	Side door S	OFF	Main Board Identifier	I
Plate detection S	ON	Lower door S	OFF	Erase Unit Identifier	II
Tray HP S	OFF	Erase button	OFF	EraseLEDBoard Temperature	23.1 deg
Insertion nip HP S	OFF	Polygon lock detection	OFF		

Start Monitoring Back

Item/button	Explanation
Command control panel	The command control key group is made up of 3 panels. Panels can be viewed by clicking their tabs. The following text provides a description for each panel. • LED: Refer to "[LED] panel (Page 8-4)" • Motor: Refer to "[Motor] panel (Page 8-5)" • Eraser Unit: Refer to "[Erase Unit] panel (Page 8-6)"
[Sensor status]	An ON/OFF status is displayed for each sensor. Sensor status right after the reader starts up is displayed in black, and any sensor status which differs from that is displayed in red. To monitor and update sensor status, click the [Start Monitoring] button. Sensor status is updated every 3 seconds. IMPORTANT Other single device operation commands cannot be sent while the sensors are being monitored. For example, sensors cannot be monitored while operating a motor. If you want to run a different single device operation command, you'll have to stop sensor monitoring first.
[Start Monitoring] button/[Stop] button	• Clicking the [Start Monitoring] button will begin sensor status monitoring. It will then change to the [Stop] button on screen. • A progress bar is displayed above the [Stop] button during monitoring. • Clicking the [Stop] button will halt sensor status monitoring.
[Back] button	Takes you back to the [Unit test] (Check) screen.

[LED] panel

Enables you to check the operation for each of the LEDs used as status lamps.



Item/button		Explanation
[Ready LED]	[ON] button	Causes the Ready Lamp to light up.
	[Blink] button	Causes the Ready Lamp to blink.
	[OFF] button	Turns off the Ready Lamp.
[Busy LED]	-	Causes the Busy Lamp to light up, blink, or turn off. (Same as [Ready LED])
[Error LED]	-	Causes the Error Lamp to light up, blink, or turn off. (Same as [Ready LED])
[Erase LED]	-	Causes the Erase Lamp to light up, blink, or turn off. (Same as [Ready LED])
[All Setting]	-	Causes all lamps to light up, blink, or turn off. (Same as [Ready LED])

[Motor] panel

Allows you to check the operation for each motor.

IMPORTANT

If you only want to confirm motor operation, select [CW] or [CCW]. If using a cassette, refer to "[Unit Test: Cassette Feeding Flow \(Page 4-4\)](#)", and follow the process described for operation.

Item/button		Explanation
[Initialization]	[Init. All] button	Sets all motors at home position.
	[Init. Transport] button	All parts within the device will be moved to safe positions for feed. (Home position applies to all motors, so clicking this button has the same effect as clicking the [Init. All] button.)
[Cassette tray move motor]		Allows you to check tray moving motor operation. 1. Select which operation you would like performed at the Operation Selection column. 2. When selecting [CW] or [CCW] for operation, enter the number of rotations you want the motor to perform in the Set Rotations column. 3. Click the [Start] button. IMPORTANT The [CW] of the Cassette tray move motor does not work if the Tray HP Sensor is turned off. [CW] also stops if the Tray HP Sensor is turned off while it is working.
[Insertion nip motor]		Allows you to check loading NIP motor operation. 1. Select which operation you would like performed at the Operation Selection column. 2. When selecting [CW] or [CCW] for operation, enter the number of rotations you want the motor to perform in the Set Rotations column. 3. Click the [Start] button.
[Insertion feed motor]		Allows you to check loading/feeding motor operation. 1. Select which operation you would like performed at the Operation Selection column. 2. When selecting [CW] or [CCW] for operation, enter the number of rotations you want the motor to perform in the Set Rotations column. 3. Select [Erase Speed] to set the speed of the erase operation. (When selecting the [Transport Plate (Return)] operation.) 4. Select [Resolution] to set the resolution used during the image read operation. (When selecting the [Transport Plate (Send)] operation.) 5. Select [ElectroMag clutch] to set whether or not the magnetic clutch is operated. (When selecting the [Transport Plate (Send)] or [Transport Plate (Return)] operation.) • ON: Operates the magnetic clutch. • OFF: Halts magnetic clutch operation. (The magnetic clutch status will stay as it was when selected with the [ElectroMag clutch] button.) 6. Use [Cassette Size] to select the cassette size. (When selecting the [Transport Plate (Return)] operation.) 7. Click the [Start] button.

Item/button	Explanation
[ElectroMag clutch]	Allows you to check magnetic clutch operation. - Clicking the [ON] button will activate the magnetic clutch. - Clicking the [OFF] button will deactivate the magnetic clutch.
[Polygon]	Allows you to check polygon operation. - Clicking the [ON] button will rotate the polygon. - Clicking the [OFF] button will halt the polygon.
[PolygonLaser]	Allows you to check polygon and laser operations. - Clicking the [ON] button will rotate the polygon and cause the laser to emit light. - Clicking the [OFF] button will halt the polygon and deactivate the laser.
[SubScan feed motor]	Allows you to check sub-scan feeding motor operation. - Select which operation you would like performed at the Operation Selection column. - When selecting [CW] or [CCW] for operation, enter the number of rotations you want the motor to perform in the Set Rotations column. - Select [Erase Speed] to set the speed of the erase operation. (When selecting the [Transport Plate (Return)] operation.) - Select [Resolution] to set the resolution used during the image read operation. (When selecting the [Transport Plate (Send)] operation.) - Use [Cassette Size] to select the cassette size. (When selecting the [Transport Plate (Return)] operation.) - Click the [Start] button.
[SubScan nip motor]	Allows you to check sub-scan NIP motor operation. - Select which operation you would like performed at the Operation Selection column. - When selecting [CW] or [CCW] for operation, enter the number of rotations you want the motor to perform in the Set Rotations column. - Select [Resolution] to set the resolution used during the image read operation. (When selecting the [Transport Plate (Send) 1] or [Transport Plate (Send) 2] operation.) - Click the [Start] button.

[Erase Unit] panel

Enables you to check operation of the Eraser LED and Eraser LED cooling fan.

Item/button	Explanation
[Erase LED & Erase FAN]	Enables you to check operation of the Eraser LED and Eraser LED cooling fan. [Duration of turning on] enables you to set how many seconds the Eraser LED will stay lit. - Click the [ON] button. The Eraser LED will stay lit, and the Eraser LED cooling fan will continue operating for the designated number of seconds. (Clicking [OFF] during operation can immediately halt operation even if the designated number of seconds hasn't elapsed.)
[Erase FAN]	Allows you to check Eraser LED cooling fan operation. - Clicking the [ON] button will start Eraser LED cooling fan operation. - Clicking the [OFF] button will halt Eraser LED cooling fan operation.

8.1.3 [Error Hist] Screen

Error Hist Screen Display Process

Click the [Disp Error Hist] button at the [Unit test] screen (Page 8-2).



This can also be displayed from the "Reader Log" screen.

Refer to the ImagePilot Installation & Service Manual for details.

[Error Hist] Screen

Date	Time	Error code
2011-02-10	19:39:06	7810
2011-02-10	18:17:58	7810
2011-02-10	14:25:55	28100
2011-02-10	11:53:21	7812
2011-02-10	11:52:28	7812
2011-02-10	11:45:02	7810
2011-02-10	11:44:34	7003
2011-02-10	11:43:55	7810
2011-02-10	11:24:19	7810
2011-02-10	11:22:54	7812
2011-02-10	11:22:25	7810
2011-02-10	11:22:22	7811
2011-02-10	11:22:18	7810
2011-02-10	11:15:17	7810
2011-02-10	11:14:46	7811
2011-02-10	11:14:45	7810

Prev Next Accum. error list Exit

Item/button	Explanation
List	A chronological log of error occurrences is displayed here.
[Error Information]	A list of selected error information (messages displayed on the ImagePilot) is displayed.
[Prev] button	Available when the error history is displayed on more than one page, allowing you to cycle through pages.
[Next] button	
[Accum. error list] button	Frequency of error occurrence can be checked for each error code. The [Accum. error list] screen (Page 8-7) is displayed.
[Exit] button	Exits the [Error Hist] screen.

[Accum. error list] screen

Frequency of error occurrence is displayed for each error code.

Error ID	count
7000	1
7003	2
7810	24
7811	3
7812	11
24000	1
27002	6

Back Save Table

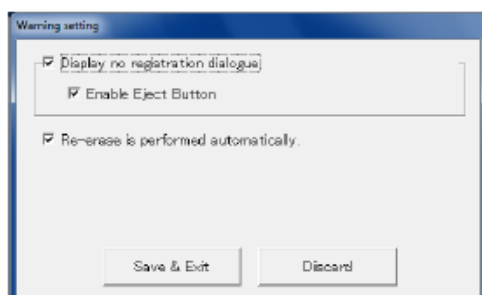
Item/button	Explanation
[Back] button	Exits the [Accum. error list] screen.
List	Frequency of error occurrence is displayed for each error code.
[Save Table] button	Enables you to output an "Accum. error list" file in text format (.txt). Pressing this button will display the [Save As] dialog, so choose any name to save the file as.

8.1.4 [Warning setting] Screen

Warning setting Screen Display Process

Click the [Setting for Warning] button at the "Unit test" screen (Page 8-2).

[Warning setting] Screen



Item/button	Explanation
[Display no registration dialogue]	Sets whether or not the "Study is not registered." dialog is displayed when loading a cassette into the device without putting the ImagePilot in image capture standby mode (without clicking the CR Modality button). By filling in the checkbox, this dialog will be displayed. (Default: checkbox filled in)
[Enable Eject Button]	Sets whether or not a cassette [Eject] button will be displayed, enabling the user to eject the cassette, along with the "Study is not registered." dialog. (By using the [Eject] button, the cassette can be removed without using the lever inside the upper door.) By filling in the checkbox, this button will be displayed. (Default: checkbox filled in)

IMPORTANT

When installing an ImagePilot client, removing the check mark is recommended. This dialog will be displayed for both the ImagePilot server and ImagePilot client, so if the checkbox is filled in to display the [Eject] button, the server's [Eject] button might be clicked before the CR modality button is clicked.

Item/button	Explanation
[Re-erase is performed automatically.]	Erase speed is determined from the image signal. When a high dose of X-rays are used for exposure, and the signal value is at 4095 or above, it is unknown how many mR of X-rays have hit the plate, so one of the following 2 controls can be selected. (1) Automatically run the erase process a second time. No message display. (2) Display a message and select whether or not to run the erase process a second time. The [Continue] button and [Eject] button are displayed together with this message: "Excessive X-ray dosage at examination. Need to be re-erased." Here you can select to run the erase process again, or eject the cassette. When filling in the checkbox, Control (1) will be performed. (Default: checkbox filled in)
[Save & Exit] button	Saves the selected settings and exits the [Warning Settings] screen. The change in the settings are reflected after restarting the Image Pilot.
[Discard] button	Exits the [Warning Settings] screen without saving settings.

HINT

This setting can be changed for both the service tool and user tool.

8.1.5 [Report of reader] Screen

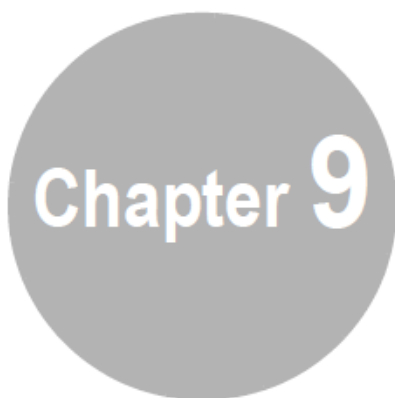
Report of reader Screen Display Process

Click the [Confirm reader status] button at the "Unit test" screen (Page 8-2).

[Report of reader] Screen

Cassette type	Resolution (um)	Cassette Size	Normal	Low	High
Regular	175	14x17			
Regular	87.5	14x17			
Cassette type	Resolution (um)	Normal	Low	High	
Regular	175	1685	1408	1989	
Regular	87.5	1760	1498	2125	

Item/button	Explanation
[Back] button	Exits the [Report of reader] screen.
Chart	Shows adjustment results set to the reader.
[Save table] button	Enables you to output a file of the displayed content in text format (.txt). Pressing this button will display the "Save as..." dialog, so choose any name to save the file as.



Chapter 9

Revision History

9.1	Manual revision history.....	9-2
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9.1 Manual revision history

This manual was revised as follows.

Version	Issue Date	Content
A	Apr. 19, 2012	• New (The same manual is used for both REGIUS SIGMA and SIGMA2)
01	Apr. 1, 2013	• Change of the company name.

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